

A Unified Formula Atlas for the Thue–Morse Sequence

Digital, valiative, Pascal, automatic, finite-difference, spectral, and analytic representations

Merged and expanded companion report to the Fabius–Rvachev formal development

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Abstract

Four draft reports in the Fabius-function research-frontier directory developed partly overlapping collections of formulae for the Thue–Morse sequence. This document merges them into one proof-oriented atlas, removes duplicated arguments, reconciles notation and sign conventions, and retains the strongest version of each genuinely different result. The signed sequence $\varepsilon_n = (-1)^{\text{wt}_2(n)}$ and the zero–one sequence $\tau(n) = \text{wt}_2(n) \bmod 2$ are treated simultaneously.

The resulting network begins with binary recursion, block concatenation, XOR characters, floor and fractional-part sums, trigonometric signs, factorial and central-binomial valuations, Catalan valuations, and Kummer carry cocycles. It then develops the Pascal-row viewpoint in several directions: Glaisher’s odd entry count, the existing exact binomial–logarithm formula, a logarithm-free modulo-three formula, sparse power-of-two probes, Boolean Möbius inversion, multinomial support counts, and a sparse Prouhet theorem on the odd support of an arbitrary Pascal row.

The lacunary product

$$\sum_{n \geq 0} \varepsilon_n z^n = \prod_{j \geq 0} (1 - z^{2^j})$$

serves as the central analytic engine. From it we obtain Mahler equations, cyclotomic factorizations, a ruler-function Euler transform, convolution, partition, Bell-polynomial, and Hessenberg-determinant formulae, an algebraic formal series over $\mathbb{F}_2$, an integral algebraic lift, exact finite differences, all dyadic power moments, Walsh and ordinary Fourier transforms, Riesz products, autocorrelation recurrences, iterated prefix sums, and the bridge back to the Fabius and Rvachev functions.

New material added in the unification includes an explicit dyadic weight-distribution chapter, a systematic centered-moment parity theorem, and a Mellin–Dirichlet theory. In particular, the shifted Dirichlet series

$$D(s, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n + a)^s}$$

continues to an entire function, satisfies an exact dyadic parameter equation, and vanishes at every nonpositive integer. The proof is driven by the super-polynomial flatness

$$\log \prod_{j \geq 0} (1 - e^{-2^j t}) = -\frac{\log^2(1/t)}{2 \log 2} + \mathcal{O}(\log(1/t)) \quad (t \downarrow 0).$$

A second consolidation pass merged three parallel sibling atlases into this document. It contributed the weighted digit enumerator $\prod_j (1 + uz^{2^j})$ and the subtraction/borrow cocycle; self-contained proofs that the unit circle is a natural boundary and that the Hankel determinants never vanish, with the sharp irrationality exponent $\mu(\Theta_b) = 2$ of the Thue–Morse–Mahler numbers stated from the literature; a Mahler functional equation for the autocorrelation series; a chapter on the combinatorics and dynamics of the infinite word (uniform recurrence, aperiodicity, Thue’s

overlap theorem, critical exponent two, linear factor complexity); and a chapter on rarefied sums – exact residue-class sums modulo three, Gelfond’s optimal exponent $\log_4 3$, a matrix recursion for odd moduli, and Newman’s positivity phenomenon with its modern extensions along primes, squares, and cubes.

Every elementary assertion is proved in the text; deeper historical results about transcendence, substitution spectra, and evaluated products are clearly identified and cited. No mathematical priority is claimed for identities merely derived here, and no formula is silently represented as already formalized in Lean.

Contents

1	Four-draft merge	5
1.1	Two normalizations	5
1.2	The Pascal-row formula being complemented	5
1.3	Merge policy	5
1.4	A compact formula index	6
2	Binary character and automata	7
2.1	Recursion and uniqueness	7
2.2	Binary concatenation and reflection	8
2.3	Exclusive-or is the natural group law	8
2.4	Finite automata, matrices, and tensor powers	9
2.5	The sequence is primitive recursive	10
2.6	Exact weight distribution in a dyadic block	11
2.7	Evil and odious enumerators	11
3	Digit and trigonometric formulae	12
3.1	Extracting one binary digit	12
3.2	Dyadic floor and fractional-part sums	13
3.3	Square waves and a sine-sign product	14
3.4	The sign of the Fabius sinc product	14
3.5	The ruler transition	16
4	Valuations and carries	16
4.1	Legendre’s formula	16
4.2	A single central binomial coefficient	17
4.3	Catalan valuations	17
4.4	Kummer’s carry cocycle	18
4.5	Subtraction and borrow counts	18
5	Pascal support and Boolean inversion	19
5.1	Lucas’s theorem as a submask criterion	19
5.2	A logarithm-free Pascal-row formula modulo three	20
5.3	A one-parameter row statistic	20
5.4	Sparse Pascal probes for individual bits	21
5.5	Odd multinomial coefficients and exact r -nomial logarithms	21
5.6	Boolean weight enumerators and Möbius inversion	22

6	Generating products	24
6.1	The Boolean-cube mother identity	24
6.2	The finite Prouhet polynomial	24
6.3	Reciprocity and cyclotomic multiplicities	25
6.4	The infinite product and Mahler equation	26
6.5	A weighted digit enumerator	27
6.6	Thue–Morse constants in every base	28
7	Euler transforms and algebraic series	28
7.1	The logarithmic ruler transform	28
7.2	A nonlocal convolution recurrence	29
7.3	Partition and Bell-polynomial formulae	29
7.4	A Hessenberg determinant	30
7.5	The zero–one series over $\mathbb{F}_2$	31
7.6	An integral algebraic lift	32
8	Natural boundary and Hankel rigidity	34
8.1	The unit circle is a natural boundary	34
8.2	Hankel determinants never vanish	35
8.3	Transcendental values and the sharp irrationality exponent	36
9	Finite differences and moments	36
9.1	The mixed-difference master identity	36
9.2	Exact multiple-integral remainder	37
9.3	Arbitrary translation weights	38
9.4	Prouhet cancellation and the first surviving moment	38
9.5	The finite exponential transform	39
9.6	All power moments: a finite composition formula	40
9.7	Bell-polynomial compression and cumulants	40
9.8	Centered moments and a parity selection rule	41
9.9	All binomial-basis moments	42
10	Sparse Prouhet identities	43
11	Prefix sums and the Fabius bridge	45
11.1	The first prefix discrepancy	45
11.2	Iterated inclusive prefixes	46
11.3	Terminal zero runs at dyadic boundaries	47
11.4	Why this is the discrete spine of the Fabius theory	47
12	Fourier transforms	48
12.1	A finite Fourier product	48
12.2	Sine and cosine products as Thue–Morse sums	48
12.3	The exact dyadic discrete Fourier transform	49
13	Walsh, Riesz, and autocorrelation	50
13.1	A one-point Walsh spectrum	50
13.2	Rademacher step functions	50
13.3	Finite Riesz products	50
13.4	Finite and limiting autocorrelation	51
13.5	A Mahler equation for the autocorrelation series	53

14 Word combinatorics and dynamics	53
14.1 Uniform recurrence and letter frequencies	53
14.2 Aperiodicity	54
14.3 Overlap-freeness and the critical exponent	55
14.4 Complement and reversal closure	55
14.5 Linear factor complexity	55
15 Rarefied sums and Newman’s phenomenon	57
15.1 Root-of-unity filters at dyadic endpoints	57
15.2 Complete evaluation modulo three	58
15.3 Gelfond’s uniform exponent	58
15.4 A matrix recursion for odd moduli	59
15.5 The global Newman theorem and the arithmetic frontier	60
16 Mellin–Dirichlet continuation	61
16.1 Quadratic logarithmic decay near one	61
16.2 The shifted Dirichlet family	62
16.3 Mellin representation and entire continuation	63
16.4 The harmonic kernel	64
17 Thue–Morse exponent products	64
17.1 Pair reduction	64
17.2 A two-parameter master product	64
17.3 The Woods–Robbins product	66
17.4 Three dyadic block-product limits	67
18 Base-q generalization	68
19 Formula selection and Lean roadmap	69
19.1 Formalized modules (August 2026)	69
19.2 Which representation is best?	74
19.3 A compact verification script	75
19.4 A dependency-aware formalization order	75
19.5 Proof-dependency graph	77
20 Worked examples	77
20.1 One index through many representations	77
20.2 An all-order dyadic moment	78
20.3 A Fourier sample	78
20.4 Autocorrelation from the recursion	78
20.5 A complete four-bit block	79
20.6 A carry-corrected addition	79
21 Conclusion	79
A First 64 terms	80
B Identity dictionary	80
C Draft concordance	81

1 Scope, conventions, and the four-draft merge

1.1 Two normalizations

Write the binary expansion of $n \in \mathbb{N} = \{0, 1, 2, \dots\}$ as

$$n = \sum_{j \geq 0} b_j(n) 2^j, \quad b_j(n) \in \{0, 1\}, \quad (1.1)$$

with only finitely many nonzero digits. Its binary weight and the two standard normalizations of the Thue–Morse sequence are

$$\text{wt}_2(n) = \sum_{j \geq 0} b_j(n), \quad \tau(n) = \text{wt}_2(n) \bmod 2 \in \{0, 1\}, \quad \varepsilon_n = (-1)^{\text{wt}_2(n)} = 1 - 2\tau(n). \quad (1.2)$$

Thus

$$\tau(n) = \frac{1 - \varepsilon_n}{2}, \quad \varepsilon_n = 1 - 2\tau(n). \quad (1.3)$$

The signed normalization is usually cleaner for products, transforms, and cancellation; the zero–one normalization is natural for automata, base expansions, and the formula in the source exposition.

We use $a \oplus b$ for bitwise exclusive-or and $k \preceq_2 n$ when every one-bit of k is also a one-bit of n . Empty products are 1, empty sums are 0, and $0^0 = 1$ when it occurs in a finite moment identity. The valuation $\nu_2(m)$ is the exponent of 2 in a positive integer m , with $\nu_2(1) = 0$.

1.2 The Pascal-row formula being complemented

Section 11.8 of the main Fabius–Rvachev exposition defines

$$\mathcal{X}(n) = \sum_{k=0}^n (-1)^{\binom{n}{k}} \quad (1.4)$$

and proves the exact-power identity

$$n + 1 - \mathcal{X}(n) = 2^{\text{wt}_2(n)+1}. \quad (1.5)$$

Consequently

$$\tau(n) = \frac{1 + (-1)^{\log_2 \left(n+1 - \sum_{k=0}^n (-1)^{\binom{n}{k}} \right)}}{2}. \quad (1.6)$$

The logarithm is exact because its argument was first proved to be a power of two. The formula hides the binary expansion inside the parity pattern of a whole Pascal row. The present atlas asks what other structures carry the same parity character.

1.3 Merge policy

The four input drafts were not independent papers: large parts shared the same binary, valuation, generating-product, Prouhet, and Fourier skeleton. A literal concatenation would repeat the same proof several times while obscuring the interesting differences. The unified text therefore follows three rules.

1. A theorem common to several drafts appears once, in the strongest useful form, with one proof and cross-references to its later specializations.

2. Material unique to a draft is retained when it is mathematically distinct: in particular the modulo-three Pascal formula, multinomial logarithms, sparse Pascal Prouhet identities, Boolean Möbius inversion, determinant formula, integer algebraic lift, odious/evil enumerators, autocorrelation, Dirichlet integrals, and the explicit Fabius-prefix bridge.
3. Statements added during the merge are marked by context rather than by novelty rhetoric. The report claims derivation, not historical priority.

This unified atlas was itself produced in parallel: three sibling consolidations of the same four drafts were written independently, and a second merge pass (26 August 2026) folded their distinctive material into the present document under the same three rules. The additions are listed in the abstract and in the concordance of [appendix C](#).

Proof-status convention. The repository already contains formal developments of the binary recurrence, finite generating product, Prouhet cancellations, iterated prefixes, and the binomial–logarithm evaluator. The paper proofs below make many additional results natural Lean targets, but do not assert that those declarations already exist. A dependency-aware formalization map is given in [section 19](#). Since 26 August 2026 the formalization of this atlas is underway in the same repository: nineteen new compiled Lean modules now cover the Boolean-cube master identities, the valuation and digit chapters, the enumerators, block-polynomial algebra with reciprocity and the cyclotomic table, the complete power, binomial, centered, and sparse moment composition formulas, the general finite-difference engine, the sparse Prouhet identities, Lucas’s submask criterion and the Walsh spectrum, the $\mathbb{F}_2$ Artin–Schreier layer with uniqueness and the integer-lift parity, the Euler transform of the ruler function with the prefix and discrepancy theory, rarefaction for every odd modulus, the finite autocorrelation and Mahler equations, and the words layer — aperiodicity, Thue’s overlap-freeness, and uniform recurrence with a linear bound. Statements proved in Lean carry their declaration names in the text, typeset as `Fabius.thueMorseSign`; see [section 19.1](#) for the module map. Many are proved in Lean in *more* generality than stated here, and the text records the stronger forms.

1.4 A compact formula index

The table is a navigation device rather than a substitute for hypotheses and proofs. Here $P_m(z) = \sum_{n < 2^m} \varepsilon_n z^n$, and $E(z) = \sum_{n \geq 0} \varepsilon_n z^n$.

Viewpoint	Representative exact formula
Binary recursion	$\varepsilon_{2n} = \varepsilon_n, \varepsilon_{2n+1} = -\varepsilon_n.$
Block concatenation	$\varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r$ for $0 \leq r < 2^m.$
XOR character	$\varepsilon_{a \oplus b} = \varepsilon_a \varepsilon_b.$
Digit product	$\varepsilon_n = \prod_j (1 - 2b_j(n)).$
Floors	$\varepsilon_n = (-1)^{\sum_{j \geq 0} \lfloor n/2^j \rfloor}.$
Fractional parts	$\text{wt}_2(n) = \sum_{j \geq 1} \{n/2^j\}.$
Factorials	$\varepsilon_n = (-1)^{n - \nu_2(n)!}.$
Central binomial	$\varepsilon_n = (-1)^{\nu_2 \binom{2n}{n}}.$
Carries	$\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2 \binom{a+b}{a}}.$
Pascal support	$\#\{k : \binom{n}{k} \text{ odd}\} = 2^{\text{wt}_2(n)}.$
Sparse Pascal probes	$\varepsilon_n = (-1)^{\sum_j \binom{n}{2^j}}.$
Log-free Pascal residue	$\tau(n) = (2^{\text{wt}_2(n)} \bmod 3) - 1.$
Multinomial support	$O_r(n) = r^{\text{wt}_2(n)}.$

Viewpoint	Representative exact formula
Boolean Möbius	$\mu_2(k, n) = \varepsilon_{n \oplus k}$ for $k \preceq_2 n$.
Finite product	$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j})$.
Infinite product	$E(z) = \prod_{j \geq 0} (1 - z^{2^j})$, $ z < 1$.
Mahler equation	$E(z) = (1 - z)E(z^2)$.
Euler transform	$\log E(z) = -\sum_{r \geq 1} (2^{\nu_2(r)+1} - 1) z^r / r$.
Finite differences	$\sum_{n < 2^m} \varepsilon_n f(x + n) = (-1)^m \Delta_1 \cdots \Delta_{2^{m-1}} f(x)$.
Prouhet	$\sum_{n < 2^m} \varepsilon_n p(n) = 0$ when $\deg p < m$.
Sharp moment	$\sum_{n < 2^m} \varepsilon_n n^m = (-1)^m m! 2^{\binom{m}{2}}$.
Sparse Prouhet	$\sum_{k \preceq_2 n} \varepsilon_k p(k) = 0$ when $\deg p < \text{wt}_2(n)$.
Prefix discrepancy	$\sum_{n < N} \varepsilon_n = 0$ for even N , $\varepsilon_{(N-1)/2}$ for odd N .
Walsh transform	A length- 2^m block is one Walsh frequency.
Ordinary DFT	$\hat{\varepsilon}_m(k) = \prod_{j < m} (1 - e^{-2\pi i k 2^j / 2^m})$.
Riesz density	$2^{-m} P_m(e^{2\pi i x}) ^2 = \prod_{j < m} (1 - \cos(2\pi 2^j x))$.
Characteristic two	$(1+x)^3 T(x)^2 + (1+x)^2 T(x) + x = 0$ in $\mathbb{F}_2[[x]]$.
Weighted enumerator	$\sum_n u^{\text{wt}_2(n)} z^n = \prod_j (1 + u z^{2^j})$.
Subtraction cocycle	$\varepsilon_{a-b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2 \binom{a}{b}}$ for $b \leq a$.
Autocorrelation Mahler	$2zC(z) = 1 - (1-z)^2 C(z^2)$.
Natural boundary	E continues across no arc of $ z = 1$.
Hankel rigidity	$\det(\varepsilon_{i+j})_{i,j < r} \neq 0$ for every r .
Rarefaction mod 3	$(A_{2u,r}^{(3)})_r = 3^{u-1} (2, -1, -1)$.
Gelfond exponent	$3^{m/2} \leq \sup_x P_m(e^{2\pi i x}) \leq \frac{2}{\sqrt{3}} 3^{m/2}$.
Newman positivity	$\sum_{n < X, 3 n} \varepsilon_n > 0$ for all $X \geq 1$.
Critical exponent	Overlap-free; squares occur, exponent exactly 2.
Factor complexity	$\ell + 1 \leq p(\ell) \leq 6\ell - 4$ for $\ell \geq 1$.
Dirichlet–Mellin	$D(s, a) = \Gamma(s)^{-1} \int_0^\infty t^{s-1} e^{-at} E(e^{-t}) dt$. <i>Delivered</i> : boundary flatness, convergence, the dyadic equation, the analytic infinite product, the Mellin representation, and the entire continuation (ThueMorseBoundaryFlatness.lean, ThueMorseDirichlet.lean, ThueMorseInfiniteProduct.lean, ThueMorseMellin.lean, ThueMorseEntireContinuation.lean).

2 Binary character, blocks, automata, and enumerators

2.1 Recursion and uniqueness

Theorem 2.1 (Binary recursion and uniqueness). *The signed sequence satisfies*

$$\varepsilon_0 = 1, \quad \varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n. \quad (2.1)$$

Equivalently,

$$\tau(0) = 0, \quad \tau(2n) = \tau(n), \quad \tau(2n+1) = 1 - \tau(n). \quad (2.2)$$

Either system and its initial value determines the sequence uniquely.

Proof. Appending a terminal binary digit 0 leaves the digit sum unchanged; appending a terminal 1 increases it by one. For uniqueness, every positive integer is $2q$ or $2q + 1$ with $q < n$, so strong induction eventually reduces its value to the initial value at zero. $\square$

Let $W_m = \tau(0)\tau(1) \cdots \tau(2^m - 1)$, and let $\overline{W}$ denote bitwise complement. The recurrence gives

$$W_{m+1} = W_m \overline{W_m}. \quad (2.3)$$

Thus the infinite word is the fixed point beginning with 0 of

$$0 \mapsto 01, \quad 1 \mapsto 10. \quad (2.4)$$

In signed form a block is followed by its negative.

2.2 Binary concatenation and reflection

Theorem 2.2 (Block concatenation). *For $m, q \geq 0$ and $0 \leq r < 2^m$,*

$$\text{wt}_2(2^m q + r) = \text{wt}_2(q) + \text{wt}_2(r), \quad (2.5)$$

$$\varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r, \quad (2.6)$$

$$\tau(2^m q + r) = \tau(q) \oplus \tau(r). \quad (2.7)$$

Proof. Multiplication by 2^m shifts the binary expansion of q by m places. Since $r < 2^m$, adding r fills only those vacated positions and creates no carry. The bits concatenate, so weights add and parities follow. $\square$

Corollary 2.3 (Reflection and bit-position symmetry). *For $0 \leq n < 2^m$,*

$$\varepsilon_{2^m - 1 - n} = (-1)^m \varepsilon_n, \quad \tau(2^m - 1 - n) = \tau(n) \oplus (m \bmod 2). \quad (2.8)$$

Permuting the first m binary positions leaves ε_n and $\tau(n)$ unchanged.

Proof. Within m positions, $2^m - 1 - n$ is the bitwise complement of n , so its weight is $m - \text{wt}_2(n)$. A permutation of positions leaves their sum unchanged. $\square$

Formal versions. The signed reflection (2.8) is `Fabius.thueMorseSign_dyadic_complement` (`DyadicClosedForm.lean`), proved by induction on m through the two-scale laws rather than through the weight of the complement; its only hypothesis is the stated $n < 2^m$. Only the signed form is named in Lean; the bit form is its parity reading, and the digit-permutation invariance is the invariance of wt_2 under permuting positions.

2.3 Exclusive-or is the natural group law

Theorem 2.4 (XOR character). *For all $a, b \geq 0$,*

$$\tau(a \oplus b) = \tau(a) \oplus \tau(b), \quad \boxed{\varepsilon_{a \oplus b} = \varepsilon_a \varepsilon_b}. \quad (2.9)$$

Thus $n \mapsto \varepsilon_n$ is a character of the Boolean group $\bigoplus_{j \geq 0} \mathbb{Z}/2\mathbb{Z}$.

Proof. At each position, XOR adds the two bits modulo two. Summing the coordinate identities modulo two gives $\text{wt}_2(a \oplus b) \equiv \text{wt}_2(a) + \text{wt}_2(b) \pmod{2}$. $\square$

Corollary 2.5 (XOR autocorrelation and Hamming spheres). *Fix $m \geq 0$, $0 \leq a < 2^m$, and $0 \leq r \leq m$. Then*

$$\sum_{n=0}^{2^m-1} \varepsilon_n \varepsilon_{n \oplus a} = 2^m \varepsilon_a, \quad (2.10)$$

$$\sum_{\substack{0 \leq n < 2^m \\ \text{wt}_2(n \oplus a) = r}} \varepsilon_n = \varepsilon_a (-1)^r \binom{m}{r}. \quad (2.11)$$

Proof. The first summand is termwise ε_a . For the second, put $h = n \oplus a$. Then $\varepsilon_n = \varepsilon_a \varepsilon_h = \varepsilon_a (-1)^r$, and there are $\binom{m}{r}$ masks of weight r . $\square$

Formal versions. `Fabius.sum_thueMorseSign_mul_xor` (the autocorrelation, which the Lean proof establishes termwise from the character identity, with no alignment hypothesis on a) and `Fabius.sum_thueMorseSign_hammingSphere` (the sphere sums, by reindexing along the XOR involution), in `ThueMorseEnumerators.lean`.

2.4 Finite automata, matrices, and tensor powers

The recurrence gives a two-state automaton: begin in state $+1$, read the binary digits in either order, and flip the state for each one-bit. In matrix form, let

$$M_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad e_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (2.12)$$

If $n = \sum_{j=0}^{m-1} b_j 2^j$, then

$$M_{b_0} \cdots M_{b_{m-1}} e_0 = \begin{pmatrix} 1 - \tau(n) \\ \tau(n) \end{pmatrix}. \quad (2.13)$$

The matrices commute, reflecting the fact that only the number of one-bits matters. In natural binary order,

$$(\varepsilon_0, \dots, \varepsilon_{2^m-1})^\top = ((1, -1)^\top)^{\otimes m}. \quad (2.14)$$

Hence every residue subsequence in the 2-kernel is either ε or $-\varepsilon$:

$$(\varepsilon_{2^m n + r})_{n \geq 0} = \varepsilon_r (\varepsilon_n)_{n \geq 0}. \quad (2.15)$$

This is the finite-kernel criterion for 2-automaticity in its smallest possible form.

Algorithm 2.6 (XOR-fold evaluator). For a fixed-width machine word, fold the bits by XORing with successively larger right shifts. The final least significant bit is $\tau(n)$.

```
x := n
x := x xor (x >> 1)
x := x xor (x >> 2)
x := x xor (x >> 4)
x := x xor (x >> 8)
... continue while the shift is below the word width ...
tau := x & 1
epsilon := 1 - 2*tau
```

2.5 The sequence is primitive recursive

The automaton above bounds the *memory* needed to generate the sequence. The following statement bounds the *recursion* needed to define it, and it is what licenses the fixed, computable step count of Algorithm 2.6.

Theorem 2.7 (Primitive recursiveness). *The binary weight wt_2 , the zero–one sequence τ , and the signed sequence ε are primitive recursive functions. Consequently all three are computable, and the evil and odious sets are decidable.*

This is not quite the content of (2.2). That recurrence computes $\tau(n)$ from $\tau(\lfloor n/2 \rfloor)$, whereas the official definition of a primitive recursive function permits a recursive call only at the immediate predecessor $n - 1$. A recursion that descends by halving is a *course-of-values* recursion, and the device that reduces it to the official scheme is to carry the entire history along. We isolate that device once, in the generality every base- b digit recursion needs.

Lemma 2.8 (Recursion along a primitive recursive descent). *Let d be primitive recursive with $d(n) < n$ for every $n \geq 1$, let g be primitive recursive, and let z be a constant. Then the unique function f satisfying*

$$f(0) = z, \quad f(n) = g(n, f(d(n))) \quad (n \geq 1) \quad (2.16)$$

is primitive recursive. No hypothesis is imposed on $d(0)$.

Proof. Let $F(n)$ code the list $\langle f(0), \dots, f(n-1) \rangle$ of all values below n , and let $\lambda(L, i)$ be the primitive recursive lookup that returns the i -th entry of the list coded by L when i is in range, and reports failure otherwise. Then F obeys an *ordinary* primitive recursion,

$$F(0) = \langle \rangle, \quad F(n+1) = F(n) \frown \begin{cases} g(n, \lambda(F(n), d(n))) & \text{if that lookup succeeds,} \\ z & \text{otherwise.} \end{cases}$$

For $n \geq 1$ the entry demanded carries index $d(n) < n$ and is therefore already present in the length- n list $F(n)$, so the appended value is $g(n, f(d(n))) = f(n)$; at $n = 0$ the list is empty, every lookup fails, and the appended value is $z = f(0)$. Note that the fallback must guard the *whole* step expression rather than the lookup alone – defaulting only the lookup would append $g(0, z)$ instead of z – and that this is exactly what frees the statement from any assumption on $d(0)$. Hence F is primitive recursive, and so is $f(n) = \lambda(F(n+1), n)$. $\square$

Proof of Theorem 2.7. Deleting the last binary digit of n deletes exactly that digit from the digit sum, so

$$\text{wt}_2(n) = \text{wt}_2(\lfloor n/2 \rfloor) + (n \bmod 2), \quad (2.17)$$

an identity that holds at $n = 0$ as well, both sides being zero. This is (2.16) with $d(n) = \lfloor n/2 \rfloor$, which is primitive recursive and satisfies $d(n) < n$ for $n \geq 1$; with $z = 0$; and with $g(n, s) = s + (n \bmod 2)$, primitive recursive because addition and remainder are. Lemma 2.8 then makes wt_2 primitive recursive, whence $\tau = \text{wt}_2 \bmod 2$ is primitive recursive by composition, and so is $\varepsilon = 1 - \tau$ by (1.2). Primitive recursive functions are total and computable, and a set whose characteristic function is primitive recursive is decidable. $\square$

Remark 2.9. Identity (2.17) is the unsigned form of (2.2): reducing it modulo two gives $\tau(n) = \tau(\lfloor n/2 \rfloor) \oplus (n \bmod 2)$, which is the two clauses $\tau(2n) = \tau(n)$ and $\tau(2n+1) = 1 - \tau(n)$ written as one. The bounded expression $\text{wt}_2(n) = \sum_{i < n} (\lfloor n/2^i \rfloor \bmod 2)$ yields a second, recursion-free proof, because bounded sums of primitive recursive functions are primitive recursive; the descent formulation is preferred here because it is the one that survives the passage to base q in Section 18.

Formal versions. [Theorem 2.7](#) and [Lemma 2.8](#) are fully formal (`ThueMorseComputability.lean`). The descent scheme is `Fabius.primrec_of_descent`, stated for an arbitrary primitive recursive d with $d(n) < n$ and an arbitrary Primcodable codomain — more general than the halving case needed here, which is recorded separately as `Fabius.primrec_of_halving`. The halving identity (2.17) is `Fabius.binaryWeight_div_two`, and the three primitive-recursiveness statements are `Fabius.primrec_binaryWeight`, `Fabius.primrec_thueMorseBit`, and `Fabius.primrec_thueMorseSign`; `Fabius.nat_primrec_binaryWeight` and `Fabius.nat_primrec_thueMorseBit` record the same facts in the textbook form `Nat.Primrec` for functions $\mathbb{N} \rightarrow \mathbb{N}$. The computability corollaries are `Fabius.computable_thueMorseBit` and `Fabius.computable_thueMorseSign`, and decidability of the odious set is `Fabius.primrecPred_thueMorseBool`. The signed sequence needs no integer arithmetic at all: every function out of `Bool` is primitive recursive, so ε is the Boolean word followed by $b \mapsto \pm 1$. The Boolean word itself (`Fabius.thueMorseBool`) carries the automaticity clauses `Fabius.thueMorseBool_two_mul` and `Fabius.thueMorseBool_two_mul_add_one`. Finally `Fabius.primrec_thueMorsePrefix` states that $n \mapsto \tau(0)\tau(1)\cdots\tau(n-1)$ is primitive recursive: the infinite word can be *generated*, not merely decided one letter at a time.

2.6 Exact weight distribution in a dyadic block

The four drafts repeatedly use the fact that a length- 2^m block is a Boolean cube. It is useful to record the complete distribution, not only its parity consequence.

Theorem 2.10 (Binomial weight enumerator). *For every $m \geq 0$,*

$$\sum_{n=0}^{2^m-1} u^{\text{wt}_2(n)} = (1+u)^m. \quad (2.18)$$

Consequently

$$\#\{0 \leq n < 2^m : \text{wt}_2(n) = r\} = \binom{m}{r}. \quad (2.19)$$

Under the uniform distribution on the block, $\text{wt}_2(n)$ is binomial with mean $m/2$ and variance $m/4$.

Proof. Each of the m positions independently contributes 1 or u . Multiplication gives $(1+u)^m$; coefficient extraction and the standard binomial moments give the remaining statements. $\square$

Formal versions. The enumerator is `Fabius.sum_pow_binaryWeight_eq_one_add_pow`, proved over an arbitrary commutative semiring, and the count (2.19) is `Fabius.card_filter_binaryWeight_eq`, extracted from it exactly as in the proof above: read the enumerator in $\mathbb{N}[z]$ and compare the coefficient of z^r . Both live in `ThueMorseEnumerators.lean`; the underlying Boolean-cube machinery is described after [theorem 6.2](#).

Setting $u = -1$ gives exact sign balance for $m \geq 1$. More generally, the characteristic function of the weight is

$$2^{-m} \sum_{n < 2^m} e^{it \text{wt}_2(n)} = \left(\frac{1 + e^{it}}{2} \right)^m. \quad (2.20)$$

This elementary statistical layer is useful when comparing Boolean/Walsh structure with ordinary Fourier structure later in the report.

2.7 Evil and odious enumerators

Call n *evil* when $\text{wt}_2(n)$ is even and *odious* when it is odd. Let $\mathcal{E}_n$ and $\mathcal{O}_n$ be the increasing enumerations, starting at index zero.

Theorem 2.11 (Evil and odious numbers). *For every $n \geq 0$,*

$$\boxed{\mathcal{E}_n = 2n + \tau(n), \quad \mathcal{O}_n = 2n + 1 - \tau(n).} \quad (2.21)$$

Hence

$$\mathcal{O}_n - \mathcal{E}_n = \varepsilon_n, \quad \mathcal{O}_n + \mathcal{E}_n = 4n + 1. \quad (2.22)$$

Proof. The two consecutive integers $2n$ and $2n + 1$ have parities $\tau(n)$ and $1 - \tau(n)$ of their binary weights. If $\tau(n) = 0$, then $2n$ is evil and $2n + 1$ odious; if $\tau(n) = 1$, the roles reverse. The displayed formulas choose the correct member of each consecutive pair. $\square$

Corollary 2.12 (Composition laws). *For every $n \geq 0$,*

$$\mathcal{E}_{\mathcal{E}_n} = 2\mathcal{E}_n, \quad \mathcal{O}_{\mathcal{E}_n} = 2\mathcal{E}_n + 1, \quad (2.23)$$

$$\mathcal{E}_{\mathcal{O}_n} = 2\mathcal{O}_n + 1, \quad \mathcal{O}_{\mathcal{O}_n} = 2\mathcal{O}_n. \quad (2.24)$$

Proof. By definition $\tau(\mathcal{E}_n) = 0$ and $\tau(\mathcal{O}_n) = 1$. Substitute these values into (2.21) with the new index. $\square$

Formal versions. In `ThueMorseEnumerators.lean` the enumerations are `Fabius.evileNum` and `Fabius.odiousEnum`, with landing lemmas `Fabius.thueMorseBit_evileNum`, `Fabius.thueMorseBit_odiousEnum`, strict monotonicity `Fabius.evileNum_strictMono`, `Fabius.odiousEnum_strictMono`, difference and sum `Fabius.odiousEnum_sub_evileNum`, `Fabius.evileNum_add_odiousEnum`, and the four composition laws. The Lean proof of surjectivity isolates a detail worth recording in prose: for every m with $\tau(m) = 0$ one has $\mathcal{E}_{\lfloor m/2 \rfloor} = m$ (`Fabius.evileNum_div_two`) – the same index $\lfloor m/2 \rfloor$ works whether m is even (then $\tau(m/2) = 0$ and the τ -summand vanishes) or odd (then $\tau((m-1)/2) = 1$ and the summand supplies the missing one) – and dually for odious m (`Fabius.odiousEnum_div_two`). Monotone enumerations that hit everything are complete, so no separate counting argument is needed.

3 Digit, floor, fractional-part, and trigonometric formulas

3.1 Extracting one binary digit

Lemma 3.1 (Floor extraction). *For every $j, n \geq 0$,*

$$b_j(n) = \left\lfloor \frac{n}{2^j} \right\rfloor - 2 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor. \quad (3.1)$$

Proof. Write $\lfloor n/2^j \rfloor = 2q + r$ with $r \in \{0, 1\}$. Then $q = \lfloor n/2^{j+1} \rfloor$, while r is exactly the j -th binary digit. $\square$

Since $(-1)^b = 1 - 2b$ for $b \in \{0, 1\}$, the definition factorizes over coordinates.

Theorem 3.2 (Digit product and Boolean polynomial). *For every $n \geq 0$,*

$$\boxed{\varepsilon_n = \prod_{j \geq 0} (1 - 2b_j(n)), \quad \tau(n) = \frac{1 - \prod_{j \geq 0} (1 - 2b_j(n))}{2}.} \quad (3.2)$$

Only finitely many factors differ from 1. If J is any finite set containing the support of the binary expansion, then

$$\boxed{\tau(n) = \sum_{\emptyset \neq S \subseteq J} (-2)^{|S|-1} \prod_{j \in S} b_j(n).} \quad (3.3)$$

Proof. Multiply $(-1)^{b_j} = 1 - 2b_j$ over all positions. Expanding the product in the zero–one conversion (1.3) gives (3.3). $\square$

Substitution of (3.1) gives a digit-free, floor-only product:

$$\varepsilon_n = \prod_{j \geq 0} \left(1 - 2 \left\lfloor \frac{n}{2^j} \right\rfloor + 4 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor \right). \quad (3.4)$$

Every factor is $1 - 2b_j(n) \in \{1, -1\}$.

3.2 Dyadic floor and fractional-part sums

Lemma 3.3 (Legendre floor sum). *For every $n \geq 0$,*

$$\sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = n - \text{wt}_2(n), \quad (3.5)$$

$$\sum_{j \geq 0} \left\lfloor \frac{n}{2^j} \right\rfloor = 2n - \text{wt}_2(n). \quad (3.6)$$

Proof. Insert the binary expansion and interchange finite sums:

$$\begin{aligned} \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor &= \sum_{j \geq 1} \sum_{\ell \geq j} b_\ell(n) 2^{\ell-j} \\ &= \sum_{\ell \geq 1} b_\ell(n) (2^\ell - 1) = n - \sum_{\ell \geq 0} b_\ell(n). \end{aligned}$$

Adding the $j = 0$ term gives the second identity. $\square$

Theorem 3.4 (Floor-sum formulae). *For every $n \geq 0$,*

$$\boxed{\varepsilon_n = (-1)^{\sum_{j \geq 0} \lfloor n/2^j \rfloor}, \quad \tau(n) \equiv \sum_{j \geq 0} \left\lfloor \frac{n}{2^j} \right\rfloor \pmod{2}.} \quad (3.7)$$

Equivalently,

$$\varepsilon_n = \prod_{j \geq 0} (-1)^{\lfloor n/2^j \rfloor}. \quad (3.8)$$

Proof. By (3.6), the exponent differs from $\text{wt}_2(n)$ by an even integer. $\square$

Formal versions. `ThueMorseDigits.lean` proves the additive Legendre formula $\text{wt}_2(n) + \nu_2(n!) = n$ (`Fabius.binaryWeight_add_padicValNat_factorial`), the floor sums in truncation-free form (`Fabius.binaryWeight_add_sum_div_two_pow`, `Fabius.sum_div_two_pow_add_binaryWeight`), the floor-parity formula (3.7) (`Fabius.thueMorseSign_eq_neg_one_pow_sum_div`), the bound $\text{wt}_2(n) \leq n$ (`Fabius.binaryWeight_le_self`), and the bit-extraction bridge $\text{wt}_2(n) = \sum_{j < m} b_j(n)$ (`Fabius.binaryWeight_eq_sum_testBit`) together with the digit product (3.2) (`Fabius.thueMorseSign_eq_prod_testBit`). Natural-number subtraction being awkward, every Lean statement is phrased additively; the prose forms above are recovered by moving one term across.

There is also a genuinely infinite real-valued representation whose sum is nevertheless always an integer.

Proposition 3.5 (Fractional-part digit sum). *For every $n \geq 0$,*

$$\boxed{\text{wt}_2(n) = \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\}}. \quad (3.9)$$

Consequently

$$\varepsilon_n = \cos \left(\pi \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\} \right), \quad \tau(n) = \frac{1 - \varepsilon_n}{2}. \quad (3.10)$$

Proof. The geometric series gives $\sum_{j \geq 1} n/2^j = n$. Subtract (3.5) termwise:

$$\sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\} = n - \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = \text{wt}_2(n).$$

□

3.3 Square waves and a sine-sign product

For nonintegral x ,

$$\text{sgn}(\sin \pi x) = (-1)^{\lfloor x \rfloor}. \quad (3.11)$$

The half-unit shift avoids every zero.

Theorem 3.6 (Rademacher–sine formula). *For every $n \geq 0$,*

$$\boxed{\varepsilon_n = \prod_{j \geq 0} \text{sgn} \left(\sin \frac{\pi(n + \frac{1}{2})}{2^j} \right)}. \quad (3.12)$$

All sufficiently large factors are +1. Moreover the j -th factor is exactly $(-1)^{b_j(n)}$.

Proof. The number $(n + \frac{1}{2})/2^j$ is never integral and has the same floor as $n/2^j$. Apply (3.11), then observe from (3.1) that $(-1)^{\lfloor n/2^j \rfloor} = (-1)^{b_j(n)}$. The factors are positive once $2^j > n + \frac{1}{2}$. □

3.4 The sign of the Fabius sinc product

Define $\text{sinc } y = \sin y/y$, with $\text{sinc } 0 = 1$, and

$$\Phi(x) = \prod_{j=0}^{\infty} \text{sinc} \left(\frac{\pi x}{2^j} \right). \quad (3.13)$$

The product converges locally uniformly because the logarithm of the tail is controlled by $\sum_j x^2/4^j$. It is, up to normalization conventions, the infinite Fourier product naturally associated with the Rvachev up-function.

Corollary 3.7 (Interval sign of the sinc product). *If $N \in \mathbb{N}$ and $N < x < N + 1$, then*

$$\text{sgn } \Phi(x) = \varepsilon_N. \quad (3.14)$$

In particular,

$$\varepsilon_n = \text{sgn } \Phi \left(n + \frac{1}{2} \right). \quad (3.15)$$

Proof. For $x > 0$, every sinc denominator is positive. No multiple of 2^j lies strictly between the consecutive integers N and $N + 1$, so $\lfloor x/2^j \rfloor = \lfloor N/2^j \rfloor$. The numerator signs therefore multiply to $(-1)^{\sum_j \lfloor N/2^j \rfloor} = \varepsilon_N$ by theorem 3.4. □

Formal versions. The product (3.13) is `Fabius.rvachevFourierProduct (Basic.lean)`. The interval sign (3.14) is now the exact identity

$$\Phi(x) = \varepsilon_N |\Phi(x)| \quad (N < x < N + 1),$$

proved by `Fabius.rvachevFourierProduct_eq_thueMorse_sign_mul_norm` in `ThueMorseLobeSign.lean`. In particular, positivity of the norm off the integer lattice gives the displayed sign and the half-integer sample without an extra analytic assumption.

The same sinc product now has a complete formal zero-set statement:

$$\Phi(z) = 0 \iff \exists m \in \mathbb{Z} \setminus \{0\}, z = m,$$

namely `Fabius.rvachevFourierProduct_eq_zero_iff` in `FourierProduct.lean`. Its canonical product is

$$\Phi(z) = \prod_{r=0}^{\infty} \left(1 - \frac{z^2}{(r+1)^2}\right)^{1+\nu_2(r+1)},$$

formalized as `Fabius.rvachevFourierProduct_eq_canonical` in `SincCanonicalProduct.lean`. For each positive integer m , the real punctured limit

$$\frac{|\Phi(x)|}{|x - m|^{1+\nu_2(m)}} \longrightarrow C_m > 0 \quad (x \rightarrow m, x \neq m)$$

is `Fabius.tendsto_norm_div_pow_zero_order` in `PhiZeroOrder.lean`; this is the exact positive-integer real-axis zero order, not a broader assertion about complex local multiplicity. On the real axis the origin is the unique unit-modulus point:

$$|\Phi(x)| = 1 \iff x = 0, \quad x \neq 0 \implies |\Phi(x)| < 1.$$

These are `Fabius.norm_rvachevFourierProduct_eq_one_iff` and `Fabius.norm_rvachevFourierProduct_lt_one_of_ne_zero`. For every $m \geq 1$, the modulus also has exactly one maximizer in each positive side lobe $(m, m + 1)$ and reflected negative side lobe $(-(m + 1), -m)$, by `Fabius.existsUnique_isMaxOn_lobe` and `Fabius.existsUnique_isMaxOn_neg_lobe` in `RvachevProductContinuity.lean`.

The product-level renormalization is available independently of the Fourier-transform wrapper:

$$\Phi(z) = \text{sinc}(\pi z) \Phi(z/2), \quad \Phi(z) = \left(\prod_{j < n} \text{sinc}(\pi z/2^j) \right) \Phi(z/2^n).$$

These are `Fabius.rvachevFourierProduct_scaling` and `Fabius.rvachevFourierProduct_shell` in `RenormalizationIdentity.lean`. Beyond these structural identities, `SincProductShells.lean` proves the exact dyadic shell factorization of the modulus: for real $y \neq 0$ and every $k \geq 0$,

$$|\Phi(2^k y)| = \frac{\prod_{j < k} |\sin(2^{j+1} \pi y)|}{2^{k(k+1)/2} (\pi |y|)^k} |\Phi(y)| \quad (3.16)$$

(`Fabius.norm_rvachevFourierProduct_two_pow_mul`): dilating the argument by 2^k creates exactly the k sine factors of the corresponding dyadic shell. Bounding the lacunary sine numerator by the formal Gelfond bound `Fabius.abs_prod_sin_two_pow_le` (`GelfondLogisticBound.lean`) – the sine-product form of (15.8), with the optimal rate $(\sqrt{3}/2)^k$ and the slightly larger constant $\sqrt{5/3}$ in place of $2/\sqrt{3}$ – turns (3.16) into the decay bound

$$|\Phi(2^k y)| \leq \sqrt{\frac{5}{3}} \frac{(\sqrt{3}/2)^k}{2^{k(k+1)/2} (\pi |y|)^k} |\Phi(y)| \quad (3.17)$$

(`Fabius.norm_rvachevFourierProduct_two_pow_mul_1e`), valid along every dyadic ray at once, with no hypothesis on y beyond $y \neq 0$. The triangular power $2^{-k(k+1)/2}$ left standing in the denominator is the universal $\exp(-(\log x)^2/(2 \log 2))$ factor of the Fabius–Rvachev decay theory; the accompanying geometric factor is what the companion reports convert into the spectrum of decay exponents.

The sine-product bound is in turn a corollary of a statement in which no trigonometry appears at all. Along *any* orbit of the full logistic map $v \mapsto 4v(1-v)$ whose *starting point* lies in $[0, 1]$ – a hypothesis on u_0 only, the range at every later index being supplied by invariance of $[0, 1]$ – the running product satisfies

$$\prod_{j < n} u_j \leq \frac{5}{3} \left(\frac{3}{4}\right)^n \quad (3.18)$$

(`Fabius.logistic_prod_le_of_init`, `GelfondLogisticBound.lean`), by telescoping the potential $V(v) = 1 + 2v/3$ through the exact inequality $3V(v) - 4vV(4v(1-v)) = (2v+1)(4v-3)^2/3 \geq 0$; the constant $5/3$ is the bound $V(u_0) \leq V(1) = 5/3$, so it too comes from $u_0 \leq 1$ alone. Squared sines at doubling angles form such an orbit (`Fabius.sin_sq_two_mul`), which is the only input the Gelfond bound needs. The geometric rate in (3.18) is exact already at this abstract level: $3/4$ is the interior fixed point of the map, and the constant orbit it generates has running product exactly $(3/4)^n$ (`Fabius.logistic_prod_eq_of_init_three_quarters`), so no smaller geometric rate can hold along all orbits whatever the constant in front. On the trigonometric side the same fixed point is attained by every seed with $\sin^2(\pi t) = 3/4$ (`Fabius.prod_sin_sq_two_pow_eq_of_sin_sq_eq`), the Gelfond angle $t = 1/3$ among them (`Fabius.prod_sin_sq_two_pow_third`).

3.5 The ruler transition

Incrementing n flips all trailing one-bits to zero and the next zero-bit to one. The number of erased one-bits is $\nu_2(n+1)$.

Theorem 3.8 (Successor and ruler formulae). *For every $n \geq 0$,*

$$\text{wt}_2(n+1) - \text{wt}_2(n) = 1 - \nu_2(n+1), \quad (3.19)$$

$$\varepsilon_{n+1} = (-1)^{1+\nu_2(n+1)} \varepsilon_n, \quad (3.20)$$

$$(-1)^{\nu_2(n+1)} = -\varepsilon_n \varepsilon_{n+1}. \quad (3.21)$$

Consequently

$$\varepsilon_n = \prod_{k=1}^n (-1)^{1+\nu_2(k)}. \quad (3.22)$$

Proof. The bit transition changes the weight by one minus the number of trailing ones. Exponentiate (-1) , then telescope from $\varepsilon_0 = 1$. $\square$

Formal versions. The successor law in additive form $\text{wt}_2(n+1) + \nu_2(n+1) = \text{wt}_2(n) + 1$ is `Fabius.binaryWeight_succ_add_padicValNat`; the adjacent-sign readout (3.21) is `Fabius.thueMorseSign_mul_succ`, the recursion (3.20) is `Fabius.thueMorseSign_succ_eq`, and the telescoped ruler product (3.22) is `Fabius.thueMorseSign_eq_prod_ruler`, all in `ThueMorseValuation.lean`.

4 Factorial valuations, central binomials, Catalan numbers, and carries

4.1 Legendre’s formula

The floor sum in [lemma 3.3](#) counts powers of two in a factorial.

Theorem 4.1 (Factorial valuation). *For every $n \geq 0$,*

$$\nu_2(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = n - \text{wt}_2(n). \quad (4.1)$$

Hence

$$\boxed{\varepsilon_n = (-1)^{n - \nu_2(n!)} = (-1)^{n + \nu_2(n!)}, \quad \tau(n) \equiv n + \nu_2(n!) \pmod{2}.} \quad (4.2)$$

Proof. Every multiple of 2 contributes at least one factor, every multiple of 4 contributes one additional factor, and so on. This gives the floor sum; [lemma 3.3](#) gives the digit form. Addition and subtraction agree modulo two. $\square$

The telescoping product (3.22) returns to this formula because $\sum_{k=1}^n \nu_2(k) = \nu_2(n!)$.

4.2 A single central binomial coefficient

Theorem 4.2 (Central-binomial valuation). *For every $n \geq 0$,*

$$\boxed{\nu_2 \binom{2n}{n} = \text{wt}_2(n).} \quad (4.3)$$

Consequently

$$\boxed{\varepsilon_n = (-1)^{\nu_2 \binom{2n}{n}}, \quad \tau(n) = \nu_2 \binom{2n}{n} \pmod{2}.} \quad (4.4)$$

Equivalently,

$$\tau(n) = \sin^2 \left(\frac{\pi}{2} \nu_2 \binom{2n}{n} \right), \quad \varepsilon_n = \cos \left(\pi \nu_2 \binom{2n}{n} \right). \quad (4.5)$$

Proof. Use (4.1) and $\text{wt}_2(2n) = \text{wt}_2(n)$:

$$\begin{aligned} \nu_2 \binom{2n}{n} &= \nu_2((2n)!) - 2\nu_2(n!) \\ &= (2n - \text{wt}_2(2n)) - 2(n - \text{wt}_2(n)) = \text{wt}_2(n). \end{aligned}$$

$\square$

The shortest Pascal replacement. The source formula inspects all $n + 1$ entries of row n and takes an exact logarithm. Formula (4.4) inspects the two-adic order of one entry in row $2n$, and in fact recovers the entire digit sum rather than only its parity.

4.3 Catalan valuations

Let

$$\text{Cat}_n = \frac{1}{n+1} \binom{2n}{n}. \quad (4.6)$$

Corollary 4.3 (Catalan formula). *For every $n \geq 0$,*

$$\nu_2(\text{Cat}_n) = \text{wt}_2(n+1) - 1, \quad (4.7)$$

and hence

$$\boxed{\varepsilon_{n+1} = -(-1)^{\nu_2(\text{Cat}_n)}.} \quad (4.8)$$

Proof. By [theorem 4.2](#), $\nu_2(\text{Cat}_n) = \text{wt}_2(n) - \nu_2(n+1)$. Equation (3.19) rewrites this as $\text{wt}_2(n+1) - 1$. $\square$

Formal versions. `Fabius.catalan_padicValNat_two_add_one` (stated additively as $\nu_2(\text{Cat}_n) + 1 = \text{wt}_2(n+1)$) and the sign form `Fabius.thueMorseSign_succ_eq_neg_catalan`, in `ThueMorseValuation.lean`. The Lean proof derives the positivity of Catalan numbers inline from $(n+1)\text{Cat}_n = \binom{2n}{n}$ rather than importing it.

4.4 Kummer’s carry cocycle

Theorem 4.4 (Addition with carries). *For all $a, b \geq 0$,*

$$\nu_2\binom{a+b}{a} = \text{wt}_2(a) + \text{wt}_2(b) - \text{wt}_2(a+b). \quad (4.9)$$

This is the number of carry operations in the binary addition of a and b , including propagated carries. Therefore

$$\boxed{\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2\binom{a+b}{a}}}. \quad (4.10)$$

Equivalently,

$$\tau(a+b) \equiv \tau(a) + \tau(b) + \nu_2\binom{a+b}{a} \pmod{2}. \quad (4.11)$$

Proof. Legendre’s formula gives

$$\nu_2\binom{a+b}{a} = (a+b - \text{wt}_2(a+b)) - (a - \text{wt}_2(a)) - (b - \text{wt}_2(b)),$$

which is (4.9). Rearranging modulo two and raising (-1) to both sides gives the sign formula. The carry interpretation is the binary case of Kummer’s theorem. $\square$

The no-carry case $a \& b = 0$ makes the valuation zero and recovers ordinary multiplicativity. In particular, [theorem 2.2](#) is the separated-support case.

Corollary 4.5 (Multinomial carry formula). *For $a_1, \dots, a_r \geq 0$ and $N = a_1 + \dots + a_r$,*

$$\nu_2\binom{N}{a_1, \dots, a_r} = \sum_{j=1}^r \text{wt}_2(a_j) - \text{wt}_2(N), \quad (4.12)$$

$$\varepsilon_N = \left(\prod_{j=1}^r \varepsilon_{a_j} \right) (-1)^{\nu_2\binom{N}{a_1, \dots, a_r}}. \quad (4.13)$$

Proof. Apply Legendre’s formula to $N!/(a_1! \cdots a_r!)$ and then take parity. $\square$

4.5 Subtraction and borrow counts

Kummer’s theorem has a mirror statement for subtraction. For $0 \leq b \leq a$, the same Legendre computation applied to $\binom{a}{b} = a!/(b!(a-b)!)$ gives

$$\nu_2\binom{a}{b} = \text{wt}_2(b) + \text{wt}_2(a-b) - \text{wt}_2(a), \quad (4.14)$$

which counts the borrow operations, including propagated borrows, in the binary subtraction $a - b$. Therefore

$$\boxed{\varepsilon_{a-b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2\binom{a}{b}}}. \quad (4.15)$$

Addition and subtraction thus carry the same sign correction, read from the two-adic valuation of the binomial coefficient whose top entry is the larger of the numbers involved; (4.10) is recovered from (4.15) by renaming $a + b$ as the top entry.

Formal versions. Kummer’s carry formula and cocycle were already in the corpus (`Fabius.addChoose_padicValNat_two`, `Fabius.thueMorseSign_add_valuation`, in `TwoAdic.lean`); `ThueMorseValuation.lean` adds the truncation-free additive form `Fabius.binaryWeight_add_addChoose_padicValNat`, the weight subadditivity $\text{wt}_2(a + b) \leq \text{wt}_2(a) + \text{wt}_2(b)$ (`Fabius.binaryWeight_add_le`, itself proved by a four-way parity induction), and the borrow pair `Fabius.subChoose_padicValNat_two_add`, `Fabius.thueMorseSign_sub_valuation`.

5 Pascal support, residue formulae, multinomials, and Boolean inversion

5.1 Lucas’s theorem as a submask criterion

Theorem 5.1 (Lucas parity criterion). *For $0 \leq k \leq n$,*

$$\binom{n}{k} \equiv 1 \pmod{2} \iff k \preccurlyeq_2 n. \quad (5.1)$$

Equivalently,

$$\mathbf{1}_{k \preccurlyeq_2 n} = \frac{1 - (-1)^{\binom{n}{k}}}{2}. \quad (5.2)$$

Proof. In $\mathbb{F}_2[z]$, Frobenius gives

$$(1 + z)^n = \prod_{j: b_j(n)=1} (1 + z)^{2^j} = \prod_{j: b_j(n)=1} (1 + z^{2^j}).$$

The exponents occurring in the expansion are exactly the submasks of n , and their coefficients are one modulo two. $\square$

Define the odd-support polynomial of row n by

$$Q_n(z) = \sum_{k=0}^n \mathbf{1}_{\binom{n}{k} \text{ odd}} z^k. \quad (5.3)$$

The proof gives the complete factorization

$$Q_n(z) = \prod_{j: b_j(n)=1} (1 + z^{2^j}). \quad (5.4)$$

Corollary 5.2 (Glaisher–Fine count). *The number $\nu(n)$ of odd entries in row n is*

$$\nu(n) = 2^{\text{wt}_2(n)}. \quad (5.5)$$

Proof. Set $z = 1$ in (5.4). $\square$

Formal versions. `ThueMorseLucasSupport.lean` proves [theorem 5.1](#) in three forms: `Fabius.odd_choose_iff_testBit` (every set bit of k is set in n), `Fabius.odd_choose_iff_land` ($k \& n = k$), and `Fabius.odd_choose_iff_bitSupport_subset`, by strong induction through the parity recursions of Pascal rows rather than through $\mathbb{F}_2[z]$. The supporting fact that a sum of distinct two-powers has exactly the prescribed binary support is `Fabius.testBit_sum_two_pow`,

stated for an arbitrary finite set of bit positions. The Glaisher–Fine count (5.5) has long been formal as `Fabius.card_oddBinomialIndices` (`Parity.lean`); the odd row positions are identified with the submask cube by `Fabius.oddBinomialIndices_eq_image_powerset`.

The source formula is now transparent. Since even entries contribute $+1$ and odd entries contribute -1 ,

$$\mathcal{X}(n) = n + 1 - 2\nu(n) = n + 1 - 2^{\text{wt}_2(n)+1}, \quad (5.6)$$

which is exactly (1.5).

5.2 A logarithm-free Pascal-row formula modulo three

Let $\text{rem}_3(x) \in \{0, 1, 2\}$ denote the least nonnegative residue modulo three. Powers of two alternate between one and two modulo three.

Theorem 5.3 (Odd-count residue formula). *For every $n \geq 0$,*

$$\boxed{\tau(n) = \text{rem}_3(\nu(n)) - 1, \quad \varepsilon_n = 3 - 2\text{rem}_3(\nu(n)).} \quad (5.7)$$

The residue is always 1 or 2, never 0.

Proof. By (5.5), $\nu(n) = 2^{\text{wt}_2(n)}$. Its residue is one when $\text{wt}_2(n)$ is even and two when it is odd. $\square$

A form using exactly the row signs of the source formula can be written without the odd-count notation. Put

$$S(n) = \sum_{k=1}^n (-1)^{\binom{n}{k}}, \quad S(0) = 0. \quad (5.8)$$

Since the omitted $k = 0$ term is -1 ,

$$S(n) = n + 2 - 2^{\text{wt}_2(n)+1}. \quad (5.9)$$

Corollary 5.4 (Binomial residue and sine formulae). *For every $n \geq 0$,*

$$\boxed{\tau(n) = \text{rem}_3(2n + S(n)),} \quad (5.10)$$

$$\boxed{\tau(n) = \frac{4}{3} \sin^2\left(\frac{\pi}{3}(n - S(n))\right).} \quad (5.11)$$

Proof. Equation (5.9) gives $2n + S(n) = 3n + 2 - 2^{\text{wt}_2(n)+1}$, whose residue is zero or one according as the weight is even or odd. Also $n - S(n) = 2^{\text{wt}_2(n)+1} - 2$, congruent to zero or two modulo three; the squared sine is therefore zero or $3/4$. $\square$

The logarithmic and residue formulas are two shadows of the same invariant $\nu(n)$: the first reads the exponent of a power of two; the second reads that exponent's parity from the two-cycle of powers modulo three.

5.3 A one-parameter row statistic

For a scalar u , define

$$R_u(n) = \sum_{k=0}^n u^{\binom{n}{k} \bmod 2}. \quad (5.12)$$

Exactly $\nu(n)$ exponents are one and the others are zero, so

$$\boxed{R_u(n) = n + 1 + (u - 1)2^{\text{wt}_2(n)}.} \quad (5.13)$$

Thus $R_{-1}(n) = \mathcal{X}(n)$, while $R_0(n)$ counts the even entries. For every fixed $u \neq 1$,

$$2^{\text{wt}_2(n)} = \frac{R_u(n) - (n+1)}{u-1}. \quad (5.14)$$

The signed choice $u = -1$ is elegant, but structurally it is one member of a whole exact family.

5.4 Sparse Pascal probes for individual bits

Lucas's criterion at $k = 2^j$ isolates one binary coordinate:

$$\binom{n}{2^j} \equiv b_j(n) \pmod{2}. \quad (5.15)$$

Theorem 5.5 (Power-of-two column formula). *For every $n \geq 0$,*

$$\boxed{\varepsilon_n = (-1)^{\sum_{j \geq 0} \binom{n}{2^j}}} = \prod_{j \geq 0} \cos\left(\pi \binom{n}{2^j}\right), \quad (5.16)$$

$$\tau(n) = \frac{1}{2} \left(1 - \prod_{j \geq 0} \cos\left(\pi \binom{n}{2^j}\right) \right). \quad (5.17)$$

Only the columns $2^j \leq n$ matter.

Proof. Only the parity of each binomial coefficient contributes to the exponent or cosine. Equation (5.15) turns their parity sum into $\text{wt}_2(n)$. $\square$

Unlike the source formula, this scans only $O(\log n)$ selected entries and uses no logarithm. Its price is the distinguished set of power-of-two columns.

5.5 Odd multinomial coefficients and exact r -nomial logarithms

Fix $r \geq 2$. Let $O_r(n)$ be the number of odd coefficients in $(x_1 + \cdots + x_r)^n$.

Theorem 5.6 (Multinomial support count). *For every $n \geq 0$,*

$$\boxed{O_r(n) = r^{\text{wt}_2(n)}}. \quad (5.18)$$

Consequently

$$\boxed{\varepsilon_n = (-1)^{\log_r O_r(n)}, \quad \tau(n) = \frac{1 - (-1)^{\log_r O_r(n)}}{2}}. \quad (5.19)$$

The logarithm is exact.

Proof. In characteristic two,

$$(x_1 + \cdots + x_r)^n = \prod_{j: b_j(n)=1} (x_1^{2^j} + \cdots + x_r^{2^j}). \quad (5.20)$$

For each one-bit of n , choose independently one of the r variables to receive it. Unique binary expansions make distinct assignments produce distinct exponent vectors, so there are $r^{\text{wt}_2(n)}$ nonzero coefficients modulo two. $\square$

There are $\binom{n+r-1}{r-1}$ weak compositions of n into r parts. Define

$$\mathcal{X}_r(n) = \sum_{k_1 + \cdots + k_r = n} (-1)^{\binom{n}{k_1, \dots, k_r}}. \quad (5.21)$$

Corollary 5.7 (Signed r -nomial row formula). *For every $r \geq 2$,*

$$\binom{n+r-1}{r-1} - \mathcal{X}_r(n) = 2r^{\text{wt}_2(n)}, \quad (5.22)$$

and therefore

$$\tau(n) = \frac{1 - (-1)^{\log_r \left(\frac{\binom{n+r-1}{r-1} - \mathcal{X}_r(n)}{2} \right)}}{2}. \quad (5.23)$$

Proof. Each odd multinomial coefficient contributes -1 , each even coefficient $+1$. Subtracting the signed sum from the total count gives twice the odd support size, which is $2r^{\text{wt}_2(n)}$ by [theorem 5.6](#). $\square$

Formal versions. `ThueMorseMultinomial.lean` proves [theorem 5.6](#) as `Fabius.card_filter_odd_multinomial`, for an arbitrary finite index set, resting on `**Kummer’s criterion for multinomial coefficients**` `Fabius.odd_multinomial_iff_binaryWeight`: a multinomial coefficient is odd exactly when the binary weights of its parts are additive. The proof uses no polynomial ring: Legendre’s formula converts oddness into weight additivity, and the induction that splits off one part reduces the count to the bivariate submask enumerator at $(1, r)$. The exact logarithm (5.19) is `Fabius.thueMorseSign_eq_neg_one_pow_log_card`, and the signed row formula (5.22) is `Fabius.sum_neg_one_pow_multinomial` (with the generic count-parity bridge `Fabius.sum_neg_one_pow_eq_card_sub_two_mul`); the closed value $\binom{n+r-1}{r-1}$ of the total count is left to the reader’s `Mathlib`.

5.6 Boolean weight enumerators and Möbius inversion

The submasks of n form a Boolean lattice on its set of one-bit positions.

Theorem 5.8 (Submask weight enumerator). *For every indeterminate z ,*

$$\sum_{k \preceq_2 n} z^{\text{wt}_2(k)} = (1 + z)^{\text{wt}_2(n)}, \quad (5.24)$$

$$z^{\text{wt}_2(n)} = \sum_{k \preceq_2 n} (z - 1)^{\text{wt}_2(k)}. \quad (5.25)$$

More generally,

$$\sum_{k \preceq_2 n} x^{\text{wt}_2(k)} y^{\text{wt}_2(n-k)} = (x + y)^{\text{wt}_2(n)}. \quad (5.26)$$

Proof. For each occupied bit of n , choose whether it belongs to k , to its complement $n - k$, or is omitted in the one-variable specialization. The second formula is the first with z replaced by $z - 1$. $\square$

Important specializations are

$$\#\{k : k \preceq_2 n\} = 2^{\text{wt}_2(n)}, \quad (5.27)$$

$$\sum_{k \preceq_2 n} \varepsilon_k = \begin{cases} 1, & n = 0, \\ 0, & n > 0, \end{cases} \quad (5.28)$$

$$\varepsilon_n = \sum_{k \preceq_2 n} (-2)^{\text{wt}_2(k)}. \quad (5.29)$$

Theorem 5.9 (Boolean Möbius inversion). *If $k \preceq_2 n$, then the Möbius function of the submask poset is*

$$\mu_2(k, n) = (-1)^{\text{wt}_2(n) - \text{wt}_2(k)} = \varepsilon_{n \oplus k}. \quad (5.30)$$

Thus

$$g(n) = \sum_{k \preceq_2 n} f(k) \iff \boxed{f(n) = \sum_{k \preceq_2 n} \varepsilon_{n \oplus k} g(k)}. \quad (5.31)$$

Proof. The interval $[k, n]$ is the Boolean lattice on the bits of n absent from k ; its rank is $\text{wt}_2(n) - \text{wt}_2(k) = \text{wt}_2(n \oplus k)$. The usual Boolean Möbius function is $(-1)^{\text{rank}}$, and standard Möbius inversion applies. $\square$

Formal versions. `ThueMorseBooleanMöbius.lean` proves the whole lattice layer. The enumerators (5.24) and (5.26) are `Fabius.sum_oddBinomialIndices_pow_binaryWeight` and `Fabius.sum_oddBinomialIndices_pow_pow`, over any commutative semiring; (5.25) and the reconstruction (5.29) are `Fabius.pow_binaryWeight_eq_sum_oddBinomialIndices` and `Fabius.thueMorseSign_eq_sum_oddBinomialIndices_neg_two_pow`; the count (5.27) and the cancellation (5.28) were already `Fabius.card_oddBinomialIndices` and `Fabius.sum_thueMorseSign_oddBinomialIndices`. Möbius inversion itself is proved in two forms: on finsets for any additive commutative group (`Fabius.sum_powerset_mobius`, with the interval sum `Fabius.sum_filter_superset_neg_one_pow`), and numerically on submasks with the Möbius function $\varepsilon_{n \oplus k}$ (`Fabius.submask_mobius_inversion`); the underlying lattice isomorphism is `Fabius.bitSupport_sum_two_pow` with the complement laws `Fabius.sub_sum_two_pow` and `Fabius.xor_sum_two_pow`. The matrix statement (5.33) is the poset form read row by row.

Let

$$P_{n,k} = \mathbf{1}_{\binom{n}{k} \text{ odd}} = \mathbf{1}_{k \preceq_2 n}. \quad (5.32)$$

As a locally finite lower-triangular matrix,

$$\boxed{(P^{-1})_{n,k} = P_{n,k} \varepsilon_{n \oplus k}}. \quad (5.33)$$

In particular, the first column of P^{-1} is the signed Thue–Morse sequence. This is a second global Pascal encoding: ε is not only a row statistic but the Möbius inverse of the entire parity incidence matrix.

Combining (5.29), Lucas’s indicator, and the central binomial valuation gives a formula involving only ordinary binomial coefficients:

Theorem 5.10 (Nested binomial formula). *For every $n \geq 0$,*

$$\boxed{\varepsilon_n = \sum_{k=0}^n \frac{1 - (-1)^{\binom{n}{k}}}{2} (-2)^{\nu_2 \binom{2k}{k}}}. \quad (5.34)$$

Proof. The first factor is $\mathbf{1}_{k \preceq_2 n}$ by (5.2), and the exponent in the second is $\text{wt}_2(k)$ by theorem 4.2. The sum is therefore exactly (5.29). $\square$

Formal versions. (5.34) is `Fabius.thueMorseSign_eq_sum_nested_binomial` (`ThueMorseBooleanMöbius.lean`), from the reconstruction formula and the central-binomial valuation `Fabius.centralChoose_padicValNat_two`.

6 Finite and infinite generating products

6.1 The Boolean-cube mother identity

A multivariate identity contains the finite generating product, the weight enumerator, and the Walsh character formula as specializations.

Theorem 6.1 (Boolean-cube product). *For commuting indeterminates $x_0, \dots, x_{m-1}$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n \prod_{j=0}^{m-1} x_j^{b_j(n)} = \prod_{j=0}^{m-1} (1 - x_j). \quad (6.1)$$

Proof. Identify n with its bit vector $b \in \{0, 1\}^m$. The sum factors over coordinates:

$$\sum_{b \in \{0,1\}^m} \prod_j (-x_j)^{b_j} = \prod_j (1 - x_j).$$

□

Setting all $x_j = -u$ recovers (2.18); setting $x_j = z^{2^j}$ gives the central finite polynomial.

6.2 The finite Prouhet polynomial

Define

$$P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n. \quad (6.2)$$

Theorem 6.2 (Complete finite factorization). *For every $m \geq 0$,*

$$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (6.3)$$

The empty product for $m = 0$ is 1.

Proof. Use theorem 6.1 with $x_j = z^{2^j}$. Equivalently, expanding the product chooses a subset of powers of two; its sum is an integer $n < 2^m$, uniquely, and the coefficient is $(-1)^{\#\text{chosen bits}} = \varepsilon_n$. □

Corollary 6.3 (Coefficient and block formulae). *For every $n \geq 0$,*

$$\varepsilon_n = [z^n] \prod_{j \geq 0} (1 - z^{2^j}), \quad (6.4)$$

where only factors with $2^j \leq n$ affect the coefficient. Moreover, for $q, m \geq 0$,

$$\sum_{r=0}^{2^m-1} \varepsilon_{2^m q + r} z^r = \varepsilon_q P_m(z). \quad (6.5)$$

Proof. The first statement is coefficient stabilization in the finite products. The second is theorem 2.2 summed over r . □

Formal versions. The corpus proves the finite factorization as `Fabius.thueMorseBlockPolynomial_eq_product` (`ThueMorseGenerating.lean`); `ThueMorseBooleanCube.lean` now proves it over an *arbitrary* commutative ring as `Fabius.prod_one_sub_pow_eq_sum_thueMorseSign`, as the specialization $u = -1$ of a two-parameter identity

$$\prod_{j < m} (1 + u z^{2^j}) = \sum_{n < 2^m} u^{\text{wt}_2(n)} z^n \quad (\text{Fabius.prod_one_add_mul_pow}),$$

valid in any commutative semiring with no convergence hypothesis. Its engine is a reindexing kernel identifying the dyadic block with the powerset of $\{0, \dots, m-1\}$ (`Fabius.sum_power_set_two_pow`) together with the digit dictionary `Fabius.binaryWeight_sum_two_pow`, `Fabius.sum_two_pow_lt_two_pow`: exactly the Boolean-cube structure this section exploits informally.

6.3 Reciprocity and cyclotomic multiplicities

The reflection symmetry becomes a reciprocal-polynomial identity.

Proposition 6.4 (Reciprocity). *For every $m \geq 0$,*

$$z^{2^m-1} P_m(z^{-1}) = (-1)^m P_m(z). \quad (6.6)$$

Thus P_m is reciprocal for even m and anti-reciprocal for odd m .

Proof. Reverse the coefficients and apply (2.8); alternatively, apply $z^{2^j}(1 - z^{-2^j}) = z^{2^j} - 1 = -(1 - z^{2^j})$ to every factor. $\square$

Repeatedly factor $1 - z^{2^j}$ by difference of squares:

$$P_m(z) = (1 - z)^m \prod_{q=0}^{m-2} (1 + z^{2^q})^{m-1-q}. \quad (6.7)$$

If Φ_d is the d -th cyclotomic polynomial, then equivalently

$$P_m(z) = (-1)^m \prod_{q=0}^{m-1} \Phi_{2^q}(z)^{m-q}. \quad (6.8)$$

Here $\Phi_1(z) = z - 1$. In particular, a primitive 2^q -th root of unity is a zero of multiplicity $m - q$ when $0 \leq q < m$. This multiplicity table simultaneously explains Prouhet cancellation at $z = 1$ and the systematic zeros in the dyadic discrete Fourier transform.

Formal versions. `ThueMorseBlockAlgebra.lean` proves the reciprocity (6.6) in two forms: as the coefficient-word reversal `Fabius.reverse_thueMorseBlockPolynomial` ($P_m^{\text{rev}} = (-1)^m P_m$ through `Polynomial.reverse`) and as the evaluation identity `Fabius.sum_thueMorseSign_mul_pow_reflect` over an arbitrary commutative ring, which is (6.6) cleared of the denominator z^{2^m-1} . The degree data feeding it — $\deg P_m = 2^m - 1$, top and leading coefficient $(-1)^m$, $P_m \neq 0$ — are `Fabius.natDegree_thueMorseBlockPolynomial`, `Fabius.coeff_thueMorseBlockPolynomial_top`, `Fabius.leadingCoeff_thueMorseBlockPolynomial`, and `Fabius.thueMorseBlockPolynomial_ne_zero`. The difference-of-squares ladder (6.7) is `Fabius.prod_one_sub_two_pow_eq_ladder`, valid over any commutative ring and built on the telescope $(1 - z) \prod_{i < j} (1 + z^{2^i}) = 1 - z^{2^j}$ (`Fabius.one_sub_mul_prod_one_add_two_pow`); the resulting Prouhet divisibility $(1 - X)^m \mid P_m$ is `Fabius.one_sub_X_pow_dvd_thueMorseBlockPolynomial`. The cyclotomic table (6.8) is `Fabius.thueMorseBlockPolynomial_eq_cyclotomic_prod`, resting on $\Phi_{2^{q+1}} = 1 + z^{2^q}$ (`Fabius.cyclotomic_two_pow_succ`, over any commutative ring).

6.4 The infinite product and Mahler equation

For $|z| < 1$, define

$$E(z) = \sum_{n \geq 0} \varepsilon_n z^n, \quad T(z) = \sum_{n \geq 0} \tau(n) z^n. \quad (6.9)$$

Theorem 6.5 (Infinite lacunary product). *For $|z| < 1$, and coefficientwise as a formal power series,*

$$E(z) = \prod_{j=0}^{\infty} (1 - z^{2^j}). \quad (6.10)$$

The product converges uniformly on compact subsets of the unit disk. The zero–one series is

$$T(z) = \frac{1}{2(1-z)} - \frac{1}{2}E(z). \quad (6.11)$$

Proof. For $|z| \leq \rho < 1$, the sum $\sum_j |z|^{2^j}$ converges, so the product converges uniformly on the compact disk. Every coefficient stabilizes in the finite products P_m , where [theorem 6.2](#) identifies it as ε_n . The formula for T follows from $\tau(n) = (1 - \varepsilon_n)/2$. $\square$

Peeling off factors gives the Mahler and dissection equations.

Corollary 6.6 (Mahler equations and dissections). *For $|z| < 1$,*

$$E(z) = (1 - z)E(z^2), \quad (6.12)$$

$$E(z) = P_m(z)E(z^{2^m}), \quad (6.13)$$

$$T(z) = (1 - z)T(z^2) + \frac{z}{1 - z^2}. \quad (6.14)$$

Proof. Separate the first factor, or the first m factors, in (6.10). Substitute (6.11) for the final identity; it also follows directly from (2.2). $\square$

Formal versions. `ThueMorseMahler.lean` proves the Mahler equation (6.12) exactly, as an identity of formal power series under `PowerSeries.expand` (`Fabius.thueMorseSeries_mahler`), and the dissection (6.13) at every level with the substitution realized as the m -th iterate of $z \mapsto z^2$ (`Fabius.thueMorseSeries_dissection`, `Fabius.coe_thueMorseBlockPolynomial_series`). The bridge (6.11) is formal in the cleared form $(1 - z)(2T(z) + E(z)) = 1$ (`Fabius.one_sub_X_mul_bridge`: the combination $2T + E$ is the geometric series). The coefficientwise content of [theorem 6.5](#) has long been formal as the stabilization lemma `Fabius.coeff_finite_thueMorse_product` (`ThueMorseGenerating.lean`), and its analytic identity is now formal on the whole complex unit disc. For every $z \in \mathbb{C}$ with $\|z\| < 1$, the signed series converges absolutely (`Fabius.summable_thueMorseSign_mul_pow_complex`), the lacunary product is multipliable (`Fabius.multipliable_one_sub_pow_two_pow_complex`), and

$$\sum_{n \geq 0} \varepsilon_n z^n = \prod_{j \geq 0} (1 - z^{2^j}) \quad (\|z\| < 1), \quad (6.15)$$

by `Fabius.tsum_thueMorseSign_mul_pow_complex` (`ThueMorseInfiniteProduct.lean`). The real APIs `Fabius.summable_thueMorseSign_mul_pow`, `Fabius.multipliable_one_sub_pow_two_pow`, and `Fabius.tsum_thueMorseSign_mul_pow` are retained as specializations and compatibility forms on $-1 < x < 1$, with negative x and $x = 0$ included, together with the unordered and partial-product restatements `Fabius.hasProd_one_sub_pow_two_pow` and `Fabius.tendsto_prod_one_sub_pow_two_pow`. Both sides restrict at scale 2^m to the finite

identity (6.3) – valid over any commutative ring – and uniqueness of limits closes the argument. The exponential ray $\sum \varepsilon_n e^{-nt} = \mathcal{E}(t)$ for $t > 0$, which the Dirichlet–Mellin layer consumes, is now the one-line specialization $x = e^{-t}$ (Fabi.us.tsum_thueMorseSign_exp_eq_lacunar_yExpProduct). The holomorphic sum Fabi.us.thueMorseDiscSeries is identified with the same complex product by Fabi.us.thueMorseDiscSeries_eq_tprod; its Mahler equation and finite-level factorization $F(z) = P_k(z)F(z^{2^k})$ are Fabi.us.thueMorseDiscSeries_mahler and Fabi.us.thueMorseDiscSeries_iterate (ThueMorseDiscSeries.lean), the forms used by the boundary theory. The bit Mahler equation (6.14) and the lacunary quotient are now formal in ThueMorseLacunaryQuotient.lean: the quotient $z/(1 - z^2)$ is realized as the dyadic geometric series G_1 – with $G_j = \sum_k z^{k2^j}$ proved inverse to $1 - z^{2^j}$ (Fabi.us.one_sub_X_pow_mul_dyadicGeometric) – so that (6.14) becomes the denominator-free identity $T = (1 - z)T(z^2) + zG_1$ (Fabi.us.thueMorseBitZSeries_mahler), and theorem 6.7 holds at every finite level as $T = \sum_{j < M} P_j z^{2^j} G_{j+1} + P_M T(z^{2^M})$ (Fabi.us.thueMorseBitZSeries_lacunary_quotient), by iterating the Mahler step; only the analytic convergence statements remain informal.

Iterating the inhomogeneous bit equation yields a useful lacunary sum.

Theorem 6.7 (Lacunary quotient expansion). *Let $P_0(z) = 1$ and $P_j(z) = \prod_{h=0}^{j-1} (1 - z^{2^h})$ for $j \geq 1$. Then, for $|z| < 1$,*

$$T(z) = \sum_{j=0}^{\infty} P_j(z) \frac{z^{2^j}}{1 - z^{2^{j+1}}}. \quad (6.16)$$

Proof. After M iterations of (6.14),

$$T(z) = \sum_{j=0}^{M-1} P_j(z) \frac{z^{2^j}}{1 - z^{2^{j+1}}} + P_M(z)T(z^{2^M}).$$

On compact subdisks P_M is bounded and $T(z^{2^M}) \rightarrow T(0) = 0$. □

6.5 A weighted digit enumerator

Both E and the geometric series are specializations of one umbrella product that keeps the whole digit sum, not only its parity.

Theorem 6.8 (Weighted enumerator). *For $|z| < 1$ and any scalar u , and coefficientwise as formal series,*

$$F_u(z) := \sum_{n \geq 0} u^{\text{wt}_2(n)} z^n = \prod_{j \geq 0} (1 + uz^{2^j}), \quad (6.17)$$

with the Mahler equation

$$F_u(z) = (1 + uz)F_u(z^2). \quad (6.18)$$

The specializations $u = 1$ and $u = -1$ are

$$F_1(z) = \frac{1}{1 - z}, \quad F_{-1}(z) = E(z). \quad (6.19)$$

Proof. Expanding the product chooses a finite set of bit positions; by uniqueness of binary expansions each exponent n arises exactly once, with coefficient $u^{\text{wt}_2(n)}$. Peeling the first factor gives (6.18). At $u = 1$ every n receives coefficient one, which restates the uniqueness of binary expansions; at $u = -1$ the coefficient is ε_n . □

Differentiating (6.17) with respect to u at $u = 1$ gives the ordinary generating function of the digit sum itself:

$$\sum_{n \geq 0} \text{wt}_2(n) z^n = \frac{1}{1-z} \sum_{j \geq 0} \frac{z^{2^j}}{1+z^{2^j}}. \quad (6.20)$$

Higher derivatives at $u = 1$ generate the falling-factorial moments $\text{wt}_2(n)(\text{wt}_2(n) - 1) \cdots (\text{wt}_2(n) - r + 1)$ in the same way. The lacunary sum in (6.20) is the characteristic-zero twin of the Artin–Schreier solution (7.16) below: the same series $\sum_j z^{2^j}/(1+z^{2^j})$ computes the digit sum with integer coefficients and the digit parity in characteristic two. The umbrella product itself is formalized in full generality as `Fabius.prod_one_add_mul_pow (ThueMorseBooleanCube.lean)`; the specializations (6.19) are the $z = 1$ and $u = -1$ readings recorded there as `Fabius.sum_pow_bin`, `aryWeight_eq_one_add_pow` and `Fabius.prod_one_sub_pow_eq_sum_thueMorseSign`.

6.6 Thue–Morse constants in every base

For a real $b > 1$, put

$$\Theta_b = \sum_{n \geq 0} \frac{\tau(n)}{b^{n+1}}. \quad (6.21)$$

For integer $b \geq 2$, these are the base- b numbers whose fractional digits are the Thue–Morse bits.

Corollary 6.9 (Base- b product). *For every real $b > 1$,*

$$\Theta_b = \frac{1}{2(b-1)} - \frac{1}{2b} \prod_{j \geq 0} (1 - b^{-2^j}). \quad (6.22)$$

The signed companion is

$$\sum_{n \geq 0} \frac{\varepsilon_n}{b^{n+1}} = \frac{1}{b} \prod_{j \geq 0} (1 - b^{-2^j}). \quad (6.23)$$

Proof. Set $z = 1/b$ in (6.11) and (6.10), then multiply by $1/b$. $\square$

For $b = 2$, $\Theta_2 = 0.0110100110010110 \dots_2$ is the classical Prouhet–Thue–Morse constant. Mahler’s theory implies its transcendence; that deep arithmetic result is separate from the elementary product identity proved here [8, 12].

7 Euler transforms, coefficient recurrences, and algebraic series

7.1 The logarithmic ruler transform

Take the formal logarithm of (6.10):

$$\log E(z) = \sum_{j \geq 0} \log(1 - z^{2^j}) = - \sum_{j \geq 0} \sum_{r \geq 1} \frac{z^{r 2^j}}{r}. \quad (7.1)$$

For a fixed exponent k , the admissible j ’s satisfy $0 \leq j \leq \nu_2(k)$, and the corresponding denominator is $k/2^j$. Define

$$a_k = \sum_{j=0}^{\nu_2(k)} 2^j = 2^{\nu_2(k)+1} - 1. \quad (7.2)$$

Theorem 7.1 (Euler transform of the ruler function). *As a formal series, and analytically for $|z| < 1$,*

$$\log E(z) = - \sum_{k \geq 1} \frac{a_k}{k} z^k, \quad a_k = 2^{\nu_2(k)+1} - 1. \quad (7.3)$$

This representation contains no explicit binary digit sum. The coefficient sequence (a_k) is a simple transform of the ruler sequence.

7.2 A nonlocal convolution recurrence

Theorem 7.2 (Ruler-convolution recurrence). *With $\varepsilon_0 = 1$, for every $n \geq 0$,*

$$n\varepsilon_n = - \sum_{k=1}^n a_k \varepsilon_{n-k}. \quad (7.4)$$

Proof. Differentiate (7.3), multiply by $E(z)$, and compare the coefficient of z^{n-1} :

$$E'(z) = -E(z) \sum_{k \geq 1} a_k z^{k-1}.$$

□

The weighted convolution on the right always collapses to $\pm n$. As an evaluator it is inferior to popcount, but it is the natural recurrence attached to the Euler product.

Remark 7.3 (Combinatorial proof of the recurrence). [Theorem 7.2](#) also has a purely finite proof which explains why the convolution always collapses to $\pm n$. Write $a_k = \sum_{j \geq 0, 2^j | k} 2^j$ as the sum of the binary divisors of k and interchange the two sums:

$$\sum_{k=1}^n a_k \varepsilon_{n-k} = \sum_{j \geq 0} 2^j \sum_{\substack{1 \leq k \leq n \\ 2^j | k}} \varepsilon_{n-k} = \sum_{j \geq 0} 2^j \sum_{\substack{0 \leq m < n \\ m \equiv n \pmod{2^j}}} \varepsilon_m.$$

By block concatenation $\varepsilon_{2^j t + r} = \varepsilon_t \varepsilon_r$ ($r < 2^j$), the residue-class sum equals $\varepsilon_{n \bmod 2^j} \sum_{t < \lfloor n/2^j \rfloor} \varepsilon_t$, and the prefix sum collapses by pair cancellation to $\mathbf{1}_{\lfloor n/2^j \rfloor \text{ odd}} \varepsilon_{\lfloor n/2^{j+1} \rfloor}$. When the j -th binary digit of n is 1, the odd-successor law $\varepsilon_{2u+1} = -\varepsilon_u$ turns the product back into $-\varepsilon_n$; when it is 0, the class sum vanishes. Hence $\sum_k a_k \varepsilon_{n-k} = -\varepsilon_n \sum_{j: b_j(n)=1} 2^j = -n\varepsilon_n$: the surviving binary divisors reassemble the binary expansion of n .

Formal versions. `ThueMorseEulerTransform.lean` proves [theorem 7.2](#) as `Fabius.ruler_convolution`, following [remark 7.3](#) — no logarithm, derivative, or convergence enters, and no side condition on n is imposed: the Lean statement is hypothesis-free, the case $n = 0$ being the empty sum. The reusable pieces are the prefix collapse `Fabius.sum_thueMorseSign_range`, the residue-class collapse `Fabius.sum_thueMorseSign_progression`, the divisor-sum identification `Fabius.sum_ite_two_pow_dvd`, the binary expansion `Fabius.sum_ite_odd_div_two_pow`, and the fully general reindexing `Fabius.sum_filter_dvd_sub_eq_sum_progression`, valid for any function into any additive commutative monoid and any modulus. The Euler transform (7.3) itself remains analytic; the partition formula below is now fully formal.

7.3 Partition and Bell-polynomial formulae

Exponentiating (7.3) gives a finite partition sum for every coefficient.

Theorem 7.4 (Partition formula). *For $n \geq 0$,*

$$\varepsilon_n = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \prod_{j=1}^n \frac{1}{m_j!} \left(-\frac{a_j}{j} \right)^{m_j}. \quad (7.5)$$

Proof. Write $E(z) = \prod_{j \geq 1} \exp(-a_j z^j / j)$, expand each exponential, and collect terms of total degree n . $\square$

Let $\mathcal{B}_n$ denote the complete exponential Bell polynomial, defined by

$$\exp \left(\sum_{j \geq 1} x_j \frac{z^j}{j!} \right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (7.6)$$

Corollary 7.5 (Bell-polynomial coefficient). *For every $n \geq 0$,*

$$\varepsilon_n = \frac{1}{n!} \mathcal{B}_n(-0!a_1, -1!a_2, \dots, -(n-1)!a_n). \quad (7.7)$$

Proof. Use $x_j = -(j-1)!a_j$ in (7.6). $\square$

Although the right sides are rational-looking sums, they always equal ± 1 . This integrality collapse is itself encoded by the original lacunary product.

Formal versions. [Theorem 7.4](#) is `Fabius.thueMorse_partition_formula` and [corollary 7.5](#) is `Fabius.thueMorse_bell_formula` (`ThueMorsePartition.lean`), both over $\mathbb{Q}$ with the weighted partitions realized as the finset `Fabius.weightedPartitions` of multiplicity vectors. The formalization replaces the analytic exponentiation argument by finite combinatorics: the partition sum obeys the *marked-part recurrence* $nP(n) = \sum_{j \leq n} jx_j P(n-j)$ – proved for an arbitrary coefficient sequence into any characteristic-zero field (`Fabius.partitionExpSum_recurrence`) by distributing the weight $n = \sum jm_j$ over the parts and removing the marked part – while the ruler convolution ([theorem 7.2](#)) says the sign sequence obeys the same recurrence at $x_j = -a_j/j$; the division-by- n uniqueness lemma `Fabius.eq_of_ruler_recurrence` then identifies the two. No power series, exponentials, or convergence enter the proof.

7.4 A Hessenberg determinant

For $n \geq 0$, let

$$H_n = \begin{pmatrix} a_1 & 1 & 0 & \cdots & 0 \\ a_2 & a_1 & 2 & \ddots & \vdots \\ a_3 & a_2 & a_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & n-1 \\ a_n & a_{n-1} & \cdots & a_2 & a_1 \end{pmatrix}. \quad (7.8)$$

For $n = 0$ this is the empty matrix, with $\det H_0 = 1$.

Theorem 7.6 (Determinant formula). *For every $n \geq 0$,*

$$\varepsilon_n = \frac{(-1)^n}{n!} \det H_n. \quad (7.9)$$

In particular,

$$\det H_n = (-1)^n n! \varepsilon_n \in \{n!, -n!\}, \quad |\det H_n| = n!. \quad (7.10)$$

Proof. Put $d_0 = 1$ and $d_n = \det H_n$. Expansion of a lower Hessenberg determinant along its first column gives

$$d_n = \sum_{k=1}^n (-1)^{k-1} \frac{(n-1)!}{(n-k)!} a_k d_{n-k}. \quad (7.11)$$

Set $d_n = (-1)^n n! c_n$. Then (7.11) reduces to

$$n c_n = - \sum_{k=1}^n a_k c_{n-k},$$

with $c_0 = 1$. By uniqueness of the recurrence in [theorem 7.2](#), $c_n = \varepsilon_n$. $\square$

Remark 7.7 (An expansion-free proof). The formalization (`Fabius.det_thueMorseHessenberg`, `ThueMorseHessenberg.lean`) avoids the Hessenberg expansion (7.11) altogether. Write the ruler convolution for $\varepsilon(1), \dots, \varepsilon(n)$ as a linear system $T_n \mathbf{x} = \mathbf{c}$ whose matrix T_n is lower triangular with diagonal $1, 2, \dots, n$, so $\det T_n = n!$ (`Fabius.det_thueMorseTriangular`). Cramer’s rule – in the reusable form `cramer_A(Ax) = \det A \cdot x`, proved for any commutative ring (`Matrix.cramer_mulVec`, `ToeplitzHessenberg.lean`) – evaluates $n! \varepsilon(n)$ as the determinant of T_n with its last column replaced by $\mathbf{c}$; one cyclic column rotation ($\text{sign } (-1)^{n-1}$) and one column negation turn that matrix into H_n . Hence $\det H_n = (-1)^n n! \varepsilon(n)$ with no minor ever expanded, for every $n \geq 0$; that $\det H_n = \pm n!$ is `Fabius.det_thueMorseHessenberg_eq_or`, and the sharper $|\det H_n| = n!$ is `Fabius.natAbs_det_thueMorseHessenberg`.

None of this is special to the ruler coefficients. The argument is carried out in `ToeplitzHessenberg.lean` for arbitrary band data (d_m) and (a_k) over an arbitrary commutative ring: the triangular matrix `Fabius.triBand`, of size N and with $\det = \prod_{i < N} d_i$ (`Fabius.det_triBand`); the lower Toeplitz–Hessenberg matrix `Fabius.hessBand`; and the determinant formula $\det = (-1)^{n+1} (\prod_{i \leq n} d_i) x_n$ for a matrix of size $n+1$, read off from *any* solution x of the convolution system $T\mathbf{x} = -(a_1, \dots, a_{n+1})$ (`Fabius.det_hessBand`), together with the scalar form `Fabius.det_hessBand_of_recurrence` that the Thue–Morse instance consumes. No invertibility, no positivity, and no uniqueness of the solution is assumed anywhere.

7.5 The zero–one series over $\mathbb{F}_2$

The signed series loses all information modulo two because $-1 = 1$. The bit series remains nontrivial. Work in $\mathbb{F}_2[[z]]$ and write

$$\Theta(z) = \sum_{n \geq 0} \tau(n) z^n. \quad (7.12)$$

Theorem 7.8 (Artin–Schreier equation). *In $\mathbb{F}_2[[z]]$,*

$$\Theta(z) = (1+z)\Theta(z^2) + \frac{z}{1+z^2}, \quad (7.13)$$

$$\boxed{(1+z)^3 \Theta(z)^2 + (1+z)^2 \Theta(z) + z = 0.} \quad (7.14)$$

If $Y(z) = (1+z)\Theta(z)$, then

$$\boxed{Y(z)^2 + Y(z) = \frac{z}{1+z}.} \quad (7.15)$$

Proof. Split the series into even and odd coefficients and use $\tau(2n) = \tau(n)$, $\tau(2n+1) = 1 + \tau(n)$ in characteristic two:

$$\Theta(z) = \Theta(z^2) + z \sum_{n \geq 0} (1 + \tau(n)) z^{2n} = (1+z)\Theta(z^2) + \frac{z}{1+z^2}.$$

Frobenius gives $\Theta(z^2) = \Theta(z)^2$, while $1 + z^2 = (1+z)^2$. Clear the denominator and substitute $Y = (1+z)\Theta$. $\square$

Corollary 7.9 (Explicit Artin–Schreier sum). *The unique zero-constant-term solution is*

$$Y(z) = \sum_{j=0}^{\infty} \left(\frac{z}{1+z} \right)^{2^j} = \sum_{j=0}^{\infty} \frac{z^{2^j}}{1+z^{2^j}}, \quad (7.16)$$

$$\Theta(z) = \frac{1}{1+z} \sum_{j=0}^{\infty} \frac{z^{2^j}}{1+z^{2^j}}. \quad (7.17)$$

Proof. Let $A = z/(1+z)$. In the z -adic topology, $Y = \sum_{j \geq 0} A^{2^j}$ is well defined and $Y^2 + Y = A$ by index shifting. Frobenius gives $A^{2^j} = z^{2^j}/(1+z^{2^j})$. A zero-constant solution of the homogeneous equation $Z^2 + Z = 0$ must vanish, proving uniqueness. $\square$

Christol’s theorem says that a finite-field sequence is automatic exactly when its generating series is algebraic. Here the degree-two certificate is derived directly; it is an algebraic shadow of the two-state automaton [13].

Formal versions. `ThueMorseArtinSchreier.lean` proves the functional equation (7.13) in the cleared form $\Theta = (1+z)\Theta^2 + zG^2$ with $G = \sum z^n$ (`Fabius.thueMorseBitSeries_functional`, `Fabius.one_add_X_mul_geometricSeries`), the boxed quadratic (7.14) as `Fabius.thueMorseBitSeries_quadratic`, and the normalized form (7.15) cleared of its denominator, $(1+z)(Y^2 + Y) = z$, as `Fabius.artinSchreier_one_add_X_mul`. The Frobenius step $\Theta(z)^2 = \Theta(z^2)$ is isolated as a coefficientwise lemma valid over *any* commutative ring of characteristic two (`Fabius.coeff_sq_charTwo`: the off-diagonal convolution terms cancel in pairs), specializing over $\mathbb{F}_2$ to `Fabius.coeff_sq_zmodTwo`. Of the explicit solution (7.16) the formalization keeps the finite content: the partial sums $S_J = \sum_{j < J} a^{2^j}$ satisfy $S_J^2 + S_J = a^{2^J} + a$ in any commutative ring of characteristic two (`Fabius.sum_pow_two_pow_sq_add`), so they solve the Artin–Schreier equation up to the single tail term, and no z -adic limit is needed for any identity in this subsection.

The quadratic’s *entire* solution set is now formal as well, and it is a two-element set: in $\mathbb{F}_2[[z]]$,

$$(1+z)^3 Y^2 + (1+z)^2 Y + z = 0 \iff Y = \Theta \text{ or } Y = \Theta + G \quad (7.18)$$

(`Fabius.artinSchreier_solution_iff`), with no hypothesis on Y whatever – in particular none on its constant term. The difference $D = Y - \Theta$ of two solutions satisfies the factored identity $(1+z)^2 D((1+z)D + 1) = 0$; since $\mathbb{F}_2[[z]]$ is a domain and $1+z \neq 0$, either $D = 0$ or $(1+z)D = 1$, and the second alternative is not a contradiction but the second root: it says $D = G$, because $(1+z)G = 1$. The Artin–Schreier symmetry $Y \mapsto Y + 1$ of (7.15) is exactly the translation $\Theta \mapsto \Theta + G$ after the substitution $Y = (1+z)\Theta$, and $\Theta + G$ is verified to be a root independently (`Fabius.thueMorseBitSeries_add_geometricSeries_quadratic`).

7.6 An integral algebraic lift

The fourth draft records a rather different integer sequence whose parity is Thue–Morse. Define $c_0 = 0$, $c_1 = 1$, and for $n \geq 2$,

$$nc_n = (5 - 2n)c_{n-1} + 3(n-1)c_{n-2} + 1. \quad (7.19)$$

Its first terms are

$$0, 1, 1, 2, 1, 4, -2, 13, -23, 68, -164, 439, -1146, 3067, \dots$$

Theorem 7.10 (Integer algebraic lift). *Let $C(z) = \sum_{n \geq 0} c_n z^n$. Then*

$$C(z) = \frac{1}{2(1-z)} \left(\sqrt{\frac{1+3z}{1-z}} - 1 \right), \quad (7.20)$$

$$(1 - z)^3 C(z)^2 + (1 - z)^2 C(z) = z. \quad (7.21)$$

All c_n are integers and

$$c_n \equiv \tau(n) \pmod{2}, \quad \varepsilon_n = (-1)^{c_n}. \quad (7.22)$$

Proof. Multiplying (7.19) by z^{n-1} , summing, and including the initial terms gives a first-order differential equation whose zero-constant solution is (7.20); squaring yields (7.21).

Conversely, the coefficient of z^n in $(1 - z)^2 C(z)$ contains c_n with coefficient one, while the quadratic term uses only earlier coefficients because $C(0) = 0$. Thus (7.21) recursively determines integers c_n .

Modulo two, (7.21) becomes precisely (7.14). That equation has a unique zero-constant solution in $\mathbb{F}_2[[z]]$, so $C(z) \equiv \Theta(z) \pmod{2}$. $\square$

Remark 7.11. The lift grows and changes sign; it is not a disguised digit sum. Its parity is Thue–Morse because an integral algebraic equation reduces to the automatic sequence’s Artin–Schreier equation. This suggests a reusable discovery method: search for integral algebraic lifts of finite-field automatic series.

Remark 7.12 (Closed form and a purely arithmetic parity proof). Expanding (7.20) with $u = z/(1 - z)$ against the Catalan generating function gives the explicit coefficients

$$c_n = \sum_{k=1}^n (-1)^{k-1} C_{k-1} \binom{n}{k}, \quad (7.23)$$

where C_m is the m -th Catalan number – manifestly integers, with no recursion needed. This representation yields a parity proof that bypasses power series entirely: Catalan numbers are odd exactly at the Mersenne indices $m = 2^j - 1$ (fold the convolution $C_{m+1} = \sum C_i C_{m-i}$ at its middle and recurse), so modulo two the sum (7.23) retains only $k = 2^j$; by Lucas’s theorem $\binom{n}{2^j}$ is odd exactly when the j -th bit of n is set; and the bits sum to $s_2(n)$. Hence $c_n \equiv s_2(n) \equiv \tau(n) \pmod{2}$.

Formal versions. The uniqueness step is `Fabius.artinSchreier_solution_unique` (`ThueMorseArtinSchreier.lean`), and it is now a one-line consequence of the complete solution set (7.18): the quadratic has exactly the two roots Θ and $\Theta + G$, and of these only Θ has constant term zero, because G has constant term one. The statement itself is unchanged – a zero-constant-term solution of the quadratic equals Θ – only its proof now factors through the stronger classification. The parity statement (7.22) is formal in conditional-universal form: *any* integer sequence with $c_0 = 0$ whose generating series satisfies (7.21) has Thue–Morse parity (`Fabius.integerLift_parity`, `Fabius.integerLift_modEq`). The sequence itself is now *constructed* in `ThueMorseIntegerLift.lean` via the closed form (7.23) (`Fabius.integerLift`), and the boxed parity claims are proved unconditionally along the route of [remark 7.12](#): Catalan parity is `Fabius.catalan_odd_iff` (via `Fabius.catalan_mod_two`, reusable facts about C_m independent of this atlas), and $c_n \equiv \tau(n) \pmod{2}$, $\varepsilon_n = (-1)^{c_n}$ are `Fabius.integerLift_cast_zmod`, `Fabius.integerLift_emod_two`, and `Fabius.thueMorseSign_eq_neg_one_pow_integerLift`. The algebraic equation (7.21) is now proved for the constructed series (`Fabius.integerLiftSeries_algebraic`, `ThueMorseIntegerLiftEquation.lean`) by pure finite combinatorics: the second difference of the lift equals the substituted alternating Catalan series (`Fabius.integerLift_delta_bridge`, double Pascal), the interior product sum collapses through a reusable *diagonal Vandermonde* identity (`Fabius.sum_range_choose_mul_choose`), the substituted series satisfies the quadratic convolution identity $\sum_{p+q=m} h_p h_q = h_m - h_{m+1} - [m=0]$ (`Fabius.catalanSeriesDelta_conv`) – the coefficient form of $H = 1 - uH^2$ – and cancelling the non-zero-divisor $1 - z$ in the domain $\mathbb{Z}[[z]]$ yields the boxed equation. Finally, the defining recurrence (7.19) is `Fabius.integerLift_recurrence` (`ThueMorseIntegerLiftRecurrence.lean`): differentiating the algebraic equation and linearizing with one certificate produces the first-order equation $(1 + 3z)(1 - z)^2 C' = 1 + 3(1 - z^2)C$ (`Fabius.integerLiftSeries_ode`), whose coefficients give a four-term relation that telescopes onto the recurrence by induction. Every clause of [theorem 7.10](#) is now formal and unconditional.

8 Natural boundary, Hankel determinants, and Diophantine consequences

The algebraic picture of the previous section is completed by three analytic-arithmetic facts: the series E cannot be continued across any boundary point, its Hankel determinants never vanish, and its values at algebraic points are transcendental with the smallest possible irrationality exponent. The first two facts are proved here in full – and both are formalized; the Diophantine statements are deep literature theorems, stated with references and not reproved.

8.1 The unit circle is a natural boundary

A *natural boundary* of a power series is a curve through which no analytic continuation across any arc is possible.

Theorem 8.1 (Natural boundary). *Every point of the unit circle is a singular point of E ; the circle is a natural boundary. Consequently E and T are transcendental over the field $\mathbb{C}(z)$ of rational functions.*

Proof. Let ζ be a root of unity of two-power order, say $\zeta^{2^r} = 1$. For $0 < \rho < 1$, split the product (6.10) at index r : the head has r factors, each of modulus at most 2, while every tail factor $1 - (\rho\zeta)^{2^j} = 1 - \rho^{2^j}$ with $j \geq r$ is a real number in $(0, 1)$. Hence

$$|E(\rho\zeta)| \leq 2^r (1 - \rho^{2^r}) \xrightarrow[\rho \uparrow 1]{} 0.$$

Such roots of unity are dense on the circle. If E continued analytically through some open boundary arc, the continued function would vanish on a set with an accumulation point inside its domain, hence would vanish identically by the identity theorem, contradicting $E(0) = 1$. An algebraic function over $\mathbb{C}(z)$ has only finitely many singular points on the Riemann sphere, so it cannot have a natural boundary; and T differs from $-E/2$ by the rational function in (6.11). $\square$

Formal versions. The natural-boundary core of Theorem 8.1 is fully formal. The series $F(z) = \sum \varepsilon_n z^n$ is realized as a holomorphic function on the open unit disc by locally uniform convergence of its polynomial partial sums (`Fabius.thueMorseDiscSeries`, `Fabius.thueMorseDiscSeries_differentiableOn`, `ThueMorseDiscSeries.lean`), satisfying the analytic Mahler equation $F(z) = (1 - z)F(z^2)$ and its iterate $F(z) = \prod_{j < k} (1 - z^{2^j}) F(z^{2^k})$ (`Fabius.thueMorseDiscSeries_mahler`, `Fabius.thueMorseDiscSeries_iterate`), with the real-ray bound $|F(x)| \leq 1 - x$ on $[0, 1)$ (`Fabius.thueMorseDiscSeries_real_1e`) – the first factor of the lacunary product already vanishes at the boundary. The radial vanishing at every two-power root of unity is `Fabius.tendsto_thueMorseDiscSeries_radial_zero`, and the theorem itself is `Fabius.thueMorse_natural_boundary`: no function holomorphic on any disc centered at a point of the unit circle agrees with F on the overlap with the unit disc – the hypothetical extension vanishes at the dyadic roots accumulating at the center, hence identically by the identity theorem, forcing $F \equiv 0$ on the disc against $F(0) = 1$. Only the algebraic coda – that an algebraic function over $\mathbb{C}(z)$ has finitely many singular points, whence the transcendence of E and T over $\mathbb{C}(z)$ – remains informal.

Remark 8.2. Two softer routes to the same conclusion deserve mention. The Pólya–Carlson theorem [20] says that a power series with integer coefficients and radius of convergence one is either rational or has the unit circle as a natural boundary, and E is visibly not rational because its coefficient sequence is not eventually periodic (section 14). For Mahler functions there is also a general rational-or-natural-boundary dichotomy [21]. The radial-zero argument above is preferred here because it is self-contained.

The contrast with [section 7.5](#) is worth displaying:

$$\left. \begin{array}{l} \text{over } \mathbb{F}_2(z) \\ \text{over } \mathbb{C}(z) \end{array} \right| \begin{array}{l} \Theta(z) \text{ is algebraic of degree two,} \\ T(z) \text{ is transcendental with a natural boundary.} \end{array} \quad (8.1)$$

Automaticity is algebraicity in the matching finite characteristic, not in characteristic zero.

8.2 Hankel determinants never vanish

For a sequence $c = (c_n)_{n \geq 0}$, the order- r *Hankel determinant* is the determinant of the $r \times r$ matrix whose (i, j) -entry is c_{i+j} :

$$H_r(c) = \det(c_{i+j})_{0 \leq i, j < r}. \quad (8.2)$$

For the signed sequence the first values are

$$H_1, H_2, \dots = 1, -2, 4, 8, -16, -32, -64, 128, -256, -1536, \dots \quad (8.3)$$

Unlike the Hessenberg matrix of [theorem 7.6](#), which was engineered from the ruler weights precisely so that its determinant would telescope, these matrices sample the raw signs and have no visible triangular structure.

Theorem 8.3 (Allouche–Peyrière–Wen–Wen). *For every $r \geq 1$,*

$$\boxed{H_r(\varepsilon) \neq 0}. \quad (8.4)$$

Proof. We show the exact two-adic valuation $H_r(\varepsilon) = 2^{r-1} \cdot (\text{odd})$, which is visible in (8.3) – for instance $H_{10} = -1536 = -2^9 \cdot 3$. Interleave the rows and the columns of H_{2n} by parity (the same permutation on both sides, so no sign). By $\varepsilon_{2k} = \varepsilon_k$, $\varepsilon_{2k+1} = -\varepsilon_k$ the four blocks are $\begin{bmatrix} A & -A \\ -A & K \end{bmatrix}$ with A the order- n Hankel matrix and $K_{ij} = \varepsilon_{i+j+1}$; adding the top block row to the bottom one triangularizes, and the Schur block has entries $\varepsilon_{i+j+1} - \varepsilon_{i+j} = 2m_{i+j}$, where $m_k = \tau(k) - \tau(k+1) \in \{-1, 0, 1\}$ is the half-increment. Hence

$$H_{2n}(\varepsilon) = 2^n H_n(\varepsilon) H_n(m), \quad H_{2n+1}(\varepsilon) = 2^n H_{n+1}(\varepsilon) H_n(m),$$

the odd case by the same elimination against the truncated top block. The determinants $H_n(m)$ are *odd*: modulo two, $|m|$ is the period-doubling sequence p with $p(2k) = 1$, $p(2k+1) = 1 + p(k)$, and the two mod-two determinant families – Hankel determinants $D(n)$ of $p(k)$ and $Y(n)$ of $p(\lfloor k/2 \rfloor)$ – close under the same parity interleaving:

$$D(2n) = D(n), \quad D(2n+1) = Y(n), \quad Y(2n) = D(n)Y(n), \quad Y(2n+1) = D(n+1)Y(n),$$

using $p(k) + p(k+1) = p(\lfloor k/2 \rfloor)$ and the vanishing of all permutation signs in characteristic two. Every value is 1 by strong induction. Feeding the oddness back into the two factorizations gives $v_2(H_{2n}) = n + v_2(H_n)$ and $v_2(H_{2n+1}) = n + v_2(H_{n+1})$, whence $v_2(H_r) = r - 1$ exactly and $H_r \neq 0$. $\square$

The original proof runs a closed finite system of determinant recurrences [22]; later proofs include Han’s Hankel continued-fraction calculus [23] and a combinatorial argument of Bugeaud and Han [24]. The two-adic argument above is in the spirit of all three, compressed to a pair of block factorizations and one mod-two induction. The statement is a strong rigidity property: no finite window of the sequence ever satisfies an accidental low-order linear relation of Hankel type.

Nonvanishing has a standard approximation-theoretic reading. A *Padé approximant of type $[r - 1/r]$* to E is a rational function $A(z)/B(z)$ with $\deg A \leq r - 1$ and $\deg B \leq r$ whose Taylor expansion agrees with E to the maximal possible order. When $H_r(\varepsilon) \neq 0$, this approximant exists

and the agreement is exact through order z^{2r-1} with a nonzero discrepancy coefficient at z^{2r} ; the whole diagonal Padé table of E is therefore normal, with no degenerate blocks.

Formal versions. [Theorem 8.3](#) is fully formal, with the proof above. The mod-two engine is `PeriodDoublingHankel.lean`: the period-doubling bit and its recursions (`Fabius.pdBit`), the four determinant halvings, and the strong induction `Fabius.pdHankel_det_eq_one` — every Hankel determinant of the period-doubling sequence over $\mathbb{F}_2$ equals 1. The reusable block machinery (`Fabius.det_fromBlocks_of_eq_mul`, `Fabius.det_fromBlocks_symm`, the parity interleavings) is proved over an arbitrary commutative ring. The integer level is `ThueMorseHankel.lean`: the doubling factorizations (`Fabius.hankelSign_det_two_mul`, `Fabius.hankelSign_det_two_mul_add_one`), the mod-two transfer (`Fabius.mseqHankel_det_odd`), the exact valuation $H_r = 2^{r-1} \cdot \text{odd}$ (`Fabius.hankelSign_det_eq_two_pow_mul_odd`), and the boxed nonvanishing (8.4) (`Fabius.thueMorseHankel_ne_zero`).

8.3 Transcendental values and the sharp irrationality exponent

Beyond transcendence of the function, Mahler’s method gives transcendence of its values. The Mahler equation (6.12) relates z to z^2 , and for an algebraic α with $0 < |\alpha| < 1$ the forward orbit $\alpha, \alpha^2, \alpha^4, \dots$ stays in the unit disk and never meets the zero of the rational cofactor $1 - z$.

Theorem 8.4 (Mahler; literature theorem). *If α is algebraic and $0 < |\alpha| < 1$, then $E(\alpha)$ is transcendental; consequently so is $T(\alpha)$ [8].*

In particular, every base- b Thue–Morse constant Θ_b of (6.21) is transcendental for integer $b \geq 2$. How well such a constant can be approximated by rationals is measured by its *irrationality exponent* $\mu(\xi)$: the supremum of all real μ such that $|\xi - p/q| < q^{-\mu}$ holds for infinitely many fractions p/q . Every irrational number has $\mu \geq 2$, by Dirichlet’s pigeonhole theorem, and Lebesgue-almost every real number has $\mu = 2$.

Theorem 8.5 (Bugeaud; literature theorem). *For every integer $b \geq 2$,*

$$\boxed{\mu(\Theta_b) = 2.} \tag{8.5}$$

Thus the Thue–Morse–Mahler numbers are transcendental yet approximate as badly as possible for irrational numbers. Bugeaud’s proof [25] runs the Padé approximants of the previous subsection through the Mahler equation: iterating $z \mapsto z^2$ transports one good rational approximation at $z = 1/b$ to a whole geometric family, and the nonvanishing Hankel determinants guarantee that the approximation orders have no long gaps. Earlier partial bounds and the general determinant framework are surveyed in [26, 27]. This is the cleanest place where the determinant rigidity of [theorem 8.3](#) pays an explicitly Diophantine dividend.

9 Finite differences, Prouhet cancellation, and all dyadic moments

9.1 The mixed-difference master identity

For a function f , let

$$(\Delta_h f)(x) = f(x + h) - f(x), \quad (T_h f)(x) = f(x + h). \tag{9.1}$$

Thus $I - T_h = -\Delta_h$.

Theorem 9.1 (Dyadic finite-difference factorization). *For every $m \geq 0$, every scalar h , and every function for which the values are defined,*

$$\boxed{\sum_{n=0}^{2^m-1} \varepsilon_n f(x + nh) = \prod_{j=0}^{m-1} (I - T_{2^j h}) f(x)} \tag{9.2}$$

$$= (-1)^m (\Delta_h \Delta_{2h} \cdots \Delta_{2^{m-1}h} f)(x). \quad (9.3)$$

Proof. Expand the operator product. A subset $S \subseteq \{0, \dots, m-1\}$ contributes the shift by $h \sum_{j \in S} 2^j$ and sign $(-1)^{|S|}$. Binary expansion identifies the subset sum with a unique $n < 2^m$, and the sign is ε_n . $\square$

This theorem is stronger than any one moment identity: it is an inclusion–exclusion formula valid for arbitrary test functions.

Formal versions. `ThueMorseMixedDifference.lean` proves [theorem 9.1](#) in full generality: the argument space is any additive commutative monoid, the value space any additive commutative group, and f an arbitrary function between them. The operator $\prod_{j < m} (I - T_{2^j h})$ is `Fabius.dyadicMixedDifference`, the boxed identity (9.2) is `Fabius.sum_thueMorseSign_smul_eq_mixedDifference`, and the expanded inclusion–exclusion form is `Fabius.sum_thueMorseSign_smul_eq_sum_powerset_smul`. The polynomial and exponential specializations recover the Prouhet and moment layers formalized earlier.

9.2 Exact multiple-integral remainder

Corollary 9.2 (Integral cubature formula). *Suppose $h \geq 0$ and $f \in C^m$ on an interval containing all relevant points. Then*

$$\sum_{n < 2^m} \varepsilon_n f(x + nh) = (-1)^m \int_0^h \int_0^{2h} \cdots \int_0^{2^{m-1}h} f^{(m)}(x + t_0 + \cdots + t_{m-1}) dt_{m-1} \cdots dt_0 \quad (9.4)$$

$$= (-1)^m 2^{\binom{m}{2}} h^m \int_{[0,1]^m} f^{(m)}\left(x + h \sum_{j=0}^{m-1} 2^j u_j\right) d\mathbf{u}. \quad (9.5)$$

Consequently

$$\left| \sum_{n < 2^m} \varepsilon_n f(x + nh) \right| \leq 2^{\binom{m}{2}} h^m \sup_{x \leq y \leq x + (2^m - 1)h} |f^{(m)}(y)|. \quad (9.6)$$

Proof. Apply $(I - T_a)f(x) = -\int_0^a f'(x+t) dt$ successively. Rescale $t_j = 2^j h u_j$ for the unit-cube form. The box volume is $h^m 2^{0+1+\cdots+(m-1)}$. $\square$

Corollary 9.3 (Sign for completely monotone kernels). *If $(-1)^m f^{(m)} \geq 0$ on the interval, then*

$$\sum_{n < 2^m} \varepsilon_n f(x + nh) \geq 0. \quad (9.7)$$

For $a, s > 0$,

$$\sum_{n < 2^m} \frac{\varepsilon_n}{(a+n)^s} = (s)_m \int_0^1 \int_0^2 \cdots \int_0^{2^{m-1}} \frac{dt_{m-1} \cdots dt_0}{(a + t_0 + \cdots + t_{m-1})^{s+m}} > 0, \quad (9.8)$$

where $(s)_m = s(s+1) \cdots (s+m-1)$. For $m \geq 1$,

$$\sum_{n < 2^m} \varepsilon_n \log(a+n) < 0, \quad 0 < \prod_{n < 2^m} (a+n)^{\varepsilon_n} < 1. \quad (9.9)$$

Proof. Use $f(y) = (a+y)^{-s}$, for which $f^{(m)}(y) = (-1)^m (s)_m (a+y)^{-s-m}$. For the logarithm, $f^{(m)}(y) = (-1)^{m-1} (m-1)! / (a+y)^m$, so the integral is strictly negative. Exponentiation gives the product inequality. $\square$

9.3 Arbitrary translation weights

The dyadic steps $h, 2h, 4h, \dots$ give the identities above their Thue–Morse reading, but the cancellation mechanism does not use them. The formalization isolates the general statement, which is worth recording: it subsumes the ordinary and sparse Prouhet identities simultaneously and explains exactly where the powers of two matter (only in the value of the sharp moment, never in the vanishing).

Theorem 9.4 (Arbitrary-weight Prouhet identities). *Let R be a commutative ring, S a finite index set, $w: S \rightarrow R$ arbitrary weights, and $x \in R$. For every polynomial $p \in R[y]$ with $\deg p < |S|$,*

$$\sum_{T \subseteq S} (-1)^{|T|} p\left(x + \sum_{j \in T} w_j\right) = 0, \quad (9.10)$$

and at the first surviving degree,

$$\sum_{T \subseteq S} (-1)^{|T|} \left(x + \sum_{j \in T} w_j\right)^{|S|} = (-1)^{|S|} |S|! \prod_{j \in S} w_j. \quad (9.11)$$

Proof. For (9.10), induct on S . If $S = \emptyset$ there is nothing of degree below zero. For $S \cup \{a\}$ with $a \notin S$, split the subsets by membership of a and pair T with $T \cup \{a\}$:

$$\sum_{T \subseteq S \cup \{a\}} (-1)^{|T|} p(x + w_T) = \sum_{T \subseteq S} (-1)^{|T|} q(x + w_T), \quad q(y) := p(y) - p(y + w_a),$$

where $w_T := \sum_{j \in T} w_j$. The translation $p(y + w_a)$ has the same degree and the same leading coefficient as p , so $\deg q < \deg p$ (and $q = 0$ when p is constant). The induction hypothesis applied to q finishes.

For (9.11), induct on S again with $p(y) = y^{s+1}$, $s = |S|$. The same pairing produces $q(y) = y^{s+1} - (y + w_a)^{s+1} = -\sum_{k \leq s} \binom{s+1}{k} w_a^{s+1-k} y^k$ by the binomial theorem, the top term cancelling. Summing over $T \subseteq S$ and exchanging the two finite sums, every row $k < s$ vanishes by (9.10), and the row $k = s$ contributes $-\binom{s+1}{s} w_a \cdot (-1)^s s! \prod_{j \in S} w_j$ by the induction hypothesis. Since $\binom{s+1}{s} = s+1$, the total is $(-1)^{s+1} (s+1)! w_a \prod_{j \in S} w_j$, as claimed. $\square$

Formal versions. `Fabius.sum_powerset_neg_one_pow_eval` and `Fabius.sum_powerset_neg_one_pow_pow_card`, in `ThueMorseSparseProuhet.lean`; the degree drop is `Mathlib's Polynomial.taylor` bookkeeping. Specializing $w_j = 2^j h$ over $S = \{0, \dots, m-1\}$ and transporting along the Boolean-cube kernel yields the Thue–Morse forms over any commutative ring: `Fabius.sum_thueMorseSign_mul_affine_pow_eq_zero` and `Fabius.sum_thueMorseSign_mul_affine_pow_card`, the latter with the familiar constant $\prod_{j < m} 2^j h = 2^{\binom{m}{2}} h^m$.

9.4 Prouhet cancellation and the first surviving moment

Theorem 9.5 (Prouhet polynomial annihilation). *If p is a polynomial of degree less than m , then*

$$\sum_{n=0}^{2^m-1} \varepsilon_n p(x + nh) = 0. \quad (9.12)$$

If $p(y) = cy^m + \text{lower-degree terms}$, then

$$\sum_{n=0}^{2^m-1} \varepsilon_n p(x + nh) = (-1)^m m! 2^{\binom{m}{2}} h^m c. \quad (9.13)$$

Proof. Every m -fold finite difference annihilates a polynomial of degree below m . At degree exactly m , its m -th derivative is the constant $m!c$, so (9.5) gives the sharp value. $\square$

For $h = 1$, let

$$A_m = \{0 \leq n < 2^m : \tau(n) = 0\}, \quad B_m = \{0 \leq n < 2^m : \tau(n) = 1\}. \quad (9.14)$$

Then

$$\sum_{n \in A_m} n^r = \sum_{n \in B_m} n^r, \quad 0 \leq r < m, \quad (9.15)$$

which is Prouhet’s classical equal-sums-of-like-powers construction. At $r = m$, the difference is nonzero, so the cancellation order is exact.

Remark 9.6 (The Prouhet–Tarry–Escott problem). Equation (9.15) is the historical origin of the *Prouhet–Tarry–Escott problem*: given a degree d , find two disjoint multisets $A \neq B$ of integers of equal size with $\sum_{a \in A} a^r = \sum_{b \in B} b^r$ for all $0 \leq r \leq d$. A solution is called *ideal* when its size is $d + 1$, the smallest conceivable. Prouhet’s partition (9.14) solves degree $d = m - 1$ with the much larger size 2^{m-1} , but it does so uniformly for every degree – still the only completely elementary construction that scales [2, 17, 18]. Ideal solutions are known for all $d \leq 9$ and for $d = 11$; their existence for every degree is open. The sparse Prouhet theorem of section 10 produces further nonconsecutive solutions: for any n with $\text{wt}_2(n) = s$, splitting the submasks of n by sign parity gives two sets of size 2^{s-1} with equal power sums through degree $s - 1$, supported on the odd positions of Pascal row n rather than on an interval.

9.5 The finite exponential transform

Define

$$M_{m,r} = \sum_{n=0}^{2^m-1} \varepsilon_n n^r. \quad (9.16)$$

Substitute $z = e^t$ in (6.3):

$$\sum_{r \geq 0} M_{m,r} \frac{t^r}{r!} = \sum_{n < 2^m} \varepsilon_n e^{nt} = \prod_{j=0}^{m-1} (1 - e^{2^j t}). \quad (9.17)$$

The order- m zero at $t = 0$ immediately gives

$$M_{m,r} = 0 \quad (r < m), \quad M_{m,m} = (-1)^m m! 2^{\binom{m}{2}}. \quad (9.18)$$

Formal versions. The vanishing and sharp value (9.18) have long been formal – for an *arbitrary* rational translation – as `Fabius.thueMorseTranslatedPowerSum_eq_zero_of_1t` and `Fabius.thueMorseTranslatedPowerSum_self` (`ThueMorseExponential.lean`); the raw moments $M_{m,r}$ are the translation $c = 2^m$ of that interface, `Fabius.thueMorseTranslatedPowerSum_two_pow` (`ThueMorseMoments.lean`). The transform (9.17) itself is formal at the coefficient level: the dictionary $[t^r] \sum_{n < 2^m} \varepsilon_n e^{nt} = M_{m,r}/r!$ is `Fabius.coeff_sum_thueMorseSign_exp_pow`, and the right-hand product is the algebraic master product evaluated at $z = e^t \in \mathbb{Q}[[t]]$ – no convergence argument is used anywhere.

The normalized translated series itself has the exact finite product

$$\begin{aligned} \mathcal{S}_{c,k}(X) &:= \sum_{n \geq 0} \frac{\sum_{r < 2^k} \varepsilon_r (r - 2^k + c)^{n+k}}{(n+k)!} X^n, \\ \mathcal{S}_{c,k}(X) &= (-1)^k 2^{\binom{k}{2}} e^{(c-1)X} \prod_{j < k} \frac{e^{-2^j X} - 1}{-2^j X} \quad (c \in \mathbb{Q}, k \in \mathbb{N}). \end{aligned} \quad (9.19)$$

This is an identity in $\mathbb{Q}[[X]]$, where each displayed quotient denotes the removable formal series with constant coefficient one. It is `Fabius.thueMorseTranslatedShiftedPowerSeries_eq_exp1_product`; the quotient series is `Fabius.rationalExp1DivSeries`, both in `ThueMorseExponential.lean`.

9.6 All power moments: a finite composition formula

Theorem 9.7 (Complete moment composition). *For $m \geq 1$ and $r \geq m$,*

$$M_{m,r} = (-1)^m r! 2^{\binom{m}{2}} \sum_{\substack{q_0, \dots, q_{m-1} \geq 0 \\ q_0 + \dots + q_{m-1} = r-m}} \prod_{j=0}^{m-1} \frac{2^{jq_j}}{(q_j + 1)!}. \quad (9.20)$$

Equivalently,

$$M_{m,r} = (-1)^m r! \sum_{\substack{p_0, \dots, p_{m-1} \geq 1 \\ p_0 + \dots + p_{m-1} = r}} \frac{2^{\sum_{j=0}^{m-1} jp_j}}{p_0! \cdots p_{m-1}!}. \quad (9.21)$$

Proof. Factor the lowest-order term from each factor in (9.17):

$$\begin{aligned} \prod_{j=0}^{m-1} (1 - e^{2^j t}) &= (-1)^m 2^{\binom{m}{2}} t^m \prod_{j=0}^{m-1} \frac{e^{2^j t} - 1}{2^j t} \\ &= (-1)^m 2^{\binom{m}{2}} t^m \prod_{j=0}^{m-1} \sum_{q \geq 0} \frac{2^{jq} t^q}{(q+1)!}. \end{aligned}$$

Extract t^r and multiply by $r!$. Set $p_j = q_j + 1$ for the second form. $\square$

Formal versions. `ThueMorseMoments.lean` proves [theorem 9.7](#) as `Fabius.sum_thueMorseSign_mul_pow_add`: for all $d \geq 0$,

$$\sum_{n < 2^m} \varepsilon_n n^{m+d} = (-1)^m (m+d)! \sum_{\substack{q_0, \dots, q_{m-1} \geq 0 \\ q_0 + \dots + q_{m-1} = d}} \prod_{j=0}^{m-1} \frac{(2^j)^{q_j+1}}{(q_j + 1)!},$$

with the compositions realized as finitely supported functions (`Finset.finsuppAntidiag`). Factoring $(2^j)^{q_j+1} = 2^j \cdot 2^{jq_j}$ across the product recovers the boxed constant $2^{\binom{m}{2}}$ of (9.20), and $p_j = q_j + 1$ gives (9.21). The substitution $z = e^t$ is performed once and for all on an *arbitrary* support of bit positions — each factor becomes $-t$ times an explicit entire series (`Fabius.one_sub_exp_pow`, specialized in `Fabius.one_sub_exp_two_pow`, and collected in `Fabius.prod_one_sub_exp_two_pow`) — and the $t^{|S|+d}$ coefficient is extracted through `PowerSeries.coeff_prod` inside `Fabius.sum_powerset_thueMorseSign_mul_pow_add`. The block statement is then its case $S = \{0, \dots, m-1\}$, transported across the Boolean-cube reindexing kernel `Fabius.sum_powerset_two_pow`.

9.7 Bell-polynomial compression and cumulants

Let $B_{2\ell}$ be the Bernoulli numbers with $B_2 = 1/6$, and define

$$\kappa_1^{(m)} = \frac{2^m - 1}{2}, \quad (9.22)$$

$$\kappa_{2\ell}^{(m)} = \frac{B_{2\ell}}{2\ell} \frac{2^{2\ell m} - 1}{2^{2\ell} - 1} \quad (\ell \geq 1), \quad (9.23)$$

$$\kappa_{2\ell+1}^{(m)} = 0 \quad (\ell \geq 1). \quad (9.24)$$

Theorem 9.8 (Bell-polynomial moments). *For $m \geq 1$ and $r \geq 0$,*

$$M_{m,m+r} = (-1)^m 2^{\binom{m}{2}} \frac{(m+r)!}{r!} \mathcal{B}_r \left(\kappa_1^{(m)}, \dots, \kappa_r^{(m)} \right). \quad (9.25)$$

Proof. The classical formal expansion

$$\log \left(\frac{e^y - 1}{y} \right) = \frac{y}{2} + \sum_{\ell \geq 1} \frac{B_{2\ell}}{2\ell(2\ell)!} y^{2\ell} \quad (9.26)$$

turns the normalized product into

$$\prod_{j=0}^{m-1} (1 - e^{2^j t}) = (-1)^m 2^{\binom{m}{2}} t^m \exp \left(\sum_{r \geq 1} \kappa_r^{(m)} \frac{t^r}{r!} \right). \quad (9.27)$$

Use the defining generating function (7.6) and compare coefficients. $\square$

The first three nonzero layers are

$$M_{m,m} = (-1)^m 2^{\binom{m}{2}} m!, \quad (9.28)$$

$$M_{m,m+1} = (-1)^m 2^{\binom{m}{2}} (m+1)! \frac{2^m - 1}{2}, \quad (9.29)$$

$$M_{m,m+2} = (-1)^m 2^{\binom{m}{2}} (m+2)! \left(\frac{(2^m - 1)^2}{8} + \frac{4^m - 1}{72} \right). \quad (9.30)$$

9.8 Centered moments and a parity selection rule

Let

$$c_m = \frac{2^m - 1}{2}. \quad (9.31)$$

The centered exponential transform is especially symmetric.

Theorem 9.9 (Centered sinh product). *For every $m \geq 0$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n e^{(n-c_m)t} = (-2)^m \prod_{j=0}^{m-1} \sinh(2^{j-1}t). \quad (9.32)$$

Consequently

$$\sum_{n < 2^m} \varepsilon_n (n - c_m)^r = 0 \quad (9.33)$$

whenever $r < m$ or $r \not\equiv m \pmod{2}$. At $r = m$,

$$\sum_{n < 2^m} \varepsilon_n (n - c_m)^m = (-1)^m m! 2^{\binom{m}{2}}. \quad (9.34)$$

Proof. Use $1 - e^u = -2e^{u/2} \sinh(u/2)$ in (9.17). The product of exponential factors is $e^{c_m t}$, which disappears after centering. The right side is a product of m odd functions, hence has parity m , and each factor starts with a linear term. Coefficient comparison proves the selection rule and the first coefficient. $\square$

Remark 9.10 (Reflection proof of the selection rule). The parity cancellation (9.33) needs no exponential transform. Reindex $n \mapsto 2^m - 1 - n$ in $S_r = \sum_{n < 2^m} \varepsilon_n (n - c_m)^r$: the reflection formula (2.8) gives $\varepsilon_{2^m-1-n} = (-1)^m \varepsilon_n$, while $(2^m - 1 - n) - c_m = -(n - c_m)$. Hence $S_r = (-1)^{m+r} S_r$, and $S_r = 0$ whenever $m + r$ is odd. The same reindexing at an *arbitrary* translation x yields the functional equation

$$\sum_{n < 2^m} \varepsilon_n (x + n)^r = (-1)^{m+r} \sum_{n < 2^m} \varepsilon_n (n - x - (2^m - 1))^r,$$

valid over any commutative ring; the selection rule is its fixed point at the self-reflective translation $x = -c_m$.

Formal versions. The reflection functional equation is `Fabius.sum_thueMorseSign_mul_add_pow_reflect` (`ThueMorseMoments.lean`, any commutative ring); the selection rule (9.33) for odd $m + r$ is `Fabius.sum_thueMorseSign_mul_midpoint_pow_eq_zero`, proved by the reflection argument of [remark 9.10](#) rather than through (9.32); the complementary case $r < m$ of either parity is `Fabius.thueMorseTranslatedPowerSum_eq_zero_of_lt`. The first coefficient (9.34) is `Fabius.sum_thueMorseSign_mul_midpoint_pow_self`.

This extra parity cancellation is invisible in the uncentered Prouhet statement. It is a consequence of combining dyadic reflection with moment cancellation.

9.9 All binomial-basis moments

Define

$$B_{m,r} = \sum_{n=0}^{2^m-1} \varepsilon_n \binom{n}{r}. \quad (9.35)$$

Theorem 9.11 (Binomial moment product). *For $\ell \geq 0$,*

$$B_{m,m+\ell} = (-1)^m 2^{\binom{m}{2}} [u^\ell] \prod_{j=0}^{m-1} \frac{(1+u)^{2^j} - 1}{2^j u}. \quad (9.36)$$

For $r < m$, $B_{m,r} = 0$; in particular,

$$B_{m,m} = (-1)^m 2^{\binom{m}{2}}. \quad (9.37)$$

Proof. Expand $P_m(1+u)$ in two ways:

$$P_m(1+u) = \sum_{n < 2^m} \varepsilon_n (1+u)^n = \sum_{r \geq 0} B_{m,r} u^r,$$

while

$$P_m(1+u) = (-1)^m 2^{\binom{m}{2}} u^m \prod_{j=0}^{m-1} \frac{(1+u)^{2^j} - 1}{2^j u}.$$

Compare coefficients. □

Corollary 9.12 (Integer composition form). *For all $m, \ell \geq 0$,*

$$B_{m,m+\ell} = (-1)^m \sum_{\substack{q_0, \dots, q_{m-1} \geq 0 \\ q_0 + \dots + q_{m-1} = \ell}} \prod_{j=0}^{m-1} \binom{2^j}{q_j + 1}. \quad (9.38)$$

Proof. In (9.36) each normalized factor is, up to its constant 2^j , the geometric sum $((1+u)^{2^j} - 1)/u = \sum_{i < 2^j} (1+u)^i$, whose q -th coefficient is $\sum_{i < 2^j} \binom{i}{q} = \binom{2^j}{q+1}$ by the hockey-stick identity. Multiplying the m factors and collecting the coefficient of u^ℓ gives (9.38). The single composition at $\ell = 0$ recovers (9.37), since $\binom{2^j}{1} = 2^j$ multiplies to $2^{\binom{m}{2}}$. Unlike the power-moment composition (9.20), every summand here is an integer. $\square$

Formal versions. `ThueMorseBlockAlgebra.lean` proves the substitution identity behind (9.36) as `Fabius.sum_thueMorseSign_mul_one_add_X_pow` (over any commutative ring), the vanishing $B_{m,r} = 0$ for $r < m$ as `Fabius.sum_thueMorseSign_mul_choose_eq_zero`, and the sharp value (9.37) as `Fabius.sum_thueMorseSign_mul_choose_self`, each with an integer specialization. The complete composition form (9.38) is `Fabius.sum_thueMorseSign_mul_choose_add` (`ThueMorseMoments.lean`), with the hockey-stick column sum $\sum_{i < N} \binom{i}{q} = \binom{N}{q+1}$ isolated as the reusable `Fabius.sum_range_choose_eq_choose_succ`.

The binomial basis is often preferable in discrete formalization because finite differences obey the simple lowering rule $\Delta \binom{x}{r} = \binom{x}{r-1}$.

10 Sparse Prouhet identities on arbitrary Pascal supports

The ordinary Prouhet block $[0, 2^m)$ is the submask lattice of $2^m - 1$. Lucas’s theorem suggests replacing this all-ones mask by an arbitrary n .

Define

$$J(n) = \{j \geq 0 : b_j(n) = 1\}, \quad s = |J(n)| = \text{wt}_2(n), \quad \beta(n) = \sum_{j \in J(n)} j. \quad (10.1)$$

Theorem 10.1 (Signed Pascal-support product). *For every $n \geq 0$,*

$$R_n(z) := \sum_{k \preceq_2 n} \varepsilon_k z^k = \prod_{j \in J(n)} (1 - z^{2^j}). \quad (10.2)$$

Equivalently,

$$R_n(z) = \sum_{k=0}^n \varepsilon_k \mathbf{1}_{\binom{n}{k} \text{ odd}} z^k. \quad (10.3)$$

Proof. Expanding the product chooses a subset of the occupied bit positions of n . Its exponent is the corresponding submask k , and its sign is $(-1)^{\text{wt}_2(k)} = \varepsilon_k$. Lucas’s criterion gives the second form. $\square$

Corollary 10.2 (Balance on odd Pascal positions). *For every $n > 0$,*

$$\sum_{\substack{0 \leq k \leq n \\ \binom{n}{k} \text{ odd}}} \varepsilon_k = 0. \quad (10.4)$$

Thus exactly half of the odd positions in every nonzero Pascal row have even binary weight and half have odd binary weight.

Proof. Set $z = 1$ in (10.2). There are $2^{\text{wt}_2(n)}$ support positions by corollary 5.2. $\square$

Theorem 10.3 (Sparse Prouhet theorem). *Let $n > 0$, $s = \text{wt}_2(n)$. For every polynomial p of degree below s ,*

$$\sum_{k \preceq_2 n} \varepsilon_k p(k) = 0. \quad (10.5)$$

The first surviving power moment is

$$\sum_{k \preceq 2n} \varepsilon_k k^s = (-1)^s s! 2^{\beta(n)}. \quad (10.6)$$

Proof. Every factor in (10.2) has a simple zero at $z = 1$, so R_n has a zero of exact order s , with local leading term

$$R_n(z) = (-1)^s 2^{\beta(n)} (z - 1)^s + \mathcal{O}((z - 1)^{s+1}).$$

The derivative/falling-factorial proof of [theorem 9.5](#) applies with the sum restricted to submasks. $\square$

Formal versions. In `ThueMorseSparseProuhet.lean` the submasks of n are parametrized by the subsets of the bit support `Fabius.bitSupport` (with dictionary `Fabius.mem_bitSupport`, `Fabius.sum_two_pow_bitSupport`, `Fabius.card_bitSupport`); the sparse cancellation and the sharp moment are `Fabius.sum_submask_neg_one_pow_eval` and `Fabius.sum_submask_neg_one_pow_pow`, both instances of the arbitrary-weight engine of [theorem 9.4](#) with $w_j = 2^j h$ running over $J(n)$, over any commutative ring. The generating identity (10.2) itself is now `Fabius.prod_one_sub_pow_bitSupport` (`ThueMorseLucasSupport.lean`), the submasks running over the powerset of the bit support — faithfully, by `Fabius.sum_two_pow_injOn_powerset`; through Lucas’s criterion this index set is exactly `Fabius.oddBinomialIndices`, and the balance [corollary 10.2](#) is `Fabius.sum_thueMorseSign_oddBinomialIndices`.

Corollary 10.4 (Sparse operator identity). *For every function f ,*

$$\sum_{k \preceq 2n} \varepsilon_k f(x + hk) = \prod_{j \in J(n)} (I - T_{2^j h}) f(x). \quad (10.7)$$

If $f \in C^s$ and $h \geq 0$, its absolute value is at most

$$2^{\beta(n)} h^s \sup_{x \leq y \leq x + nh} |f^{(s)}(y)|. \quad (10.8)$$

Proof. Expand the product over the set-bit positions and apply the fundamental theorem of calculus once for each step. $\square$

The whole sparse moment sequence also has a direct finite formula.

Corollary 10.5 (All sparse moments). *For $r \geq s$,*

$$\sum_{k \preceq 2n} \varepsilon_k k^r = (-1)^s r! 2^{\beta(n)} \sum_{\substack{(q_j)_{j \in J(n)} \geq 0 \\ \sum_{j \in J(n)} q_j = r - s}} \prod_{j \in J(n)} \frac{2^{jq_j}}{(q_j + 1)!}. \quad (10.9)$$

Proof. Substitute $z = e^t$ in (10.2), factor $-2^j t$ from every occupied bit, and extract coefficients exactly as in [theorem 9.7](#). $\square$

Formal versions. `ThueMorseSparseMoments.lean` proves [corollary 10.5](#) as `Fabius.sum_submask_thueMorseSign_mul_pow_add`, and that is now all it contains: the corollary is the special case $S = J(n)$ of a statement on an arbitrary finite set S of bit positions (`Fabius.sum_powerset_t_thueMorseSign_mul_pow_add`), which also subsumes the dyadic block $S = \{0, \dots, m - 1\}$ of [theorem 9.7](#). That arbitrary-support statement and its inputs live one level down, in `ThueMorseMoments.lean` — an ancestor module cannot import a descendant, and [theorem 9.7](#) is itself derived from it: the master product on an arbitrary support is `Fabius.prod_one_sub_pow_powerset`, over any commutative ring, resting on the hypothesis-free weight law `Fabius.binaryWeight_sum_two_pow_eq_card` and its sign companion `Fabius.thueMorseSign_sum_two_pow`.

Example 10.6. For $n = 13 = (1101)_2$, $J(13) = \{0, 2, 3\}$, $s = 3$, and $\beta(13) = 5$. The odd positions in row 13 are

$$0, 1, 4, 5, 8, 9, 12, 13.$$

Their signed sums against $1, k, k^2$ vanish, while

$$\sum_{k \preccurlyeq_2 13} \varepsilon_k k^3 = (-1)^3 3! 2^5 = -192.$$

This is a Prouhet partition on a sparse, Pascal-determined grid rather than a consecutive interval.

11 Exact prefix sums, iterated antiderivatives, and the Fabius bridge

11.1 The first prefix discrepancy

Define the exclusive signed prefix and the number of one-bits in the sequence prefix by

$$S(N) = \sum_{n=0}^{N-1} \varepsilon_n, \quad A(N) = \sum_{n=0}^{N-1} \tau(n). \quad (11.1)$$

Theorem 11.1 (Exact prefix formula). *For every $q \geq 0$,*

$$\boxed{S(2q) = 0, \quad S(2q+1) = \varepsilon_q.} \quad (11.2)$$

Equivalently,

$$S(N) = \frac{1 - (-1)^N}{2} \varepsilon_{\lfloor N/2 \rfloor}. \quad (11.3)$$

Therefore

$$\boxed{A(N) = \frac{N - S(N)}{2}} \quad (11.4)$$

and explicitly

$$A(2q) = q, \quad A(2q+1) = q + \tau(q). \quad (11.5)$$

Proof. Every adjacent pair cancels: $\varepsilon_{2j} + \varepsilon_{2j+1} = \varepsilon_j - \varepsilon_j = 0$. An odd-length prefix leaves the unpaired term $\varepsilon_{2q} = \varepsilon_q$. The count formula follows from $\varepsilon_n = 1 - 2\tau(n)$. $\square$

The inclusive sum through N is

$$\sum_{n=0}^N \varepsilon_n = \frac{1 + (-1)^N}{2} \varepsilon_{\lfloor N/2 \rfloor}. \quad (11.6)$$

Corollary 11.2 (Interval discrepancy). *For $0 \leq a \leq b$,*

$$\sum_{n=a}^{b-1} \varepsilon_n = S(b) - S(a), \quad (11.7)$$

$$\left| \sum_{n=a}^{b-1} \varepsilon_n \right| \leq 2, \quad (11.8)$$

$$\left| \sum_{n=a}^{b-1} \tau(n) - \frac{b-a}{2} \right| \leq 1. \quad (11.9)$$

Both constants are attained.

Proof. The interval sum is a difference of two values in $\{-1, 0, 1\}$. Convert to zero–one values by (1.3). For sharpness, $\varepsilon_5 + \varepsilon_6 = 2$. $\square$

Formal versions. `ThueMorseEulerTransform.lean` proves the exclusive closed form (11.2) as `Fabius.sum_thueMorseSign_range` (even case `Fabius.sum_thueMorseSign_range_two_mul`), the inclusive form (11.6) as `Fabius.sum_thueMorseSign_range_succ`, the ones-count formulas (11.5) as `Fabius.sum_thueMorseBit_range_two_mul` and `Fabius.sum_thueMorseBit_range_two_mul_add_one`, and the bridge $2A(N) + S(N) = N$ as `Fabius.two_mul_sum_thueMorseBit_add_sum_thueMorseSign`. The three displayed statements of corollary 11.2 are `Fabius.sum_thueMorseSign_Ico` for (11.7), `Fabius.abs_sum_thueMorseSign_Ico_le_two` for (11.8) – both resting on the prefix bound $|S(N)| \leq 1$ (`Fabius.abs_sum_thueMorseSign_range_le_one`) – and, for the bits in doubled integer form, `Fabius.abs_two_mul_sum_thueMorseBit_Ico_sub_le_two` for (11.9). All four are inequalities or exact identities; the attainment of the two constants, checked in the proof above at $a = 5, b = 7$, is not formalized.

Corollary 11.3 (Aligned and partial dyadic blocks). *For $m \geq 1$ and $q \geq 0$,*

$$\sum_{r=0}^{2^m-1} \varepsilon_{2^m q+r} = 0. \quad (11.10)$$

More generally, for $0 \leq R \leq 2^m$,

$$\sum_{r=0}^{R-1} \varepsilon_{2^m q+r} = \varepsilon_q S(R). \quad (11.11)$$

Proof. Use (2.6) and factor out ε_q ; then apply theorem 11.1. $\square$

Formal versions. (11.11) and (11.10) are `Fabius.sum_thueMorseSign_block_prefix` and `Fabius.sum_thueMorseSign_aligned_block` (`ThueMorseEulerTransform.lean`).

The prefix generating function is unusually simple:

$$\sum_{N \geq 0} S(N) z^N = \frac{z}{1-z} E(z) = z E(z^2) = z \prod_{j \geq 1} (1 - z^{2^j}). \quad (11.12)$$

The middle equality is the Mahler equation.

Formal versions. The first and last members of (11.12) are formal: `Fabius.thueMorsePrefixSeries_eq` (`ThueMorseMahler.lean`) proves $\sum_N S(N) z^N = z E(z^2)$ directly from the exact prefix collapse, which under the dissection is ‘ $z \prod_{j \geq 1} (1 - z^{2^j})$ ’ at every truncation level.

11.2 Iterated inclusive prefixes

Define discrete antiderivatives by

$$S_n^{(0)} = \varepsilon_n, \quad S_n^{(k+1)} = \sum_{j=0}^n S_j^{(k)}. \quad (11.13)$$

Then (11.6) says

$$S_{2q}^{(1)} = \varepsilon_q, \quad S_{2q+1}^{(1)} = 0. \quad (11.14)$$

Theorem 11.4 (Binomial convolution for discrete antiderivatives). *For $k \geq 1$,*

$$S_n^{(k)} = \sum_{j=0}^n \binom{n-j+k-1}{k-1} \varepsilon_j. \quad (11.15)$$

Its ordinary generating function is

$$\sum_{n \geq 0} S_n^{(k)} z^n = \frac{E(z)}{(1-z)^k}. \quad (11.16)$$

Proof. One inclusive summation multiplies an ordinary generating function by $(1-z)^{-1}$. Iterate k times, then expand $(1-z)^{-k} = \sum_{r \geq 0} \binom{r+k-1}{k-1} z^r$ and take the Cauchy product. $\square$

Formal versions. The corpus module `ThueMorsePrefix.lean` defines `Fabius.iteratedPrefix` and proves the binomial convolution (11.15) as `Fabius.iteratedPrefix_convolution`; the generating identity (11.16) is formal below every dyadic truncation as `Fabius.coeff_thueMorseBlockPolynomial_mul_invOneSubPow_eq_iteratedPrefix` (`ThueMorseGenerating.lean`).

11.3 Terminal zero runs at dyadic boundaries

The quotient of a finite Prouhet polynomial by $(1-z)^k$ is still a polynomial when $k \leq r$:

$$\frac{P_r(z)}{(1-z)^k} = (1-z)^{r-k} \prod_{j=0}^{r-1} (1+z+\cdots+z^{2^j-1}). \quad (11.17)$$

Its degree is $2^r - k - 1$.

Theorem 11.5 (Dyadic terminal zero run). *If $1 \leq k \leq r$, then*

$$S_{2^r-k+i}^{(k)} = 0, \quad 0 \leq i < k. \quad (11.18)$$

The adjacent values are

$$S_{2^r-k-1}^{(k)} = (-1)^{r-k}, \quad S_{2^r}^{(k)} = -1. \quad (11.19)$$

Proof. Below degree 2^r , the coefficients of $E(z)/(1-z)^k$ agree with those of $P_r(z)/(1-z)^k$. The latter has degree $2^r - k - 1$, proving the zero run. Its leading coefficient is $(-1)^{r-k}$, giving the preceding value. The coefficient at 2^r comes from the first omitted factor $1 - z^{2^r}$, or directly from the prefix recurrence. $\square$

Formal versions. All three statements of [theorem 11.5](#) have long been formal in `ThueMorsePrefix.lean`: `Fabius.iteratedPrefix_dyadic_zero_run` (the run), `Fabius.iteratedPrefix_before_dyadic_run` ($S_{2^r-k-1}^{(k)} = (-1)^{r-k}$), and `Fabius.iteratedPrefix_at_dyadic` ($S_{2^r}^{(k)} = -1$), with the endpoint value `Fabius.iteratedPrefix_dyadic_endpoint`. The factorization behind (11.17) is the geometric ladder `Fabius.prod_one_sub_two_pow_eq_geom` and `Fabius.thueMorseBlockPolynomial_eq_geom_prod` (`ThueMorseBlockAlgebra.lean`), over any commutative ring.

11.4 Why this is the discrete spine of the Fabius theory

The chain connecting Thue–Morse signs to the Fabius and Rvachev functions is now algebraically visible:

$$\begin{aligned} \text{binary parity character} &\implies P_r(z) = \prod_{j < r} (1 - z^{2^j}) \\ &\implies \frac{P_r(z)}{(1-z)^k} \implies \text{iterated-prefix coefficient grids} \end{aligned} \quad (11.20)$$

The zero of order k at $z = 1$ creates the exact terminal flatness in [theorem 11.5](#); after the normalizations developed in the main repository exposition, the resulting coefficient grids and polygonal interpolants converge to the Fabius function. On the Fourier side, replacing z by a point on the unit circle turns the same finite product into a finite product of sine factors, whose scaled limit is the sinc product associated with the Rvachev up-function.

Thus the discrete-prefix and analytic-sinc appearances of Thue–Morse are not separate phenomena. They are the coefficient and Fourier-evaluation faces of the single factorization (6.3). The quantitative convergence theory belongs to the dedicated companion reports and Lean modules; the purpose here is to expose the exact common algebraic spine.

12 Ordinary Fourier transforms and product-to-sum identities

12.1 A finite Fourier product

Putting $z = e^{i\theta}$ in (6.3) gives

$$\sum_{n=0}^{2^m-1} \varepsilon_n e^{in\theta} = \prod_{j=0}^{m-1} (1 - e^{i2^j\theta}). \quad (12.1)$$

Using $1 - e^{iu} = -2ie^{iu/2} \sin(u/2)$,

$$\sum_{n=0}^{2^m-1} \varepsilon_n e^{in\theta} = (-2i)^m e^{i(2^m-1)\theta/2} \prod_{j=0}^{m-1} \sin(2^{j-1}\theta). \quad (12.2)$$

Hence

$$\left| \sum_{n=0}^{2^m-1} \varepsilon_n e^{in\theta} \right| = 2^m \prod_{j=0}^{m-1} |\sin(2^{j-1}\theta)|. \quad (12.3)$$

Formal versions. (12.1) is the master product `Fabius.prod_one_sub_pow_eq_sum_thueMorseSign` evaluated at $z = e^{i\theta} \in \mathbb{C}$. The sine form (12.2) is `Fabius.sum_thueMorseSign_exp_eq_sin_prod` (`ThueMorseSineProduct.lean`), built on the half-angle factorization $1 - e^{iu} = -2ie^{iu/2} \sin(u/2)$ (`Fabius.one_sub_exp_mul_I`, proved by pure exponential algebra). The magnitude (12.3) is carried in squared form by the Riesz factorization `Fabius.normSq_sum_thueMorseSign_exp` (`ThueMorseFourier.lean`), whose factors are $|1 - e^{i\theta}|^2 = 2 - 2\cos\theta$ (`Fabius.normSq_one_sub_exp`).

12.2 Sine and cosine products as Thue–Morse sums

A rescaling of (12.2) gives exact finite product-to-sum identities.

Theorem 12.1 (Thue–Morse trigonometric sums). *For every integer $m \geq 0$ and every complex z ,*

$$\prod_{r=0}^m \sin \frac{z}{2^r} = \frac{1}{2^m} \sum_{n=0}^{2^m-1} \varepsilon_n \sin \left(\frac{\pi m}{2} + \frac{(2n+1)z}{2^m} \right). \quad (12.4)$$

Also

$$\cos z \prod_{r=1}^m \sin \frac{z}{2^r} = \frac{1}{2^m} \sum_{n=0}^{2^m-1} \varepsilon_n \cos \left(\frac{\pi m}{2} + \frac{(2n+1)z}{2^m} \right). \quad (12.5)$$

The product from $r = 1$ to 0 is empty when $m = 0$.

Proof. Set $\theta = 2z/2^m$ in (12.2), multiply by $i^m e^{iz/2^m}$, and simplify the phase. The result is

$$\sum_{n < 2^m} \varepsilon_n \exp\left(i \left(\frac{\pi m}{2} + \frac{(2n+1)z}{2^m} \right)\right) = 2^m e^{iz} \prod_{r=1}^m \sin \frac{z}{2^r}.$$

For real z , take imaginary and real parts. Both resulting identities are identities between entire functions of z , so the identity theorem extends them to every complex z . $\square$

These identities make explicit the Fourier route from finite Thue–Morse blocks to the sinc products in the Fabius/Rvachev analysis.

12.3 The exact dyadic discrete Fourier transform

Let

$$\widehat{\varepsilon}_m(k) = \sum_{n=0}^{2^m-1} \varepsilon_n \exp\left(-\frac{2\pi i k n}{2^m}\right), \quad 0 \leq k < 2^m. \quad (12.6)$$

Theorem 12.2 (Dyadic DFT). *For every k ,*

$$\widehat{\varepsilon}_m(k) = \prod_{j=0}^{m-1} \left(1 - \exp\left(-\frac{2\pi i k 2^j}{2^m}\right) \right). \quad (12.7)$$

If $m \geq 1$, this is zero for every even k . For odd k ,

$$\widehat{\varepsilon}_m(k) = (2i)^m e^{-\pi i k (1-2^{-m})} \prod_{r=1}^m \sin \frac{\pi k}{2^r}. \quad (12.8)$$

Thus

$$|\widehat{\varepsilon}_m(k)| = 2^m \prod_{r=1}^m \left| \sin \frac{\pi k}{2^r} \right| \quad (k \text{ odd}). \quad (12.9)$$

Proof. Evaluate (6.3) at $z = e^{-2\pi i k / 2^m}$. If k is even, the factor with $j = m-1$ is $1 - (-1)^k = 0$. For odd k , apply $1 - e^{-iu} = 2ie^{-iu/2} \sin(u/2)$ factorwise and sum the phases. $\square$

Formal versions. The product form (12.7) is the master product at $z = e^{-2\pi i k / 2^m}$; the closed sine form (12.8) is the specialization of `Fabius.sum_thueMorseSign_exp_eq_sin_prod` at $x = -2\pi k / 2^m$ (the atlas phase convention conjugates it). The vanishing at even frequencies is `Fabius.sum_thueMorseSign_mul_pow_eq_zero_of_pow_eq_one` (`ThueMorseRarefied.lean`). The inversion formula (12.10) is now formal as well: `Fabius.thueMorse_dft_inversion` (`ThueMorseFourierInversion.lean`) specializes a finite Fourier inversion theorem proved over an *arbitrary* field with a primitive root of unity (`Fabius.dft_inversion`, resting on the orthogonality `Fabius.sum_pow_eq_ite`) – a geometric-series fact, no integration.

For $m \geq 1$, Fourier inversion gives a root-of-unity formula for every individual term:

$$\varepsilon_n = \frac{1}{2^m} \sum_{\substack{0 \leq k < 2^m \\ k \text{ odd}}} \widehat{\varepsilon}_m(k) e^{2\pi i k n / 2^m}, \quad 0 \leq n < 2^m. \quad (12.10)$$

This is intentionally extravagant as an evaluator, but exact and natural for spectral calculations.

13 Walsh characters, Rademacher functions, Riesz products, and autocorrelation

13.1 A one-point Walsh spectrum

For m -bit vectors $a, b \in \mathbb{F}_2^m$, write $\langle a, b \rangle = \sum_j a_j b_j \pmod{2}$, and let $\mathbf{1} = (1, \dots, 1)$. For $n < 2^m$,

$$\varepsilon_n = (-1)^{\langle \mathbf{1}, \mathbf{b}(n) \rangle}. \quad (13.1)$$

Theorem 13.1 (Walsh transform is a point mass). *For every $a \in \mathbb{F}_2^m$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n (-1)^{\langle a, \mathbf{b}(n) \rangle} = \begin{cases} 2^m, & a = \mathbf{1}, \\ 0, & a \neq \mathbf{1}. \end{cases} \quad (13.2)$$

Proof. The summand is the character indexed by $a + \mathbf{1}$. A nontrivial character sums to zero over the finite Boolean group, while the trivial character sums to its order 2^m . $\square$

Thus the broad ordinary Fourier spectrum is a basis-change phenomenon: the same block is spectrally pure in the Walsh basis.

Formal versions. `ThueMorseWalsh.lean` proves (13.1) as `Fabius.thueMorseSign_eq_neg_one_pow_land` (the inner product realized as `wt2(a & n)`), the point mass (13.2) as `Fabius.sum_t_hueMorseSign_mul_walsh`, and the underlying character orthogonality $\sum_{n < 2^m} (-1)^{\text{wt}_2(a \& n)} = 2^m \mathbf{1}_{a=0}$ as `Fabius.sum_neg_one_pow_binaryWeight_land`. The bit toolkit behind them — the all-ones digit pattern of $2^m - 1$ and the masking of two-power sums — is `Fabius.sum_range_two_pow`, `Fabius.testBit_two_pow_sub_one`, and `Fabius.land_sum_two_pow`.

13.2 Rademacher step functions

For $x \in [0, 1)$, define the right-continuous Rademacher functions

$$r_j(x) = (-1)^{\lfloor 2^{j+1}x \rfloor}. \quad (13.3)$$

If $0 \leq n < 2^m$ and $x \in [n/2^m, (n+1)/2^m)$, then the first m binary digits of x encode n , so

$$\varepsilon_n = \prod_{j=0}^{m-1} r_j(x). \quad (13.4)$$

The finite Thue–Morse block is therefore the sampling of one Walsh function on the dyadic partition.

13.3 Finite Riesz products

Define

$$\rho_m(x) = 2^{-m} |P_m(e^{2\pi i x})|^2. \quad (13.5)$$

Theorem 13.2 (Riesz-product density). *For every $m \geq 0$,*

$$\rho_m(x) = \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)). \quad (13.6)$$

Moreover,

$$\int_0^1 \rho_m(x) dx = 1, \quad (13.7)$$

so $\rho_m(x) dx$ is a probability measure.

Proof. Use $|1 - e^{i\theta}|^2 = 2(1 - \cos \theta)$ in (6.3). Parseval gives

$$\int_0^1 \rho_m(x) dx = 2^{-m} \sum_{n < 2^m} |\varepsilon_n|^2 = 1.$$

□

Formal versions. The factorization (13.6) is `Fabius.normSq_sum_thueMorseSign_exp` (`ThueMorseFourier.lean`) after normalizing by 2^m and rescaling the variable, with nonnegativity `Fabius.prod_two_sub_two_cos_nonneg`. The Parseval mass (13.7) is formal as `Fabius.integral_thueMorse_double_sum` (`ThueMorseParseval.lean`): the polynomial times its reflected conjugate integrates over a period to 2^m , by the character orthonormality `Fabius.integral_exp_two_pi_int`.

A second, entirely independent route to (13.7) – and to every Fourier coefficient of a lacunary Riesz product below its spectral floor – is formal in `LacunaryRieszIntegral.lean`, where nothing mentions Thue–Morse. For integer frequencies $m_0, m_1, \dots$, arbitrary real amplitudes a_j and phases φ_j , and an integer probe frequency K ,

$$\int_0^1 \cos(2\pi Kt + \psi) \prod_{j < n} (1 + a_j \cos(2\pi m_j t + \varphi_j)) dt = \begin{cases} \cos \psi, & K = 0, \\ 0, & K \neq 0, \end{cases} \quad (13.8)$$

under the hypothesis – exactly the one the peeling induction consumes – that

$$|K| + \sum_{i < j} m_i < m_j \quad \text{for every } j \quad (13.9)$$

(`Fabius.integral_cos_mul_rieszProduct_of_sum_lt`), so that no signed combination $\sum_j \eta_j m_j$ with $\eta_j \in \{-1, 0, 1\}$ can reach $-K$. Two corollaries carry the classical shapes: the K -free hypothesis $m_0 + \sum_{i < j} m_i \leq m_j$ with $|K| < m_0$ (`Fabius.integral_cos_mul_rieszProduct_of_add_sum_le`), and the Hadamard gap $2m_j \leq m_{j+1}$ with $|K| < m_0$ (`Fabius.integral_cos_mul_rieszProduct`), obtained from the previous one through `Fabius.add_sum_le_of_two_mul_le`. At $K = 0$ no headroom below the floor is needed, and plain super-increasingness $\sum_{i < j} m_i < m_j$ already gives the mean value 1 (`Fabius.integral_rieszProduct_of_superIncreasing`; the Hadamard-gap restatement, which still assumes $2m_j \leq m_{j+1}$ and $m_0 > 0$, is `Fabius.integral_rieszProduct`). Consequently (13.7) is the instance $m_j = 2^j$, $a_j = -1$, $\varphi_j = 0$. The companion exact identity $\int_0^1 \prod_{j < n} \sin^2(\pi m_j t) dt = 2^{-n}$ (`Fabius.integral_prod_sin_sq_of_superIncreasing`, with the geometric and dyadic specializations `Fabius.integral_prod_sin_sq_pow` and `Fabius.integral_prod_sin_sq_two_pow`) is what the Fourier-decay audit of the Rvachev up-function uses. The offset m_0 in the super-increasing hypothesis cannot be dropped when $K \neq 0$ – the frequencies 3, 4, 8, 16, $\dots$ with $K = 1$ resonate through $m_1 - m_0 = 1$ – but that counterexample is recorded in prose in the module header, not formalized.

The finite measures converge weakly to the Thue–Morse spectral measure. The formal notation

$$d\mu_{\text{TM}}(x) = \prod_{j \geq 0} (1 - \cos(2\pi 2^j x)) dx \quad (13.10)$$

means this weak limit, not pointwise multiplication of ordinary densities. The limiting measure is purely singular continuous, a deeper theorem of substitution spectral theory [14, 15].

13.4 Finite and limiting autocorrelation

For $m, k \geq 0$, define the dyadic average

$$\eta_m(k) = \frac{1}{2^m} \sum_{n=0}^{2^m-1} \varepsilon_n \varepsilon_{n+k}, \quad (13.11)$$

where the second index is not reduced modulo 2^m .

Proposition 13.3 (Finite autocorrelation recursion). *For $r \geq 0$,*

$$\eta_{m+1}(2r) = \eta_m(r), \quad (13.12)$$

$$\eta_{m+1}(2r+1) = -\frac{\eta_m(r) + \eta_m(r+1)}{2}. \quad (13.13)$$

Proof. Split the defining sum into even and odd n . For an even shift, both halves reduce to $\varepsilon_j \varepsilon_{j+r}$. For an odd shift, they reduce to $-\varepsilon_j \varepsilon_{j+r}$ and $-\varepsilon_j \varepsilon_{j+r+1}$, respectively. $\square$

Formal versions. `ThueMorseAutocorrelation.lean` proves the recursion (13.12)–(13.13) in unnormalized integer form ($A_m(k) = 2^m \eta_m(k)$): `Fabius.thueMorseAutocorrelation_succ_even` and `Fabius.thueMorseAutocorrelation_succ_odd`, with $A_m(0) = 2^m$ (`Fabius.thueMorseAutocorrelation_zero_shift`) and the closed shift values $3A_m(1) = -2^m - 2(-1)^m$ and $3A_{m+1}(2) = -2^{m+1} - 4(-1)^m$ (`Fabius.three_mul_thueMorseAutocorrelation_one`, `Fabius.three_mul_thueMorseAutocorrelation_two`) – the exact finite forms of $\eta(1) = \eta(2) = -\frac{1}{3}$.

The dyadic scaffolding of that recursion is removable. Nothing in the two-scale split uses that the window is a power of two, only that it is *even*, so the module proves the recursion first for the window sums $R_N(k) = \sum_{n < N} \varepsilon_n \varepsilon_{n+k}$ (`Fabius.thueMorseWindowAutocorrelation`) in the form

$$R_{2K}(2r) = 2R_K(r), \quad R_{2K}(2r+1) = -(R_K(r) + R_K(r+1)) \quad (13.14)$$

(`Fabius.thueMorseWindowAutocorrelation_two_mul_even`, `Fabius.thueMorseWindowAutocorrelation_two_mul_odd`), with $R_N(0) = N$ (`Fabius.thueMorseWindowAutocorrelation_zero_shift`) and the trivial bound $|R_N(k)| \leq N$ (`Fabius.abs_thueMorseWindowAutocorrelation_le`); the dyadic layer above is the case $N = 2^m$, read off through the definitional identity `Fabius.thueMorseAutocorrelation_eq_window`, and its bound $|A_m(k)| \leq 2^m$ (`Fabius.abs_thueMorseAutocorrelation_le`) puts every normalized value in $[-1, 1]$. The normalized limits at the three shifts with closed forms are formal as well – $\eta_m(0) \rightarrow 1$, $\eta_m(1) \rightarrow -\frac{1}{3}$, $\eta_m(2) \rightarrow -\frac{1}{3}$ (`Fabius.tendsto_thueMorseAutocorrelation_zero_shift`, `Fabius.tendsto_thueMorseAutocorrelation_one`, `Fabius.tendsto_thueMorseAutocorrelation_two`) – but the existence of the limit at a *general* shift k , and its spectral consequences, remain analytic frontiers.

Theorem 13.4 (Limiting autocorrelation). *For every fixed $k \geq 0$, the limit*

$$\eta(k) = \lim_{m \rightarrow \infty} \eta_m(k) \quad (13.15)$$

exists and is characterized by

$$\eta(0) = 1, \quad (13.16)$$

$$\eta(2r) = \eta(r), \quad (13.17)$$

$$\eta(2r+1) = -\frac{\eta(r) + \eta(r+1)}{2}. \quad (13.18)$$

In particular,

$$\eta(1) = -\frac{1}{3}, \quad \eta(2) = -\frac{1}{3}, \quad \eta(3) = \frac{1}{3}, \quad \eta(5) = 0. \quad (13.19)$$

Proof. Induct on k . The case $k = 0$ is constant. For $k = 1$, (13.13) gives the contraction

$$\eta_{m+1}(1) = -\frac{1 + \eta_m(1)}{2},$$

whose fixed point is $-1/3$. If $k = 2r > 1$, convergence reduces to the smaller index r . If $k = 2r+1 > 1$, both r and $r+1$ are smaller than k , so the odd recurrence and the induction hypothesis give convergence. Pass to the limit and evaluate recursively. $\square$

The numbers $\eta(k)$ are the Fourier coefficients of μ_{TM} . Thus the substitution recursion, autocorrelation, and Riesz product are exact combinatorial, coefficient, and measure-theoretic forms of the same spectral object. Extending by evenness, $\eta(-k) = \eta(k)$, the first values are

$$1, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, \frac{1}{3}, 0, -\frac{1}{3}, \frac{1}{6}, 0, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}, 0, \frac{1}{6}. \quad (13.20)$$

13.5 A Mahler equation for the autocorrelation series

The autocorrelation sequence has its own generating function, and the recursion of [theorem 13.4](#) compresses into a single functional equation of Mahler type.

Theorem 13.5 (Autocorrelation Mahler equation). *Let $C(z) = \sum_{n \geq 0} \eta(n)z^n$ for $|z| < 1$. Then*

$$\boxed{2zC(z) = 1 - (1 - z)^2 C(z^2)}. \quad (13.21)$$

Proof. Split the series by parity of the index and insert (13.17) and (13.18):

$$\begin{aligned} C(z) &= \sum_{n \geq 0} \eta(2n)z^{2n} + \sum_{n \geq 0} \eta(2n+1)z^{2n+1} \\ &= C(z^2) - \frac{z}{2} \sum_{n \geq 0} \eta(n)z^{2n} - \frac{z}{2} \sum_{n \geq 0} \eta(n+1)z^{2n} \\ &= C(z^2) - \frac{z}{2} C(z^2) - \frac{1}{2z} (C(z^2) - 1). \end{aligned}$$

Multiply by $2z$ and collect the $C(z^2)$ terms into $-(1 - z)^2 C(z^2)$. □

Equation (13.21) places the autocorrelation beside E and T in the same Mahler framework; the square $(1 - z)^2$ reflects the fact that η averages products of two signs. Since the $\eta(k)$ are also the Fourier coefficients of the spectral measure, the equation is a coefficient-level renormalization law for that singular continuous measure: the measure has no closed-form distribution function, yet its Fourier data satisfies a two-term finite recursion.

Formal versions. `ThueMorseMahler.lean` proves a *finite-level* Mahler equation sharper than (13.21): with $C_m(z) = \sum_k A_m(k)z^k$ the unnormalized dyadic autocorrelation series of [proposition 13.3](#), $zC_{m+1}(z) = 2^m - (1 - z)^2 C_m(z^2)$ holds exactly at every level (`Fabius.autocorrelationSeries_mahler`); dividing by 2^{m+1} recovers (13.21) in the limit.

14 Combinatorics and dynamics of the infinite word

The sequence read as an infinite binary word $\mathbf{t} = \tau(0)\tau(1)\tau(2)\cdots$ has a second life in combinatorics on words. A *factor* of $\mathbf{t}$ is a finite block $\tau(i)\tau(i+1)\cdots\tau(i+\ell-1)$ occurring at some position i ; the word-combinatorial theory studies which factors occur, how often, and with which repetitions. This section proves the elementary statements in full and states the two classical deep ones with references.

14.1 Uniform recurrence and letter frequencies

Recall from (2.4) that $\mathbf{t}$ is the fixed point, starting with 0, of the substitution $\mu : 0 \mapsto 01, 1 \mapsto 10$. Iterating, $\mathbf{t} = \lim_m \mu^m(0)$, and $\mu^m(0), \mu^m(1)$ are the length- 2^m blocks $W_m, \overline{W}_m$ of (2.3), called the level- m *supertiles*.

Proposition 14.1 (Uniform recurrence). *The word $\mathbf{t}$ is uniformly recurrent: for every factor w there is a length $L(w)$ such that every window of $L(w)$ consecutive letters of $\mathbf{t}$ contains an occurrence of w .*

Proof. A factor w of length ℓ occurs inside some supertile $\mu^k(a)$ with $2^k \geq \ell$ and $a \in \{0, 1\}$. Both letters occur in $\mu(0) = 01$, so every supertile of level $k+1$ contains both $\mu^k(0)$ and $\mu^k(1)$; in particular every level- $(k+1)$ supertile contains w . Since $\mathbf{t}$ is tiled by consecutive level- $(k+1)$ supertiles, every window of length $L(w) = 2^{k+2}$ contains a complete one. $\square$

Uniform recurrence says every pattern that occurs once occurs with bounded gaps. In dynamical terms, the closure of the shift orbit of $\mathbf{t}$ is a minimal, uniquely ergodic system; that standard consequence of primitive constant-length substitutions [14] is what guarantees the existence of the limiting frequencies and autocorrelations used above. The two letters have frequency exactly $\frac{1}{2}$: at dyadic lengths this is the exact balance $A(2q) = q$ of (11.5), and general lengths deviate by at most one by [corollary 11.2](#).

Formal versions. `ThueMorseRecurrence.lean` proves [proposition 14.1](#) in a form sharper than stated: the recurrence bound is explicit and *linear* — a factor of length $\ell \leq 2^k$ occurs in every window of length $15 \cdot 2^k$ (`Fabius.thueMorseBit_uniformly_recurrent`), so every factor of length ℓ recurs with gap at most 30ℓ . The route is index arithmetic rather than supertiles: the full block-concatenation laws $\text{wt}_2(q \cdot 2^k + r) = \text{wt}_2(q) + \text{wt}_2(r)$ and $\tau(q \cdot 2^k + r) = \tau(q) \oplus \tau(r)$ (`Fabius.binaryWeight_block_concat`, `Fabius.thueMorseBit_block_concat`) reduce any window to its offset and two block bits, and all four two-letter factors occur in every window of fourteen letters (`Fabius.exists_thueMorseBit_pair`).

14.2 Aperiodicity

Proposition 14.2 (No eventual period). *The sequence (ε_n) is not eventually periodic. In particular it is not periodic.*

Proof. Suppose $p \geq 1$ were a period valid from some index N_0 on. For any $r \geq 0$, choose m with $2^m > r + p$ and $2^m + r \geq N_0$. Block concatenation (2.6) gives $\varepsilon_{2^m+r} = -\varepsilon_r$ and $\varepsilon_{2^m+r+p} = -\varepsilon_{r+p}$; eventual periodicity equates the left sides, so $\varepsilon_{r+p} = \varepsilon_r$ for every $r \geq 0$, and p is a global period. Now compare reflections: by (2.8), for every m with $2^m > p$,

$$-1 = \varepsilon_{2^m} = \varepsilon_{2^m-p} = \varepsilon_{(2^m-1)-(p-1)} = (-1)^m \varepsilon_{p-1}.$$

The right side changes sign between consecutive m , while the left side is constant — a contradiction. $\square$

Formal versions. [Proposition 14.2](#) is `Fabius.thueMorseSign_not_eventually_periodic` (`ThueMorseAperiodicity.lean`), but the formal proof no longer runs the reflection argument above: aperiodicity is now a *corollary of Thue’s theorem* ([theorem 14.3](#)), through a purely word-combinatorial implication that holds over an arbitrary alphabet and knows nothing about ε . If $f : \mathbb{N} \rightarrow A$ is a sequence over any set A and satisfies $f(n+p) = f(n)$ for all $n \geq N$ with $p \geq 1$, then the window of length $2p+1$ starting at N is already an overlap of period p — every index $N+j$ with $j \leq p$ lies past the threshold, so $f(N+j) = f(N+p+j)$ (`Fabius.exists_overlap_of_eventually_periodic`); hence an overlap-free sequence over any alphabet admits no eventual period (`Fabius.not_eventually_periodic_of_overlap_free`, `ThueMorseOverlapFree.lean`). [Proposition 14.2](#) is that implication instantiated at `Fabius.thueMorseSign` and fed with `Fabius.thueMorseSign_overlap_free`; the zero-one counterpart `Fabius.thueMorseBit_not_eventually_periodic` is the same instantiation at `Fabius.thueMorseBit`. The dyadic block-flip identities on which the earlier self-contained proof ran are retained in `ThueMorseAperiodicity.lean` for their own sake — `Fabius.thueMorseSign_two_pow`, the block flip `Fabius.thueMorseSign_two_pow_add`, and the promotion of an eventual period to a global one (`Fabius.thueMorseSign_period_of_eventually`) — but they are no longer on the path to the theorem.

14.3 Overlap-freeness and the critical exponent

A *square* is a nonempty word repeated twice in immediate succession, ww ; a *cube* is www ; an *overlap* is a word of the form $axaxa$ with a a single letter and x a possibly empty word — equivalently, a square ww prolonged by one further letter of the same period. The *critical exponent* of an infinite word is the supremum, over its factors w , of $|w|/p$, where p is the smallest period of w and $|w|$ its length.

Theorem 14.3 (Thue). *The Thue–Morse word contains no overlap. Consequently it contains no cube, and no factor has exponent strictly greater than 2; since squares of arbitrarily great length do occur (apply μ^m to the square $\tau(1)\tau(2) = 11$), the critical exponent equals*

$$\boxed{2}, \tag{14.1}$$

and it is attained.

This is Thue’s foundational theorem [3]; a modern proof by desubstitution — a shortest overlap in $\mu(\mathbf{t})$ would force a shorter one in $\mathbf{t}$ after a finite boundary check — appears in [12]. Overlap-freeness is far stronger than the directly visible absence of the cubes 000 and 111, and it was the original reason for the word’s construction, decades before its automatic, spectral, and Diophantine roles emerged.

Formal versions. Thue’s theorem is no longer only a literature theorem here: `ThueMorseOverlapFree.lean` contains a complete formal proof, `Fabius.thueMorseBit_not_overlap` (with packaged forms `Fabius.thueMorseBit_overlap_free` and, for the signs, `Fabius.thueMorseSign_overlap_free`). The formal proof descends through the two-scale laws directly rather than by desubstitution: an even period halves by reading one parity class of the window (the seed fact is $\tau(2k) \neq \tau(2k+1)$, `Fabius.thueMorseBit_pair_ne`); an odd period forces $\tau(l) = \tau(l+1)$ along a run of consecutive l , which for $p \geq 5$ contradicts the absence of triples (`Fabius.thueMorseBit_no_triple`), while $p \in \{1, 3\}$ die by a Boolean analysis of the window. Cube-freeness is `Fabius.thueMorseBit_cube_free`, and the dyadic squares showing that the exponent 2 is attained are `Fabius.thueMorseBit_square_two_pow`. The module also isolates the step by which this theorem now *implies* [proposition 14.2](#): over an arbitrary alphabet an eventual period $p \geq 1$ from a threshold N on already exhibits an overlap of period p at N (`Fabius.exists_overlap_of_eventually_periodic`), so no overlap-free sequence over any alphabet is eventually periodic (`Fabius.not_eventually_periodic_of_overlap_free`).

14.4 Complement and reversal closure

Complementing every letter commutes with the substitution up to swapping the seeds $0 \leftrightarrow 1$; hence with every factor w the language of $\mathbf{t}$ also contains its bitwise complement $\bar{w}$. For reversal, the reflection formula (2.8) says that reading the length- 2^m prefix backwards reproduces it when m is even and complements it when m is odd — in signed terms this is the reciprocity of [proposition 6.4](#). Every factor sits inside some such prefix, so the factor language is closed under reversal as well. By unique ergodicity, complementary factors occur with equal frequencies; the exact prefix balance of [theorem 11.1](#) is the one-letter case.

Formal versions. Both closures are formal with fully explicit witnesses: `Fabius.thueMorseBit_exists_complement` places the complemented factor at a shift by a large power of two, and `Fabius.thueMorseBit_exists_reversal` places the reversed factor at the reflection of the window through an even-exponent dyadic block, where (2.8) carries the sign $+1$ (`ThueMorseOverlapFree.lean`).

14.5 Linear factor complexity

Let $p(\ell)$ denote the *factor complexity*: the number of distinct factors of length ℓ .

Proposition 14.4 (Elementary linear bound). *For every $\ell \geq 1$,*

$$\boxed{\ell + 1 \leq p(\ell) < 8\ell.} \quad (14.2)$$

Proof. Choose k minimal with $2^k \geq \ell$, so $2^k < 2\ell$. The word is a concatenation of level- k supertiles, so every length- ℓ factor lies inside a pair of consecutive supertiles $\mu^k(a)\mu^k(b)$. There are at most four choices of (a, b) and at most 2^k starting offsets, whence $p(\ell) \leq 4 \cdot 2^k < 8\ell$. The lower bound is the Morse–Hedlund theorem [19]: an infinite word with $p(\ell) \leq \ell$ for some ℓ is eventually periodic, which [proposition 14.2](#) forbids; hence $p(\ell) \geq \ell + 1$ for all ℓ . $\square$

The counting in that proof is wasteful: a window needs the second supertile only if it actually crosses the boundary between the two, and few offsets do. Bookkeeping the two cases separately sharpens the constant.

Proposition 14.5 (Sharpened linear bound). *For all k, ℓ with $\ell \leq 2^k$,*

$$p(\ell) \leq 2 \cdot 2^k + 2(\ell - 1), \quad (14.3)$$

and consequently, for every $\ell \geq 1$,

$$\boxed{p(\ell) \leq 6\ell - 4.} \quad (14.4)$$

Proof. Fix k with $\ell \leq 2^k$ and classify a starting position i by its offset $r = i \bmod 2^k$ in the aligned level- k block. If $r < 2^k - (\ell - 1)$ then $r + j < 2^k$ for every $j < \ell$, so the whole window lies inside that one block and is a function of the pair $(r, \tau(\lfloor i/2^k \rfloor))$: at most $2(2^k - (\ell - 1))$ windows arise this way. Otherwise r is one of the $\ell - 1$ offsets in $[2^k - \ell + 1, 2^k)$, the window straddles the boundary, and it is a function of the triple $(r, \tau(\lfloor i/2^k \rfloor), \tau(\lfloor i/2^k \rfloor + 1))$: at most $4(\ell - 1)$ windows. The two counts add to $2 \cdot 2^k + 2(\ell - 1)$, which is (14.3). Taking $k = \lceil \log_2 \ell \rceil$, so that $\ell \leq 2^k \leq 2\ell - 1$, turns it into $p(\ell) \leq 2(2\ell - 1) + 2(\ell - 1) = 6\ell - 4$. $\square$

For $\ell \geq 1$ the bound (14.4) implies (14.2), and (14.2) is retained only because it is the form the rest of the atlas quotes. Neither is the exact value: the exact piecewise-linear formula for $p(\ell)$ is a separate literature result, recalled but not reproved below.

Formal versions. [Proposition 14.4](#) and [proposition 14.5](#) are both fully formal in `ThueMorseComplexity.lean`, the sharp one being the primitive and the elementary one now its corollary. The complexity is `Fabius.thueMorseComplexity`, the cardinality of the set of length- ℓ windows. A window of length $\ell \leq 2^k$ is reconstructed from its offset in the aligned level- k block and the two adjacent block bits (`Fabius.thueMorseWindow_eq_reconstruct`), a statement that also records *where* the second bit is consulted — only for the indices j with $i \bmod 2^k + j \geq 2^k$ — which is what makes the case split of [proposition 14.5](#) available. Splitting the image accordingly, *any* collection of windows contains at most $2 \cdot 2^k + 2(\ell - 1)$ distinct ones (`Fabius.card_image_thueMorseWindow_le'`), whence (14.3) (`Fabius.thueMorseComplexity_le'`) and, at $k = \lceil \log_2 \ell \rceil$, the boxed (14.4) (`Fabius.thueMorseComplexity_le_six_mul_sub_four`). The earlier chain survives with its signatures unchanged, each entry now a one-line consequence of its sharpened version: at most $4 \cdot 2^k$ windows (`Fabius.card_image_thueMorseWindow_le`), $p(\ell) \leq 4 \cdot 2^k$ (`Fabius.thueMorseComplexity_le`), and $p(\ell) < 8\ell$ (`Fabius.thueMorseComplexity_lt`), so nothing that quoted (14.2) needs revisiting. The lower bound `Fabius.le_thueMorseComplexity` is the Morse–Hedlund argument proved from scratch: complexity is monotone (`Fabius.thueMorseComplexity_mono`), and it grows *strictly* (`Fabius.thueMorseComplexity_lt_succ`) — were $p(\ell + 1) = p(\ell)$, restriction would be a cardinality-preserving surjection and hence injective, every length- ℓ factor would extend uniquely to the right, the finitely many windows would evolve deterministically, the pigeonhole principle would force two equal windows, and the word would be eventually periodic

— contradicting the formal aperiodicity theorem. Uniform recurrence (`Fabius.thueMorseFactorSet_eq_image`) makes the factor set a computable finite image. Every bound holds in at least the stated form: the upper bounds `Fabius.thueMorseComplexity_le_six_mul_sub_four` and `Fabius.thueMorseComplexity_lt` need only $\ell \geq 1$, which is why the boxed lines (14.2) and (14.4) carry that hypothesis, while `Fabius.le_thueMorseComplexity` carries no side condition at all and holds at $\ell = 0$ too, where $p(0) = 1$ (`Fabius.thueMorseComplexity_zero`); the level form `Fabius.thueMorseComplexity_le'` needs only $\ell \leq 2^k$. No literature import is needed — and none of these statements asserts anything about the exact value of $p(\ell)$.

Linear complexity has two immediate consequences. The topological entropy, $\lim_{\ell} \ell^{-1} \log p(\ell)$, is zero. And the word is not *normal*: a normal binary word contains all 2^ℓ blocks of every length ℓ , which is incompatible with (14.2). Exact piecewise-linear formulas for $p(\ell)$ are known [12]; the elementary bound above suffices for every use in this atlas. The pointed contrast with section 15 below is worth savoring: the word itself is combinatorially rigid, yet its arithmetic subsequences behave pseudorandomly.

15 Rarefied sums, arithmetic progressions, and Newman’s phenomenon

Prefix sums of ε are bounded by theorem 11.1, and sums over aligned dyadic blocks vanish. Restricting instead to an arithmetic progression destroys this balance in a striking way: along multiples of three one sign is *always* in excess, with a power-law surplus. The finite product P_m evaluated at roots of unity controls the whole phenomenon.

15.1 Root-of-unity filters at dyadic endpoints

For a modulus $q \geq 2$ and residue $0 \leq r < q$, define

$$A_{m,r}^{(q)} = \sum_{\substack{0 \leq n < 2^m \\ n \equiv r \pmod{q}}} \varepsilon_n. \quad (15.1)$$

Let $\zeta_q = e^{2\pi i/q}$, a primitive q -th root of unity. Averaging the geometric identity $\frac{1}{q} \sum_{\ell < q} \zeta_q^{(n-r)\ell} = \mathbf{1}_{q|n-r}$ against (6.3) gives the exact finite formula

$$A_{m,r}^{(q)} = \frac{1}{q} \sum_{\ell=0}^{q-1} \zeta_q^{-r\ell} P_m(\zeta_q^\ell) = \frac{1}{q} \sum_{\ell=0}^{q-1} \zeta_q^{-r\ell} \prod_{j=0}^{m-1} (1 - \zeta_q^{\ell 2^j}). \quad (15.2)$$

For $m \geq 1$ the term $\ell = 0$ vanishes, and the exponents $\ell 2^j$ move along the orbit of multiplication by two modulo q . For *even* moduli the phenomenon collapses: powers of two are eventually $0 \pmod{2^k}$, and for example $A_{m,0}^{(2)} = \sum_{k < 2^{m-1}} \varepsilon_{2k} = S(2^{m-1}) = 0$ for $m \geq 2$ by (11.2). Odd moduli are the interesting case: there the orbit of 2 is a fixed cycle and the products in (15.2) grow geometrically.

Formal versions. The identity (15.2) itself is now formal in `ThueMorseArithmeticFilter.lean`, in the multiplied form $q A_{m,r}^{(q)} = \sum_{\ell < q} \zeta_q^{-r\ell} \prod_{j < m} (1 - \zeta_q^{\ell 2^j})$ (`Fabius.thueMorse_rarefied_filter`). It specializes `Fabius.sum_filter_modEq_mul_natCast`, the filter proved for an *arbitrary* field carrying a primitive q -th root of unity and an *arbitrary* coefficient function — stated without division so that no invertibility of q is assumed; the progression indicator arises from the geometric orthogonality `Fabius.sum_pow_eq_ite` by exponent reduction and primitive-root injectivity. The even-modulus collapse $A_{m,0}^{(2)} = 0$ for $m \geq 2$ is `Fabius.sum_thueMorseSign_even_range`. The algebraic skeleton of the filter’s vanishing is `Fabius.sum_thueMorseSign_mul_pow_eq_zero_of_pow_eq_one` (`ThueMorseRarefied.lean`): over any commutative ring the signed block sum

vanishes at any point with $z^{2^j} = 1$ for some $j < m$. Modulo three the filter is subsumed by the closed vectors below.

15.2 Complete evaluation modulo three

For $q = 3$ everything is explicit, because $2^j \equiv 1, 2, 1, 2, \dots \pmod{3}$ and, with $\omega = \zeta_3 = e^{2\pi i/3}$,

$$(1 - \omega)(1 - \omega^2) = 3. \quad (15.3)$$

Theorem 15.1 (Exact dyadic rarefaction modulo three). *For $u \geq 1$,*

$$(A_{2u,0}^{(3)}, A_{2u,1}^{(3)}, A_{2u,2}^{(3)}) = 3^{u-1} (2, -1, -1), \quad (15.4)$$

and for $u \geq 0$,

$$(A_{2u+1,0}^{(3)}, A_{2u+1,1}^{(3)}, A_{2u+1,2}^{(3)}) = 3^u (1, -1, 0). \quad (15.5)$$

Proof. At even level $m = 2u$, the factors of $P_m(\omega)$ pair into u copies of (15.3), so $P_{2u}(\omega) = P_{2u}(\omega^2) = 3^u$. At odd level one factor is left over: $P_{2u+1}(\omega) = 3^u(1 - \omega)$ and $P_{2u+1}(\omega^2) = 3^u(1 - \omega^2)$. Insert these values into (15.2) with $q = 3$ and simplify with $1 + \omega + \omega^2 = 0$. $\square$

Formal versions. `ThueMorseRarefied.lean` proves the closed vectors (15.4)–(15.5) as `Fabius.rarefiedSum_three_closed` – and does so without any root of unity: the one-step halving recursion $A_{m+1,r} = A_{m,2r \bmod 3} - A_{m,(2r+1) \bmod 3}$ (`Fabius.rarefiedSum_three_succ`; the hypothesis $r < 3$ is necessary, as the recursion is false for out-of-range residues) is iterated over paired levels by pure integer induction, so the filter argument above is subsumed rather than formalized. Newman positivity at dyadic endpoints is `Fabius.rarefiedSum_three_zero_pos`, the full-block balance is `Fabius.sum_thueMorseSign_range_two_pow`, and the even-modulus collapse of the surrounding text is `Fabius.rarefiedSum_two_zero`.

The first vectors are

$$(1, -1, 0), \quad (2, -1, -1), \quad (3, -3, 0), \quad (6, -3, -3), \quad (9, -9, 0),$$

growing by a factor of three every two levels. Writing $X = 2^m$ for the endpoint, the excess on multiples of three is

$$A_{m,0}^{(3)} = c_m X^\alpha, \quad \alpha = \log_4 3 = \frac{\log 3}{\log 4} = 0.79248\dots, \quad (15.6)$$

where the constant c_m equals $\frac{2}{3}$ for even m and $\frac{1}{\sqrt{3}}$ for odd m : a clean two-scale oscillation around the power law.

15.3 Gelfond’s uniform exponent

The rarefied sums are finitely many samples of the trigonometric polynomial $p_m(x) = P_m(e^{2\pi i x})$, whose modulus is the sine product (12.3). How large can it get? Evaluating at $x = \frac{1}{3}$ gives, by (15.3),

$$\sup_{x \in \mathbb{R}} |p_m(x)| \geq |P_m(\omega)| = 3^{m/2} \quad (m \geq 0), \quad (15.7)$$

since the factors have modulus $|1 - \omega| = |1 - \omega^2| = \sqrt{3}$ in every position. Gelfond’s theorem [28] – proved below – shows this is sharp up to a bounded factor:

$$\sup_{x \in \mathbb{R}} |p_m(x)| \leq \frac{2}{\sqrt{3}} 3^{m/2} = \frac{2}{\sqrt{3}} (2^m)^{\log_4 3}. \quad (15.8)$$

Proof of (15.8). The whole bound reduces to a single one-variable inequality. Write $\varphi(t) = 2|\sin t|$, so that $|p_m(x)| = \prod_{j < m} \varphi(2^j \pi x)$. For every real t ,

$$\varphi(t)^2 \varphi(2t) \leq 3\sqrt{3}, \quad (15.9)$$

because after squaring and substituting $u = \sin^2 t$ (so that $\sin^2 2t = 4u(1 - u)$) the claim reads $256u^3(1 - u) \leq 27$, and

$$27 - 256u^3 + 256u^4 = (4u - 3)^2((4u + 1)^2 + 2) \geq 0,$$

with equality exactly at $u = \frac{3}{4}$ – the doubling-fixed Gelfond angle $t = \pi/3$, where $\varphi = \sqrt{3}$. Multiplying (15.9) along the doubling orbit, $\prod_{j < m-1} \varphi_j^2 \varphi_{j+1} \leq (3\sqrt{3})^{m-1}$ with $\varphi_j := \varphi(2^j t)$; the regrouping

$$\left(\prod_{j < m} \varphi_j \right)^3 = \varphi_0 \varphi_{m-1}^2 \prod_{j < m-1} (\varphi_j^2 \varphi_{j+1})$$

and the crude boundary bounds $\varphi_0 \leq 2$, $\varphi_{m-1}^2 \leq 4$ then give $(\prod_{j < m} \varphi_j)^3 \leq 8 \cdot 3^{3(m-1)/2} = \left(\frac{2}{\sqrt{3}} 3^{m/2}\right)^3$. Taking cube roots yields (15.8); the constant is Gelfond’s, and by (15.7) it cannot be lowered below 1. $\square$

Formal versions. The bound (15.8) is fully formal, with the proof above (ThueMorseGelfond .lean): the key inequality (15.9) is Fabius.two_abs_sin_sq_mul_1e, proved by exactly the sum-of-squares certificate displayed; the telescoped product bound $\prod_{j < m} 2|\sin(2^j \theta)| \leq 2(\sqrt{3})^{m-1}$ is Fabius.gelfond_prod_1e; and the exponential-sum form $|\sum_{n < 2^m} \varepsilon_n e^{inx}| \leq 2(\sqrt{3})^{m-1} = \frac{2}{\sqrt{3}} 3^{m/2}$, uniform in the real frequency x (the normalization of the frequency is immaterial to a uniform bound), is Fabius.gelfond_uniform_bound, riding the corpus sine-product identity Fabius.sum_thueMorseSign_exp_eq_sin_prod.

Decomposing an arbitrary range $[0, N)$ into at most one dyadic block per binary digit of N and applying (15.8) blockwise gives

$$\sup_{x \in \mathbb{R}} \left| \sum_{n=0}^{N-1} \varepsilon_n e^{2\pi i n x} \right| = \mathcal{O}(N^{\log_4 3}), \quad (15.10)$$

so $\log_4 3$ is the optimal uniform growth exponent of Thue–Morse exponential sums [28, 29]. At $x = 0$ the sum collapses to the bounded prefix values – the supremum and the special value differ by a full power of N .

15.4 A matrix recursion for odd moduli

For a fixed odd q , the vector $\mathbf{A}_m = (A_{m,r}^{(q)})_{0 \leq r < q}$ obeys a linear recursion. Splitting (15.1) by the parity of n , and writing 2^{-1} for the inverse of 2 modulo q (which exists because q is odd),

$$A_{m+1,r}^{(q)} = A_{m,2^{-1}r \bmod q}^{(q)} - A_{m,2^{-1}(r-1) \bmod q}^{(q)}, \quad (15.11)$$

since even $n = 2n' \equiv r$ forces $n' \equiv 2^{-1}r$ with $\varepsilon_n = \varepsilon_{n'}$, and odd $n = 2n' + 1 \equiv r$ forces $n' \equiv 2^{-1}(r-1)$ with $\varepsilon_n = -\varepsilon_{n'}$. Thus $\mathbf{A}_{m+1} = M_q \mathbf{A}_m$ for a fixed matrix M_q with one entry $+1$ and one entry -1 per row; diagonalizing M_q by the discrete Fourier transform on $\mathbb{Z}/q\mathbb{Z}$ reproduces (15.2). For $q = 3$ the eigenvalues have modulus $\sqrt{3}$, which is the growth rate seen in theorem 15.1.

Formal versions. The recursion (15.11) is Fabius.rarefiedSum_succ_of_odd (ThueMorseRarefied.lean), proved for every odd modulus with the halving realized by $2^{-1} = (q+1)/2$; the invertibility of doubling modulo an odd modulus is isolated as Fabius.two_mul_mod_iff. Modulo three it specializes to Fabius.rarefiedSum_three_succ.

15.5 The global Newman theorem and the arithmetic frontier

Theorem 15.1 evaluates the excess at dyadic endpoints. Newman’s theorem extends the positivity to *every* endpoint. The positivity assertion of Newman’s theorem is proved below by a short base-four induction — and is formalized; the quantitative refinements, and the deeper equidistribution results that follow, are literature theorems, stated here to locate the atlas’s exact identities inside modern analytic number theory.

Theorem 15.2 (Newman). *For every real $X \geq 1$,*

$$N_3(X) := \sum_{\substack{0 \leq n < X \\ 3|n}} \varepsilon_n > 0. \quad (15.12)$$

Moreover $\frac{1}{20}X^{\log_4 3} \leq N_3(X) \leq 5X^{\log_4 3}$ for $X \geq 2$.

Proof of the positivity. It suffices to treat integer cutoffs $N \geq 1$. Write $T_r(N) = \sum_{n < N, n \equiv r(3)} \varepsilon_n$ and $E(N) = \sum_{n < N} \varepsilon_n$; consecutive pairs cancel, so $E(2M) = 0$ and $E(2M+1) = \varepsilon_{2M} = \pm 1$ — in particular $|E| \leq 1$ everywhere. The parity split $\varepsilon_{2j} = \varepsilon_j$, $\varepsilon_{2j+1} = -\varepsilon_j$ gives the one-step system

$$T_0(2M) = T_0(M) - T_1(M), \quad T_1(2M) = T_2(M) - T_0(M),$$

and composing two steps and eliminating $T_1(M) + T_2(M) = E(M) - T_0(M)$ collapses to

$$T_0(4M+s) = 3T_0(M) - E(M) + c_s(M), \quad s \in \{0, 1, 2, 3\},$$

where the boundary correction $c_s(M)$ collects at most three single signs: ε_M from the term $n = 4M$ (alive only when $M \equiv 0$), $-\varepsilon_M$ from the odd term of the second halving (alive only when $M \equiv 2$), and $-\varepsilon_M$ from the term $n = 4M+2$ (alive only when $M \equiv 1$, since $4M+2 \equiv M+2$, and $\varepsilon_{4M+2} = -\varepsilon_M$). The three supports are pairwise disjoint residue classes of M , so $|c_s(M)| \leq 1$. By strong induction $T_0(M) \geq 1$ for all $M \geq 1$ — the cases $N \leq 4$ are read off the signs $+, -, -, +$ — whence $T_0(4M+s) \geq 3 \cdot 1 - 1 - 1 = 1$: the three-fold amplification absorbs all boundary noise. The quantitative bounds are Newman’s refinement of the same induction and are not reproved here. $\square$

Among multiples of three below any cutoff, integers of even binary weight always outnumber those of odd binary weight [30]. Coquet determined the precise structure: $N_3(X) = X^{\log_4 3} \Psi(\log_4 X) + \mathcal{O}(1)$ with Ψ a continuous, nowhere-differentiable periodic function [31].

Formal versions. The boxed positivity (15.12) is fully formal, with the proof above (Fabi.us.newman_positivity, ThueMorseNewman.lean): the residue-class sign sums are Fabi.us.thueMorseResidueSum (defined for every modulus), the parity system is Fabi.us.thueMorseResidueSum_three_zero_two_mul and its companion, the three sums recombine to the prefix sum by Fabi.us.thueMorseResidueSum_three_add, and the prefix-sum bound $|E| \leq 1$ is Fabi.us.abs_sum_thueMorseSign_le_one. The quantitative bounds $\frac{1}{20}X^{\log_4 3} \leq N_3(X) \leq 5X^{\log_4 3}$ and Coquet’s oscillation remain literature imports. Rarefactions along other progressions and for generalized digit characters were analyzed by Drmota and Skalba [32] and Drmota and Stoll [33]; a recent automata-theoretic treatment mechanizes such evaluations [34].

Beyond arithmetic progressions, the same digit-parity character is now understood along much thinner sequences. Mauduit and Rivat proved Gelfond’s conjecture that the parity is equidistributed along the primes: $\sum_{p \leq x} \varepsilon_p = o(\pi(x))$, the sum running over primes with $\pi(x)$ their count [35]. Drmota, Mauduit, and Rivat proved that along the squares the word $(\tau(n^2))_{n \geq 0}$ is *normal* — every binary block of length L occurs with limiting frequency 2^{-L} [36] — in sharp contrast with the non-normality of the word itself (proposition 14.4). Spiegelhofer established a level-of-distribution result covering $\tau(\lfloor n^c \rfloor)$ for $1 < c < 2$ [37] and the one-symbol balance along the cubes [38]. The deterministic word is combinatorially rigid, yet its polynomial and prime subsequences behave like fair coin flips: that tension is the modern face of Gelfond’s problems, and every one of these theorems begins from the finite product and carry structure catalogued in this atlas.

16 Boundary flatness, shifted Dirichlet series, and Mellin continuation

This section develops an analytic consequence not organized systematically in the four drafts. The lacunary product is extraordinarily flat as its argument approaches one from inside the unit disk. That flatness removes every Mellin singularity and gives an entire continuation of a natural family of Thue–Morse Dirichlet series.

16.1 Quadratic logarithmic decay near one

For $t > 0$, set

$$\mathcal{E}(t) = E(e^{-t}) = \prod_{j \geq 0} (1 - e^{-2^j t}). \quad (16.1)$$

Every factor lies in $(0, 1)$, so $\mathcal{E}(t) > 0$.

Theorem 16.1 (Boundary flatness). *As $t \downarrow 0$,*

$$\log \mathcal{E}(t) = -\frac{\log^2(1/t)}{2 \log 2} + \mathcal{O}(\log(1/t)). \quad (16.2)$$

Consequently, for every $A > 0$,

$$\mathcal{E}(t) = \mathcal{O}_A(t^A) \quad (t \downarrow 0). \quad (16.3)$$

Proof. Assume $0 < t < 1/2$, put $L = \log(1/t)$, and let $m = \lfloor \log_2(1/t) \rfloor$. Then $2^m t \in (1/2, 1]$. For $0 \leq j < m$, one has $0 < 2^j t \leq 1/2$, and the elementary bounds

$$\frac{x}{2} \leq 1 - e^{-x} \leq x \quad (0 < x \leq 1) \quad (16.4)$$

show that

$$\log(1 - e^{-2^j t}) = \log t + j \log 2 + \mathcal{O}(1), \quad (16.5)$$

with an absolute $\mathcal{O}(1)$. Summing gives

$$\sum_{j=0}^{m-1} \log(1 - e^{-2^j t}) = m \log t + \frac{m(m-1)}{2} \log 2 + \mathcal{O}(m). \quad (16.6)$$

For the tail, write $2^m t = u \in (1/2, 1]$. Then

$$\sum_{j=m}^{\infty} \left| \log(1 - e^{-2^j t}) \right| = \sum_{r=0}^{\infty} \left| \log(1 - e^{-2^r u}) \right| = \mathcal{O}(1) \quad (16.7)$$

uniformly in u , because the terms decay doubly exponentially. Finally $m = L/\log 2 + \mathcal{O}(1)$; substitution in (16.6) yields (16.2). The negative quadratic term dominates $-A \log(1/t)$ for every fixed A , proving (16.3). $\square$

Remark 16.2. Sharper asymptotic expansions contain a bounded log-periodic fluctuation in $\log_2(1/t)$. The elementary estimate above is deliberately limited to the part needed for Mellin continuation: super-polynomial flatness.

Formal versions. [Theorem 16.1](#) is formal in effective two-sided form (`ThueMorseBoundaryFlatness.lean`). The product $\mathcal{E}$ is `Fabius.lacunaryExpProduct`, identified with the exponential of its log series and hence positive and at most one. The key structural fact is the *self-similarity* $\mathcal{E}(t) = \prod_{j < m} (1 - e^{-2^j t}) \cdot \mathcal{E}(2^m t)$ (`Fabius.lacunaryExpProduct_eq_prod_mul`); combining it

with the reusable sandwich $x/2 \leq 1 - e^{-x} \leq x$ on $[0, 1]$ (`Fabius.half_le_one_sub_exp_neg`, `Fabius.one_sub_exp_neg_le` – the lower half proved from $e^{-x} \leq 1/(1+x)$, no series expansion) gives, for every level m ,

$$2^{\binom{m}{2}} (t/2)^m \mathcal{E}(\tfrac{1}{2}) \leq \mathcal{E}(t) \leq 2^{\binom{m}{2}} t^m,$$

the lower bound under the dyadic matching $\frac{1}{2} \leq 2^m t \leq 2$ (`Fabius.le_lacunaryExpProduct`, `Fabius.lacunaryExpProduct_le`). Taking logarithms at the matching level recovers (16.2) with explicit constants, and the upper bound alone is (16.3) in effective form. Monotonicity in t (`Fabius.lacunaryExpProduct_mono`) and summability of the factor logarithms (`Fabius.summable_log_one_sub_exp`) are proved along the way.

16.2 The shifted Dirichlet family

For $a > 0$, define initially in the half-plane $\Re s > 0$

$$D(s, a) = \sum_{n=0}^{\infty} \frac{\varepsilon_n}{(n+a)^s}. \quad (16.8)$$

The principal real logarithm is used in $(n+a)^{-s}$.

Proposition 16.3 (Conditional convergence). *The series (16.8) converges locally uniformly in $\Re s > 0$, and hence defines a holomorphic function there.*

Proof. The partial sums $S(N) = \sum_{n < N} \varepsilon_n$ are bounded by one. On a compact set $\Re s \geq \sigma > 0$, the first difference of $(n+a)^{-s}$ is uniformly $\mathcal{O}(n^{-\sigma-1})$. Summation by parts therefore gives uniform convergence of the tails. $\square$

Formal versions. The real regime of the family is formal in `ThueMorseDirichlet.lean`: for every $\sigma > 0$ and $a > 0$ the partial sums of $D(\sigma, a)$ form a Cauchy sequence by Dirichlet’s test – the antitone shifted powers tend to zero and the sign prefix sums are bounded by one – (`Fabius.dirichletPartial_cauchy`), and the limit function `Fabius.thueMorseDirichlet` is their limit (`Fabius.tendsto_thueMorseDirichlet`); this is [proposition 16.3](#) along the real axis. The dyadic equation below is `Fabius.thueMorseDirichlet_dyadic` on that same real ray – for every real $\sigma > 0$ and $a > 0$ – proved by interleaving the even-length partial sums through $\varepsilon(2n) = \varepsilon(n)$, $\varepsilon(2n+1) = -\varepsilon(n)$ and passing to limits along the doubled subsequence; the complex form is recorded after [theorem 16.4](#) below. The Mellin representation itself is now formal in the absolutely convergent real regime (`Fabius.thueMorseDirichlet_mellin`, `ThueMorseMellin.lean`): for $\sigma > 1$ and $a > 0$, $\Gamma(\sigma)D(\sigma, a) = \int_0^\infty t^{\sigma-1} e^{-at} \mathcal{E}(t) dt$ – the integrand is rewritten through the analytic infinite-product identity, the sum and integral are interchanged by absolute convergence (the norm integrals are exactly the Gamma values, summable as a shifted p -series), and each term is a Gamma integral. For $\sigma > 1$ the conditionally defined D is identified with the absolute sum (`Fabius.thueMorseDirichlet_eq_tsum`). The entire continuation itself is now formal (`ThueMorseEntireContinuation.lean`): the function $s \mapsto \Gamma(s)^{-1} \int_0^\infty t^{s-1} e^{-at} \mathcal{E}(t) dt$ is differentiable on all of $\mathbb{C}$ (`Fabius.dirichletMellinContinuation_differentiable`) – the effective flatness bound supplies an $\mathcal{O}(t^{-b})$ estimate at zero for every b (`Fabius.mellinKernel_isBigO_rpow`), precisely realizing the maxim that boundary flatness removes every Mellin singularity, while $1/\Gamma$ is entire – and it agrees with $D(\sigma, a)$ on the ray $\sigma > 1$ (`Fabius.dirichletMellinContinuation_eq`). Only the locally-uniform holomorphy of the conditionally convergent series in the strip $0 < \Re s \leq 1$ (superseded by the continuation) remains informal.

Splitting even and odd indices gives an exact parameter equation.

Theorem 16.4 (Dyadic Dirichlet equation). *For $a > 0$ and $\Re s > 0$,*

$$D(s, a) = 2^{-s} \left(D\left(s, \frac{a}{2}\right) - D\left(s, \frac{a+1}{2}\right) \right). \quad (16.9)$$

Proof. Use $\varepsilon_{2n} = \varepsilon_n$ and $\varepsilon_{2n+1} = -\varepsilon_n$:

$$\begin{aligned} D(s, a) &= \sum_{n \geq 0} \frac{\varepsilon_n}{(2n + a)^s} - \sum_{n \geq 0} \frac{\varepsilon_n}{(2n + 1 + a)^s} \\ &= 2^{-s} \left(D\left(s, \frac{a}{2}\right) - D\left(s, \frac{a+1}{2}\right) \right). \end{aligned}$$

□

Formal versions. [Theorem 16.4](#) is now formal at *every* complex s , not only along the real ray. The carrier is the entire continuation of [theorem 16.5](#), formalized as `Fabius.dirichletMellinContinuation`: for every $a > 0$ and every $s \in \mathbb{C}$,

$$\tilde{D}(s, a) = 2^{-s} \left(\tilde{D}\left(s, \frac{a}{2}\right) - \tilde{D}\left(s, \frac{a+1}{2}\right) \right) \quad (16.10)$$

(`Fabius.dirichletMellinContinuation_dyadic`, `ThueMorseEntireContinuation.lean`), where $\tilde{D}$ denotes that continuation. The proof is the identity theorem, and it is the reason the entirety layer was built: both sides of (16.10) are entire — the left by `Fabius.dirichletMellinContinuation_differentiable` at a , the right by the same at $a/2$ and $(a+1)/2$ multiplied by the entire $s \mapsto 2^{-s}$ — and they agree on the ray $(1, \infty) \subseteq \mathbb{R} \subseteq \mathbb{C}$ (`Fabius.dirichletMellinContinuation_dyadic_ofReal`, itself `Fabius.thueMorseDirichlet_dyadic` transported through `Fabius.dirichletMellinContinuation_eq`); that ray accumulates at $s = 2$, and $\mathbb{C}$ is connected. Note what is and is not delivered: (16.10) is unconditional in s , while its identification with (16.9) — that is, $\tilde{D} = D$ — is formal only where `Fabius.dirichletMellinContinuation_eq` applies, namely on the real ray $\sigma > 1$; off that ray the identification still rests on the informal half of [proposition 16.3](#).

16.3 Mellin representation and entire continuation

Theorem 16.5 (Mellin integral). *For $a > 0$, initially when $\Re s > 1$,*

$$D(s, a) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-at} E(e^{-t}) dt. \quad (16.11)$$

The integral on the right is an entire function of s . Therefore (16.11) continues $D(s, a)$ to an entire function of $s \in \mathbb{C}$.

Proof. For $\Re s > 1$, absolute convergence permits termwise use of

$$(n + a)^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-(n+a)t} dt.$$

Summing $\varepsilon_n e^{-nt}$ gives $E(e^{-t})$.

At infinity, $E(e^{-t}) = 1 + \mathcal{O}(e^{-t})$, and the factor e^{-at} gives exponential decay. At zero, [theorem 16.1](#) gives decay faster than every positive power of t . Thus the integral and all of its s -derivatives converge uniformly on compact subsets of the entire s -plane. Multiplication by the entire function $1/\Gamma(s)$ preserves entire-ness. Equality on $\Re s > 1$, together with [proposition 16.3](#), gives the continuation by the identity theorem. □

Corollary 16.6 (Zeros at nonpositive integers). *For every $a > 0$ and every integer $r \geq 0$,*

$$D(-r, a) = 0. \quad (16.12)$$

Proof. The Mellin integral in (16.11) is finite at $s = -r$, while $1/\Gamma(s)$ has a zero there. □

These zeros are the analytic continuation of the finite Prouhet phenomenon: negative integer values formally ask for signed polynomial moments, and the flat Mellin kernel makes all of them vanish after continuation.

Formal versions. [Corollary 16.6](#) is now formal (`ThueMorseEntireContinuation.lean`): `Fabius.dirichletMellinContinuation_neg_natCast` gives $D(-r, a) = 0$ for every integer $r \geq 0$, with the origin recorded separately as `Fabius.dirichletMellinContinuation_zero`; neither statement needs any hypothesis on a , since the vanishing is supplied by the zero of $1/\Gamma$ at $s = -r$. What makes that argument legitimate is the stronger, *un-damped* statement – and it too is formal – that the Mellin factor alone is entire, with no Γ -factor damping it (`Fabius.mellin_mellinKernel_differentiable`, for $a > 0$): boundary flatness leaves no pole at $s = -r$ that could cancel that zero.

16.4 The harmonic kernel

At $s = 1$, the Mellin integral can be moved to the unit interval.

Corollary 16.7 (Dirichlet integral). *For $a > 0$,*

$$H(a) := D(1, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{n + a} \quad (16.13)$$

$$= \int_0^1 x^{a-1} E(x) dx = \int_0^1 x^{a-1} \prod_{j \geq 0} (1 - x^{2^j}) dx > 0. \quad (16.14)$$

It satisfies

$$H(a) = \frac{1}{2} H\left(\frac{a}{2}\right) - \frac{1}{2} H\left(\frac{a+1}{2}\right). \quad (16.15)$$

Proof. Set $s = 1$ in (16.11) and substitute $x = e^{-t}$. The integrand is positive on $(0, 1)$. Equation (16.15) is (16.9) at $s = 1$. $\square$

17 Infinite products with Thue–Morse exponents

17.1 Pair reduction

Suppose $R(n) \neq 0$ and the products converge. The even–odd recurrence gives the general reduction

$$\prod_{n \geq 0} R(n)^{\varepsilon_n} = \prod_{q \geq 0} \left(\frac{R(2q)}{R(2q+1)} \right)^{\varepsilon_q}. \quad (17.1)$$

A factor merely equal to $1 + \mathcal{O}(1/n)$ is often converted into a paired factor $1 + \mathcal{O}(1/n^2)$, explaining the good convergence of balanced rational products.

17.2 A two-parameter master product

For $a, b > 0$, define

$$G(a, b) = \prod_{n=0}^{\infty} \left(\frac{n+a}{n+b} \right)^{\varepsilon_n}. \quad (17.2)$$

The logarithm converges by summation by parts: the Thue–Morse partial sums are bounded and the first difference of $\log((n+a)/(n+b))$ is $\mathcal{O}(n^{-2})$.

Theorem 17.1 (Cocycle and dyadic functional equation). *For positive parameters,*

$$G(a, a) = 1, \quad G(a, b)G(b, c) = G(a, c), \quad G(b, a) = G(a, b)^{-1}, \quad (17.3)$$

$$G(a, b) = \boxed{\frac{G(a/2, b/2)}{G((a+1)/2, (b+1)/2)}}. \quad (17.4)$$

Proof. The cocycle identities are termwise. Pair indices $2n$ and $2n+1$:

$$\begin{aligned} & \left(\frac{2n+a}{2n+b} \right)^{\varepsilon_n} \left(\frac{2n+1+a}{2n+1+b} \right)^{-\varepsilon_n} \\ &= \left[\frac{n+a/2}{n+b/2} \frac{n+(b+1)/2}{n+(a+1)/2} \right]^{\varepsilon_n}. \end{aligned}$$

Multiply over n . □

Formal versions. [Theorem 17.1](#) is formal in logarithmic form (`ThueMorseMasterProduct.lean`): the sign-weighted log-series $L(a, b) = \sum \varepsilon_n \log((n+a)/(n+b))$ converges for all positive parameters by Dirichlet’s test (`Fabius.mpLog_cauchy`); the cocycle identities are `Fabius.mpLimit_self`, `Fabius.mpLimit_symm`, and `Fabius.mpLimit_cocycle`; the boxed dyadic functional equation is `Fabius.mpLimit_dyadic`, proved by the exact even-length interleaving rather than by rearrangement; and the product itself converges to $\exp L(a, b)$ (`Fabius.tendsto_masterProduct`).

The Dirichlet continuation identifies this product as a zeta-regularized derivative difference without any extra regularization being required.

Corollary 17.2 (Dirichlet derivative bridge). *For $a, b > 0$,*

$$\boxed{\log G(a, b) = D'(0, b) - D'(0, a)}. \quad (17.5)$$

Proof. The difference $D(s, a) - D(s, b)$ can be differentiated termwise at $s = 0$, because the resulting log-ratio series converges by summation by parts. Its derivative is

$$\sum_{n \geq 0} \varepsilon_n \log \frac{n+b}{n+a} = -\log G(a, b).$$

Rearrange. □

Remark 17.3 (A Fubini-free proof of the bridge). The sketch above can be completed without any uniform summation-by-parts estimate; the following route is the one the formalization takes. First, each term of the master log-series is a Frullani integral,

$$\varepsilon_n \log \frac{n+a}{n+b} = \varepsilon_n \int_0^\infty \frac{e^{-(n+b)t} - e^{-(n+a)t}}{t} dt,$$

and summing over $n < 2^m$ factors the signed exponentials into the *finite* lacunary product:

$$\sum_{n < 2^m} \varepsilon_n \log \frac{n+a}{n+b} = \int_0^\infty \frac{e^{-bt} - e^{-at}}{t} \prod_{j < m} (1 - e^{-2^j t}) dt.$$

The partial products lie in $[0, 1]$, so dominated convergence – with the integrable dominator $|a-b|e^{-\min(a,b)t}$, a consequence of $1 - e^{-x} \leq x$ – passes to the limit along the dyadic subsequence, along which the conditionally convergent series already converges:

$$\log G(a, b) = \int_0^\infty \frac{e^{-bt} - e^{-at}}{t} E(e^{-t}) dt. \quad (17.6)$$

Second, the factorization $\Gamma(s)^{-1} = s\Gamma(s+1)^{-1}$ — an identity of entire functions, valid at the poles as well — gives $(1/\Gamma)'(0) = 1$ and $\Gamma(0)^{-1} = 0$, so by the product rule the entire continuation $s \mapsto \Gamma(s)^{-1} \int_0^\infty t^{s-1} e^{-at} E(e^{-t}) dt$ of $D(s, a)$ constructed above has derivative at $s = 0$ equal to the Mellin value $\int_0^\infty t^{-1} e^{-at} E(e^{-t}) dt$ — convergent, since boundary flatness gives $E(e^{-t}) \leq t$. Subtracting the two parameters recovers (17.6).

Formal versions. [Corollary 17.2](#) is fully formal along the route of [Remark 17.3](#) (`ThueMorseDirichlet.lean`). The Frullani integral $\int_0^\infty (e^{-pt} - e^{-qt})t^{-1} dt = \log(q/p)$ is proved from scratch by differentiation under the integral sign — the $1/t$ cancels exactly against the derivative of the exponential — followed by the fundamental theorem of calculus in the parameter (`Fabius.frullani_exp`, `FrullaniIntegral.lean`; the lemma is not in `Mathlib`). The closed integral form (17.6) is `Fabius.mpLimit_eq_integral`; the derivative $(1/\Gamma)'(0) = 1$ is `Fabius.hasDerivAt_Gamma_inv_zero`, riding `Mathlib`’s entire-function factorization $\Gamma(s)^{-1} = s\Gamma(s+1)^{-1}$; the derivative of the entire continuation at zero is `Fabius.hasDerivAt_dirichletMellinContinuation_zero`; and the boxed bridge (17.5) is `Fabius.mpLimit_eq_deriv_sub`, with $D(\cdot, c)$ realized by its entire continuation.

17.3 The Woods–Robbins product

Theorem 17.4 (Woods–Robbins).

$$\boxed{\prod_{n=0}^{\infty} \left(\frac{2n+1}{2n+2} \right)^{\varepsilon_n} = \frac{1}{\sqrt{2}}.} \quad (17.7)$$

Proof. Let

$$A = \prod_{n \geq 0} \left(\frac{2n+1}{2n+2} \right)^{\varepsilon_n}, \quad B = \prod_{n \geq 1} \left(\frac{2n}{2n+1} \right)^{\varepsilon_n}. \quad (17.8)$$

Then

$$AB = \frac{1}{2} \prod_{n \geq 1} \left(\frac{n}{n+1} \right)^{\varepsilon_n}. \quad (17.9)$$

Split the product on the right into odd and even n . The odd part is A^{-1} because $\varepsilon_{2j+1} = -\varepsilon_j$, and the even part is B because $\varepsilon_{2j} = \varepsilon_j$. Hence

$$AB = \frac{1}{2} A^{-1} B. \quad (17.10)$$

Cancel B , obtaining $A^2 = 1/2$; positivity gives the stated root. $\square$

Formal versions. [Theorem 17.4](#) is fully formal (`Fabius.woods_robbins`, `ThueMorseWoodsRobbins.lean`), with the pairing argument carried out rigorously in logarithmic form: the three sign-weighted log-series converge by Dirichlet’s test (a reusable guarded variant handles weights antitone only from 1 on); the termwise combination $L_A + L_B = -\log 2 + L_C$ and the exact even-length interleaving $L_C(2N) = L_B(N) - L_A(N)$ — the formal counterpart of the odd/even split, justified along the doubled subsequence rather than by rearrangement — force $L_A = -\frac{1}{2} \log 2$ (`Fabius.tendsto_wra`), and exponentiation evaluates the partial products. That final exponentiation is itself a reusable module, `RealZPowProduct.lean`, in which nothing mentions Thue–Morse: a finite product of integer powers of positive reals is the exponential of the signed logarithm sum (`Fabius.prod_zpow_eq_exp_sum`), so a convergent logarithm sum exponentiates to a convergent product along an arbitrary filter (`Fabius.tendsto_prod_zpow_of_tendsto_sum`), and the closed values come from `Fabius.exp_neg_mul_log` with its two specializations `Fabius.exp_neg_log_log_div_two` ($e^{-\frac{1}{2} \log b} = 1/\sqrt{b}$) and `Fabius.exp_neg_three_halves_log`. Every product evaluation below reuses it. No convergence step is left informal.

A second classical evaluation is

$$\prod_{n=0}^{\infty} \left(\frac{4n+1}{4n+3} \right)^{\varepsilon_n} = \frac{1}{2}. \quad (17.11)$$

It is the specialization $G(1/4, 3/4)$; it appears in the broader Thue–Morse gamma-product calculus of Allouche, Riasat, and Shallit [16], but it also follows from the Woods–Robbins evaluation alone, by one more parity split.

Proof of (17.11). In the notation of (17.8), split the product A by the parity of the index. Using $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$,

$$A = \prod_{j \geq 0} \left(\frac{4j+1}{4j+2} \right)^{\varepsilon_j} \prod_{j \geq 0} \left(\frac{4j+3}{4j+4} \right)^{-\varepsilon_j} = \prod_{j \geq 0} \left(\frac{4j+1}{4j+3} \right)^{\varepsilon_j} \prod_{j \geq 0} \left(\frac{4j+4}{4j+2} \right)^{\varepsilon_j},$$

the second step regrouping the four linear factors. Since $(4j+4)/(4j+2) = (2j+2)/(2j+1)$, the last product is A^{-1} , so the quarter product equals $A^2 = 1/2$ by [Theorem 17.4](#): *the quarter product is the square of the Woods–Robbins product*. As with all such rearrangements, the manipulation is an exact identity at the level of even-length partial sums of the logarithmic series, and passes to the limit because both series converge. $\square$

Formal versions. The evaluation (17.11) is fully formal, with the proof above (ThueMorseQuarterProduct.lean): the parity split is the exact partial-sum identity “an even-length Woods–Robbins partial sum equals the quarter-ratio partial sum minus the Woods–Robbins partial sum” (Fabius.wrA_two_mul), whence $L(1/4, 3/4) = -\log 2$ (Fabius.mpLimit_quarter) and the product converges to $1/2$ (Fabius.quarter_product').

17.4 Three dyadic block-product limits

Dyadic truncation respects exact block recursion. The following limits are convenient fingerprints of the same product theory:

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} \left(k + \frac{1}{2} \right)^{\varepsilon_k} = \frac{1}{2}, \quad (17.12)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{\varepsilon_k} = \frac{1}{\sqrt{2}}, \quad (17.13)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{(-1)^k \varepsilon_k} = \frac{1}{2\sqrt{2}}. \quad (17.14)$$

These limits are now *proved*, not merely recorded: over a dyadic block the even/odd split collapses the unbounded logarithms into the convergent sign-weighted ratio series. That split is isolated as a pair of hypothesis-free primitives over an arbitrary additive commutative group,

$$\sum_{k < 2N} \varepsilon_k g(k) = \sum_{j < N} \varepsilon_j (g(2j) - g(2j+1)), \quad \sum_{k < 2N} (-1)^k \varepsilon_k g(k) = \sum_{j < N} \varepsilon_j (g(2j) + g(2j+1))$$

(Fabius.sum_zsmul_thueMorseSign_two_mul and Fabius.sum_zsmul_alternating_thueMorseSign_two_mul, ThueMorseEulerTransform.lean); the alternating weight is exactly the one whose two halves add instead of cancelling, and the $\mathbb{Z}$ -module formulation makes the same two lines serve real logarithms, complex values, and formal coefficients alike. Applied to $g(k) = \log(k+1)$ they give the exact partial-sum identity $\sum_{k < 2N} \varepsilon_k \log(k+1) = L_A\text{-partial}(N)$

(`Fabius.block_log_one`), whence (17.13) follows from the Woods–Robbins limit (`Fabius.tendsto_block_product_one`); the alternating version telescopes into two Woods–Robbins partials – the $\log 2$ -contributions vanish by dyadic balance – giving (17.14) (`Fabius.tendsto_block_product_mixed`); and the half-shifted block sum is the master series at $(\frac{1}{4}, \frac{3}{4})$, so the limit in (17.12) is $\exp L(\frac{1}{4}, \frac{3}{4})$ (`Fabius.tendsto_block_product_half`, `ThueMorseBlockProducts.lean`); its numerical value $1/2$ is the quarter product (17.11), proved above, and the evaluated limit is formal as well (`Fabius.tendsto_block_product_half`). Pairing adjacent factors reduces the first two to $G(1/4, 3/4)$ and $G(1/2, 1)$, respectively; the third factors into their product. The evaluations are thus (17.11) and (17.7).

The dyadic scaffolding is removable, and the shift is free. Nothing in the argument uses that the block length is a power of two, and nothing uses that the shift is 1 or $1/2$. For every real $c > 0$ and every M ,

$$\sum_{k < 2M} \varepsilon_k \log(k + c) = \sum_{j < M} \varepsilon_j \log \frac{j + \frac{c}{2}}{j + \frac{c+1}{2}}, \quad (17.15)$$

the order- M partial sum of $L(\frac{c}{2}, \frac{c+1}{2})$ (`Fabius.block_log_shift`): the pairing $2j \mapsto 2j + 1$ uses only $2j + c = 2(j + \frac{c}{2})$ and $2j + 1 + c = 2(j + \frac{c+1}{2})$, so the two $\log 2$ terms cancel inside each pair and every unbounded term disappears at once. Exponentiating through `RealZPowProduct.lean` turns this into

$$\lim_{M \rightarrow \infty} \prod_{k=0}^{2M-1} (k + c)^{\varepsilon_k} = \exp L\left(\frac{c}{2}, \frac{c+1}{2}\right) \quad (c > 0), \quad (17.16)$$

a limit along *every* even block length, with no dyadic subsequence (`Fabius.tendsto_block_product_even`). The three fingerprints above are subsequences of (17.16) and of its alternating companion: (17.13) and (17.12) are the cases $c = 1$ and $c = 1/2$ at $M = 2^{m-1}$ (`Fabius.tendsto_block_product_one_even`, `Fabius.tendsto_block_product_half_even`), and the evaluated `Fabius.tendsto_block_product_half_even`, while (17.14) is the case $M = 2^{m-2}$ of the mixed product along every multiple of four (`Fabius.block_log_mixed_four`, `Fabius.tendsto_block_product_mixed_four`). The three original statements survive verbatim, each a three-line subsequence corollary of its general form.

At $c = 1$ the general collapse (17.15) lands on $(\frac{1}{2}, 1)$, and there the master series is *termwise* the Woods–Robbins series, since $\log \frac{n+1/2}{n+1} = \log \frac{2n+1}{2n+2}$ (`Fabius.mpLog_one_half_eq_wrA`). Hence

$$L(\tfrac{1}{2}, 1) = -\tfrac{1}{2} \log 2, \quad (17.17)$$

that is, $G(1/2, 1) = 1/\sqrt{2}$ (`Fabius.mpLimit_one_half_one`): [theorem 17.4](#) is the master log-series at $(\frac{1}{2}, 1)$, not merely a relative of it. Together with $L(1/4, 3/4) = -\log 2$ (`Fabius.mpLimit_quarter`) this evaluates (17.16) at both shifts the atlas uses.

18 Base- q digit characters and generalized Prouhet identities

Let $q \geq 2$, let $s_q(n)$ be the sum of the base- q digits of n , and let $\zeta \neq 1$ satisfy $\zeta^q = 1$. Define

$$u_n = \zeta^{s_q(n)}. \quad (18.1)$$

Theorem 18.1 (Generalized digit product). *For $0 \leq r < q$,*

$$u_{qn+r} = \zeta^r u_n. \quad (18.2)$$

For every $m \geq 0$,

$$\sum_{n=0}^{q^m-1} u_n z^n = \prod_{j=0}^{m-1} \left(\sum_{r=0}^{q-1} \zeta^r z^{r q^j} \right). \quad (18.3)$$

Proof. Appending the base- q digit r increases the digit sum by r . In the product, choose one digit independently at each position; the exponent is the corresponding base- q integer and the coefficient is ζ raised to its digit sum. $\square$

Each factor vanishes at $z = 1$, since $1 + \zeta + \cdots + \zeta^{q-1} = 0$.

Theorem 18.2 (Generalized Prouhet cancellation and sharp moment). *If p is a polynomial of degree less than m , then*

$$\sum_{n=0}^{q^m-1} \zeta^{s_q(n)} p(n) = 0. \quad (18.4)$$

At the first surviving degree,

$$\sum_{n=0}^{q^m-1} \zeta^{s_q(n)} n^m = m! \frac{q^{m(m+1)/2}}{(\zeta - 1)^m}. \quad (18.5)$$

Proof. Set $z = e^t$ in (18.3). Every factor has a zero at $t = 0$, proving cancellation below degree m . Its derivative at zero is

$$q^j \sum_{r=0}^{q-1} r \zeta^r = q^j \frac{q}{\zeta - 1} = \frac{q^{j+1}}{\zeta - 1}, \quad (18.6)$$

where the finite geometric sum was differentiated. Multiplying the m linear terms yields $q^{1+2+\cdots+m}/(\zeta - 1)^m$; coefficient extraction contributes $m!$. $\square$

For $q = 2$ and $\zeta = -1$, the sequence is $u_n = \varepsilon_n$ and (18.5) becomes $(-1)^m m! 2^{\binom{m}{2}}$, exactly (9.18). The binary theory is therefore the first member of a root-of-unity digit-character family, while the particularly rich valuation, carry, and Pascal-row formulas remain genuinely base-two features.

19 Formula selection, computational checks, and a Lean roadmap

19.1 Formalized modules (August 2026)

The first tranches of the formalization now exist as sixty-four compiled Lean modules in `Analysis/FabiusFunction`, complementing the corpus modules (`Parity.lean`, `TwoAdic.lean`, `ThueMorsePrefix.lean`, `ThueMorseGenerating.lean`, `ThueMorseExponential.lean`, `ThueMorseBinomialLog.lean`) that already covered the recursion, XOR, reflection, Lucas/Glaisher, binomial–logarithm, modulo-three formula, generating products, Prouhet moments, and prefix theory. The count is exactly the number of rows of the table below, one row per module; the proposal table of [Section 19.4](#) re-lists some of these same modules under the boundaries that were originally proposed for them, and is a roadmap rather than a second inventory:

Module	Contents
<code>ThueMorseBooleanCube.lean</code>	Powerset reindexing kernel, digit dictionary, two-parameter master product over any commutative semiring, signed product over any commutative ring, weight enumerator.
<code>ThueMorseValuation.lean</code>	Successor law, weight subadditivity, ruler product, Catalan valuation, additive Kummer, borrow cocycle.
<code>ThueMorseDigits.lean</code>	Additive Legendre, floor sums, floor-parity formula, ‘testBit’ bridge, digit product.
<code>ThueMorseEnumerators.lean</code>	Weight distribution, evil/odious enumerations and composition laws, XOR autocorrelation, signed Hamming spheres.

Module	Contents
<code>ThueMorseSparseProuhet.lean</code>	Arbitrary-weight Prouhet annihilation and sharp moment (theorem 9.4), Thue–Morse specializations, bit supports, sparse submask identities.
<code>ThueMorseRarefied.lean</code>	Modulo-three rarefaction recursion and closed vectors, Newman positivity at dyadic endpoints, block balance, even-modulus collapse.
<code>ThueMorseAperiodicity.lean</code>	Block-flip identities, promotion of an eventual period to a global one, and non-eventual-periodicity – the last now deduced from overlap-freeness rather than from the reflection argument.
<code>ThueMorseBlockAlgebra.lean</code>	Telescope and multiplicity ladder, degree and leading data of P_m , reciprocity (reverse and evaluation forms), cyclotomic multiplicity table, binomial-basis vanishing and sharp moment over any commutative ring.
<code>ThueMorseMoments.lean</code>	Hockey-stick column sum, complete binomial moment composition (corollary 9.12), reflection functional equation, midpoint parity selection rule, sharp centered moment; hypothesis-free weight and sign laws, the master products on arbitrary supports (including the arbitrary-exponent form), the exponential substitution, the complete sparse moment composition on an arbitrary support, and the complete power-moment composition formula as its dyadic-block case.
<code>ThueMorseArtinSchreier.lean</code>	Characteristic-two power-series Frobenius, the $\mathbb{F}_2$ bit series with its functional equation, Artin–Schreier quadratic and normalized forms, the complete two-element solution set $\{\Theta, \Theta + G\}$ with zero-constant-term uniqueness as its corollary, finite Artin–Schreier telescope.
<code>ThueMorseEulerTransform.lean</code>	Prefix collapse and ones-count formulas, general residue-class reindexing, ruler-convolution recurrence by finite combinatorics, interval discrepancy bounds.
<code>ThueMorseLucasSupport.lean</code>	Binary support of two-power sums, Lucas’s submask criterion (bit, $\&\&\&$, and support forms), odd Pascal positions as the submask cube, signed Pascal-support product, balance on odd positions.
<code>ThueMorseWalsh.lean</code>	All-ones digit patterns and masked two-power sums, ε as a Walsh character, character orthogonality on the block, the one-point Walsh spectrum.
<code>ThueMorseSparseMoments.lean</code>	The atlas’s all-sparse-moments formula on the submask cube (corollary 10.5), as the $S = J(n)$ instance of the arbitrary-support composition of <code>ThueMorseMoments.lean</code> .
<code>ThueMorseMixedDifference.lean</code>	The dyadic mixed-difference operator and the finite-difference factorization for arbitrary functions from an additive commutative monoid into an additive commutative group, with the powerset expansion.
<code>ThueMorseAutocorrelation.lean</code>	Finite autocorrelation recursion in exact integer form along <i>any even</i> window, $R_{2K}(2r) = 2R_K(r)$ and $R_{2K}(2r + 1) = -(R_K(r) + R_K(r + 1))$, with the dyadic layer $A_m(k) = R_{2^m}(k)$ as its power-of-two case: zero-shift value, trivial bound, closed values at shifts one and two, and the normalized limits at shifts zero, one and two.
<code>ThueMorseOverlapFree.lean</code>	Thue’s overlap-freeness theorem with cube-freeness, the attained critical exponent two, the alphabet-free implication “overlap-free $\Rightarrow$ not eventually periodic”, and constructive complement and reversal closure of the factor language.
<code>ThueMorseRecurrence.lean</code>	Full block-concatenation laws for weight and bit, all two-letter factors in every fourteen-letter window, uniform recurrence with an explicit linear bound.

Module	Contents
<code>ThueMorseMahler.lean</code>	Mahler equation and dyadic dissection for the signed series, cleared zero–one bridge, prefix generating function, finite-level autocorrelation Mahler equation.
<code>ThueMorseBooleanMobius.lean</code>	Lattice isomorphism of the two-power encoding, submask weight enumerators, reconstruction and inversion-polynomial identities, nested binomial formula, Boolean Möbius inversion on finsets and on submasks.
<code>ThueMorseMultinomial.lean</code>	Kummer’s criterion for multinomial coefficients, the multinomial support count $O_r(n) = r^{\text{wt}_2(n)}$, the signed r -nomial row formula, the exact r -nomial logarithm.
<code>ThueMorseFourier.lean</code>	The circle factor $ 1 - e^{i\theta} ^2 = 2 - 2 \cos \theta$ and the finite Riesz product with its nonnegativity.
<code>ThueMorseSineProduct.lean</code>	The half-angle factorization and the sine-product closed form of the Thue–Morse trigonometric polynomial.
<code>ThueMorseFourierInversion.lean</code>	Root-of-unity orthogonality and finite Fourier inversion over any field with a primitive root, specialized to the dyadic DFT over $\mathbb{C}$.
<code>ThueMorseParseval.lean</code>	Character orthonormality on $[0, 1]$ and the Parseval mass $\int_0^1 P_m(e^{2\pi i x}) ^2 dx = 2^m$.
<code>ThueMorseComplexity.lean</code>	Factor complexity: the supertile-pair window parametrization with the boundary-crossing case split, uniform-recurrence finiteness, and the sharpened two-sided bound $\ell + 1 \leq p(\ell) \leq 6\ell - 4$ – with $p(\ell) \leq 2 \cdot 2^k + 2(\ell - 1)$ at every covering level and the older $4 \cdot 2^k, p(\ell) < 8\ell$ kept as corollaries – and the Morse–Hedlund strict-growth argument.
<code>ThueMorseArithmeticFilter.lean</code>	The root-of-unity filter for arithmetic progressions over any field (division-free form), the boxed rarefied formula, and the even-modulus collapse.
<code>ToeplitzHessenberg.lean</code>	Cramer’s rule for a known solution over any commutative ring; the triangular and the lower Toeplitz–Hessenberg band matrices of arbitrary band data; the triangular determinant $\prod_{i < n} d_i$; and the expansion-free Toeplitz–Hessenberg determinant formula, read off from any solution of the convolution system, in matrix and in scalar-recurrence form.
<code>ThueMorseHessenberg.lean</code>	The ruler convolution as an instance of that band system: the triangular ruler system with determinant $n!$, the Hessenberg formula $\det H_n = (-1)^n n! \varepsilon_n$ for every $n \geq 0$ by one column rotation, and the sharp absolute value $ \det H_n = n!$.
<code>ThueMorsePartition.lean</code>	Weighted partitions, the marked-part recurrence of $\exp(\sum x_j z^j)$ over any characteristic-zero field, recurrence uniqueness, and the partition and Bell formulae for ε_n .
<code>ThueMorseIntegerLift.lean</code>	The integral algebraic lift constructed in closed Catalan–binomial form; Catalan parity at Mersenne indices; the unconditional parity theorem $c_n \equiv \tau(n), \varepsilon_n = (-1)^{c_n}$.
<code>ThueMorseBoundaryFlatness.lean</code>	The lacunary product $\mathcal{E}(t)$ as an infinite product: self-similarity, positivity, monotonicity, the effective upper bound $\mathcal{E}(t) \leq 2^{\binom{m}{2}} t^m$ at every level m , and the matching lower bound $2^{\binom{m}{2}} (t/2)^m \mathcal{E}(\frac{1}{2}) \leq \mathcal{E}(t)$ under the dyadic matching $\frac{1}{2} \leq 2^m t \leq 2$.
<code>ThueMorseDirichlet.lean</code>	The shifted Dirichlet series in the real regime: conditional convergence for every $\sigma > 0$ by Dirichlet’s test, and the dyadic parameter equation $D(\sigma, a) = 2^{-\sigma} (D(\sigma, \frac{a}{2}) - D(\sigma, \frac{a+1}{2}))$.

Module	Contents
<code>ThueMorseCatalanSeries.lean</code>	The diagonal Vandermonde identity $\sum_i C(i, p)C(M-i, q) = C(M+1, p+q+1)$ and the second-difference bridge from the lift to the substituted Catalan series.
<code>ThueMorseCatalanConvolution.lean</code>	The interior Catalan convolution and the quadratic identity $\sum_{p+q=m} h_p h_q = h_m - h_{m+1} - [m=0]$ of the substituted series.
<code>ThueMorseIntegerLiftEquation.lean</code>	The boxed algebraic equation $(1-z)^3 C^2 + (1-z)^2 C = z$ in $\mathbb{Z}[[z]]$ for the constructed lift, by identification with the substituted series and cancellation.
<code>ThueMorseIntegerLiftRecurrence.lean</code>	The linear ODE $(1+3z)(1-z)^2 C' = 1+3(1-z^2)C$ by differentiation and one linearization certificate, and the three-term recurrence by telescoping its coefficients.
<code>ThueMorseLacunaryQuotient.lean</code>	The dyadic geometric series and its inversion of $1-z^{2^j}$; the denominator-free bit Mahler equation; the lacunary quotient expansion at every finite level.
<code>ThueMorseInfiniteProduct.lean</code>	The analytic infinite-product identity $\sum \varepsilon_n z^n = \prod_j (1-z^{2^j})$ on the complex unit disc $\ z\ < 1$, including absolute convergence and multipliability; the real interval $ x < 1$ and exponential ray $\sum \varepsilon_n e^{-nt} = \mathcal{E}(t)$, $t > 0$, are retained as specializations.
<code>ThueMorseMellin.lean</code>	The Mellin representation $\Gamma(\sigma)D(\sigma, a) = \int_0^\infty t^{\sigma-1} e^{-at} \mathcal{E}(t) dt$ for $\sigma > 1$: kernel integrability, absolute interchange, and termwise Gamma evaluation.
<code>RealZPowProduct.lean</code>	Finite products of integer powers of positive reals through the exponential: the signed-logarithm identity, the limit transfer from logarithm sums to products along an arbitrary filter, and the closed-form value computations; nothing in it mentions Thue–Morse.
<code>ThueMorseWoodsRobbins.lean</code>	The Woods–Robbins evaluation $\prod ((2n+1)/(2n+2))^{\varepsilon_n} = 2^{-1/2}$: Dirichlet-test convergence of the three log-series, the termwise and interleaving identities, and exponentiation.
<code>ThueMorseMasterProduct.lean</code>	The master product $G(a, b)$: convergence for all positive parameters, the cocycle identities, the dyadic functional equation, and $G = \exp L$.
<code>ThueMorseBlockProducts.lean</code>	Block-product limits along <i>every even</i> block length and for an arbitrary positive shift c : the shift-general log collapse onto the master partials at $(\frac{c}{2}, \frac{c+1}{2})$, the identification of Woods–Robbins as the master series at $(\frac{1}{2}, 1)$, the alternating telescoping along every multiple of four, and the three dyadic limits as subsequence corollaries.
<code>ThueMorseEntireContinuation.lean</code>	The entire continuation: flatness of every order at zero, exponential decay, local integrability, entirety of $\Gamma^{-1} \cdot \text{Mellin}$, the trivial zeros at the nonpositive integers, agreement with $D(\sigma, a)$ on $\sigma > 1$, and the dyadic parameter equation at <i>every</i> complex s by the identity theorem.
<code>FrullaniIntegral.lean</code>	The Frullani integral $\int_0^\infty (e^{-pt} - e^{-qt}) t^{-1} dt = \log(q/p)$, by differentiation under the integral sign; not in Mathlib.
<code>ThueMorseGDirichlet.lean</code>	The derivative bridge $\log G(a, b) = D'(0, b) - D'(0, a)$: the closed integral form of $\log G$, $(1/\Gamma)'(0) = 1$, and the derivative of the entire continuation at zero.
<code>ThueMorseQuarterProduct.lean</code>	The quarter product $G(1/4, 3/4) = 1/2$ from Woods–Robbins by parity splitting, and the value $1/2$ of the half-shifted block product.

Module	Contents
<code>ThueMorseDiscSeries.lean</code>	The Thue–Morse power series as a holomorphic function on the unit disc: the complex-disc product wrapper $F(z) = \prod_{j=0}^{\infty} (1 - z^{2^j})$, locally uniform convergence, the analytic Mahler equation and its iterate, and the real-ray bound $ F(x) \leq 1 - x$.
<code>ThueMorseNaturalBoundary.lean</code>	The natural-boundary theorem: radial vanishing at dyadic roots of unity, two-power roots approximating every circle point, and the double identity-theorem argument.
<code>ThueMorseComputability.lean</code>	Primitive recursiveness: the descent-recursion scheme for course-of-values recursions, the halving identity for the binary weight, and the primitive recursive and computable forms of wt_2 , τ , and ε .
<code>ThueMorseNewman.lean</code>	Newman’s positivity theorem: residue-class sign sums, the modulo-three parity system, the base-four collapse with disjoint boundary corrections, and the strong induction.
<code>ThueMorseGelfond.lean</code>	Gelfond’s sharp uniform bound: the one-variable sum-of-squares inequality $\varphi^2(t)\varphi(2t) \leq 3\sqrt{3}$, the telescoped doubling-orbit product bound, and the exponential-sum form with constant $2/\sqrt{3}$.
<code>PeriodDoublingHankel.lean</code>	The mod-two Hankel engine: reusable block-determinant eliminations, parity interleavings, the period-doubling bit, and the four determinant halvings forcing every $\mathbb{F}_2$ -determinant to equal one.
<code>ThueMorseHankel.lean</code>	The Allouche–Peyrière–Wen–Wen theorem: the doubling factorizations $H_{2n} = 2^n H_n H_n(m)$, $H_{2n+1} = 2^n H_{n+1} H_n(m)$, oddness of the difference determinants, and the exact valuation $v_2(H_r) = r - 1$.
<code>LacunaryRieszIntegral.lean</code>	The exact orthogonality theory of lacunary Riesz products, with no Thue–Morse content: the frequency detector, the master theorem $\int_0^1 \cos(2\pi Kt + \psi) \prod_{j < n} (1 + a_j \cos(2\pi m_j t + \varphi_j)) dt$ under the dissociation hypothesis $ K + \sum_{i < j} m_i < m_j$, its super-increasing and Hadamard-gap corollaries, the mean value 1, and the exact identity $\int_0^1 \prod_{j < n} \sin^2(\pi m_j t) dt = 2^{-n}$ with its geometric and dyadic specializations.
<code>GelfondLogisticBound.lean</code>	The Gelfond rate from the logistic map, again with no Thue–Morse content: invariance of $[0, 1]$, the potential telescope giving $\prod_{j < n} u_j \leq \frac{5}{3}(\frac{3}{4})^n$ along any orbit from an initial value in $[0, 1]$, exactness of the rate at the fixed point $3/4$, the squared and unsquared dyadic sine-product bounds, and the L^1 lower bracket for the sine product.
<code>FourierProduct.lean</code>	The Rvachev sinc product and its identification with the Fourier transform, including the exact complex zero set $\Phi(z) = 0 \iff z \in \mathbb{Z} \setminus \{0\}$.
<code>RenormalizationIdentity.lean</code>	The standalone product identity $\Phi(z) = \text{sinc}(\pi z)\Phi(z/2)$ and its exact n -fold shell form.
<code>SincProductShells.lean</code>	Exact dyadic shell factorization of $ \Phi $ and its Gelfond-rate bound along every nonzero real dyadic ray.
<code>SincCanonicalProduct.lean</code>	The canonical product $\prod_{m \geq 1} (1 - z^2/m^2)^{1+\nu_2(m)}$, obtained by regrouping the absolutely convergent double Euler product.
<code>PhiZeroOrder.lean</code>	The exact positive-integer real-axis zero order $1 + \nu_2(m)$, expressed by a punctured norm quotient converging to a positive constant.
<code>ThueMorseLobeSign.lean</code>	The lobe identity $\Phi(x) = \varepsilon_m \Phi(x) $ for $m < x < m + 1$, from the closed lattice-point count.

Module	Contents
<code>RvachevProductContinuity.lean</code>	Real-axis continuity of Φ and $ \Phi $; the origin as the unique unit-modulus point, with strict inequality elsewhere; and existence and uniqueness of the modulus maximum on every positive lobe $(m, m+1)$ and reflected negative lobe $(-(m+1), -m)$, $m \geq 1$.

Wherever the Lean statement is stronger than the prose – arbitrary weights, arbitrary coefficient rings, arbitrary moduli and supports – the text above records the stronger form. The original proposal table further below is retained with its planned module boundaries for comparison; most rows have since been delivered under the module names above, as marked row by row.

19.2 Which representation is best?

The formulas are equivalent mathematically but not proof-theoretically or computationally. The following table records the most economical interface for common tasks.

Goal	Recommended representation	Reason
Executable evaluation	Recurrence or machine popcount	Linear in the bit length; no large intermediate integers.
Dyadic reindexing	Block concatenation (2.6)	Quotient and remainder separate without carries.
Bitwise algebra	XOR character (2.9)	Makes ε a genuine group character.
Ordinary addition	Carry cocycle (4.10)	Isolates the exact obstruction to XOR multiplicativity.
Number-theoretic disguise	Central binomial valuation (4.4)	One classical integer stores the full binary weight.
Pascal-row interpretation	Glaisher count or modulo-three residue	Uses the odd support of one row; the residue form avoids logarithms.
Higher-dimensional Pascal	Multinomial count (5.18)	Works with any number of variables.
Formal power series	Finite/infinite products	Coefficients, Mahler equations, and zeros become immediate.
Nonlocal coefficient formulas	Euler/Bell/determinant forms	Eliminate explicit digit notation and expose ruler convolutions.
Polynomial cancellation	Mixed finite differences	Proves all Prouhet moments in one operator identity.
Arbitrary Pascal row	Sparse signed product (10.2)	Restricts finite differences to the odd row support.
Finite harmonic analysis	Walsh character	The entire block is one pure frequency.
Ordinary spectral analysis	DFT sine product and Riesz density	Exposes roots of unity, amplitudes, and correlations.
Analytic continuation	Mellin integral (16.11)	Boundary flatness removes every Mellin singularity.
Infinite products	Master product $G(a, b)$	Pair reduction and cocycle laws organize evaluations.

Goal	Recommended representation	Reason
Fabius approximation	Iterated-prefix generating function	Coefficient grids arise from $E(z)/(1-z)^k$.

19.3 A compact verification script

The following pseudocode is not a proof, but it is useful for regression tests when translating the atlas into Lean, a CAS, or a library implementation.

```

for n from 0 to N:
  s := popcount(n)
  eps := 1 - 2*(s mod 2)

  assert v2(factorial(n)) == n - s
  assert v2(binomial(2*n,n)) == s

  oddPascal := count k in [0,n] with binomial(n,k) odd
  assert oddPascal == 2^s
  assert (oddPascal mod 3) - 1 == (s mod 2)

  sparseParity := sum binomial(n,2^j) over all 2^j <= n
  assert (-1)^sparseParity == eps

for m from 0 to M:
  P := sum_{n=0}^{2^m-1} eps(n)*z^n
  assert P == product_{j=0}^{m-1} (1-z^(2^j))

  for r from 0 to m-1:
    assert sum_{n=0}^{2^m-1} eps(n)*n^r == 0

  assert sum_{n=0}^{2^m-1} eps(n)*n^m
    == (-1)^m * factorial(m) * 2^(m*(m-1)/2)

```

19.4 A dependency-aware formalization order

A conservative Lean implementation should formalize the finite algebraic spine before introducing analysis.

1. **Binary layer.** Prove recurrence, block concatenation, XOR, reflection, and the two-kernel from a chosen binary-weight primitive.
2. **Valuation layer.** Establish the base-two Legendre formula, central-binomial valuation, Catalan valuation, and Kummer carry identity.
3. **Pascal/Frobenius layer.** Formalize Lucas support, Glaisher count, sparse probes, the modulo-three consequence, multinomial support, and Boolean Möbius inversion.
4. **Polynomial layer.** Define P_m , prove the finite product, reciprocity, cyclotomic multiplicities, and signed Pascal-support products over a suitable coefficient ring.
5. **Difference and moment layer.** Derive mixed finite differences, Prouhet cancellation, sharp moments, composition formulas, and sparse moments. Most statements can be done over $\mathbb{Q}$ before any topology is imported.

6. **Formal-series layer.** Add the Mahler equations, Euler transform, ruler convolution, Bell/determinant forms, and the Artin–Schreier equation over $\mathbb{F}_2[[z]]$.
7. **Analytic layer.** Only then introduce compact-uniform convergence, Fourier evaluation, Riesz measures, Mellin transforms, and infinite products.

Possible module boundaries are listed below. Entries marked *Delivered* name modules that already appear in the inventory of [Section 19.1](#); they are repeated here to record which proposed boundary each one realizes, and are not additional modules.

Proposed module	Main declarations
<code>ThueMorseValuation.lean</code>	Factorial, central-binomial, Catalan, successor, and carry formulae. <i>Delivered</i> .
<code>ThueMorsePascal.lean</code>	Lucas submasks, Glaisher count, sparse probes, modulo-three formula. <i>Delivered</i> (in <code>Parity.lean</code> , <code>ThueMorseBinomialLog.lean</code> , <code>ThueMorseLucasSupport.lean</code>).
<code>ThueMorseMultinomial.lean</code>	Frobenius factorization and $O_r(n) = r^{\text{wt}_2(n)}$. <i>Delivered</i> (via Kummer’s criterion rather than Frobenius).
<code>ThueMorseBooleanMobius.lean</code>	Submask incidence algebra and inverse parity matrix. <i>Delivered</i> (poset form; the matrix form is its row-by-row reading).
<code>ThueMorseGeneratingProduct.lean</code>	P_m , cyclotomic multiplicities, Mahler equations. <i>Delivered</i> (<code>ThueMorseBlockAlgebra.lean</code> , <code>ThueMorseMahler.lean</code>).
<code>ThueMorseProuhetMoments.lean</code>	Difference operators, all moments, centered parity. <i>Delivered</i> (<code>ThueMorseMoments.lean</code> , <code>ThueMorseMixedDifference.lean</code>).
<code>ThueMorseSparseProuhet.lean</code>	Arbitrary Pascal-support products and sparse moments. <i>Delivered</i> (plus <code>ThueMorseSparseMoments.lean</code>).
<code>ThueMorseFormalSeriesF2.lean</code>	Artin–Schreier equation and explicit solution. <i>Delivered</i> (<code>ThueMorseArtinSchreier.lean</code> , with uniqueness and the integer lift).
<code>ThueMorseFourier.lean</code>	Finite DFT, Walsh transform, product-to-sum identities. <i>Delivered</i> (Walsh, Riesz factorization and mass, sine-product closed form, and Fourier inversion are all formal).
<code>ThueMorseDirichlet.lean</code>	Boundary flatness, Mellin integral, entire continuation. <i>Delivered</i> (<code>ThueMorseBoundaryFlatness.lean</code> , <code>ThueMorseDirichlet.lean</code> , <code>ThueMorseMellin.lean</code> , <code>ThueMorseEntireContinuation.lean</code>).
<code>ThueMorseRarefied.lean</code>	Root-of-unity filters, modulo-three closed forms, odd-modulus matrix recursion. <i>Delivered in full</i> (<code>ThueMorseArithmeticFilter.lean</code> , <code>ThueMorseRarefied.lean</code>).
<code>ThueMorseWords.lean</code>	Uniform recurrence, aperiodicity, and the elementary complexity bounds. <i>Delivered in full</i> (<code>ThueMorseAperiodicity.lean</code> , <code>ThueMorseOverlapFree.lean</code> , <code>ThueMorseRecurrence.lean</code> , <code>ThueMorseComplexity.lean</code>).

Every planned layer has now been delivered, in most cases in greater generality than proposed.

19.5 Proof-dependency graph

The central dependencies can be summarized as

$$\begin{aligned}
 \text{binary digits} &\implies \begin{cases} \text{recursion and XOR,} \\ \text{Lucas/Frobenius support,} \\ \text{finite product } P_m, \end{cases} \\
 P_m &\implies \begin{cases} \text{Mahler and Euler equations,} \\ \text{Fourier/Walsh products,} \\ \text{order-}m \text{ zero at 1,} \end{cases} \\
 \text{order-}m \text{ zero} &\implies \begin{cases} \text{Prouhet cancellation,} \\ \text{all finite moments,} \\ \text{iterated-prefix flatness,} \\ \text{Riesz-product normalization.} \end{cases}
 \end{aligned} \tag{19.1}$$

Legendre/Kummer form a parallel arithmetic branch; Lucas and the signed support product join that branch to the finite-difference branch at sparse Prouhet identities. Boundary flatness of the infinite product then opens the Mellin/Dirichlet branch.

20 Worked examples and consistency checks

20.1 One index through many representations

Take $n = 13 = (1101)_2$. The occupied positions are 0, 2, 3, so

$$\text{wt}_2(13) = 3, \quad \tau(13) = 1, \quad \varepsilon_{13} = -1. \tag{20.1}$$

The floor, factorial, and central-binomial routes give

$$\sum_{j \geq 1} \left\lfloor \frac{13}{2^j} \right\rfloor = 6 + 3 + 1 = 10, \tag{20.2}$$

$$(-1)^{13 - \nu_2(13!)} = (-1)^3 = -1, \tag{20.3}$$

$$\nu_2 \binom{26}{13} = 3. \tag{20.4}$$

The distinguished Pascal entries are

$$\binom{13}{1} = 13, \quad \binom{13}{2} = 78, \quad \binom{13}{4} = 715, \quad \binom{13}{8} = 1287. \tag{20.5}$$

Their parities are 1, 0, 1, 1, exactly the four low bits of 13, and their sum 2093 is odd. Row 13 has

$$\nu(13) = 2^3 = 8 \tag{20.6}$$

odd entries, so $\nu(13) \bmod 3 = 2$ and the logarithm-free formula gives $\tau(13) = 2 - 1 = 1$. In three variables, exactly

$$O_3(13) = 3^3 = 27 \tag{20.7}$$

coefficients of $(x + y + z)^{13}$ are odd.

The submasks are

$$0, 1, 4, 5, 8, 9, 12, 13, \tag{20.8}$$

and their signed support polynomial is

$$R_{13}(z) = (1 - z)(1 - z^4)(1 - z^8). \quad (20.9)$$

Its derivatives of orders 0, 1, 2 at one vanish, and

$$\sum_{k \preccurlyeq_2 13} \varepsilon_k k^3 = -3! 2^{0+2+3} = -192. \quad (20.10)$$

All these evaluations recover the same three occupied bits through different algebraic interfaces.

20.2 An all-order dyadic moment

For $m = 3$ and $r = 5$, direct summation gives

$$M_{3,5} = \sum_{n=0}^7 \varepsilon_n n^5 = -6720. \quad (20.11)$$

The positive compositions of 5 into three parts are

$$(1, 1, 3), (1, 2, 2), (1, 3, 1), (2, 1, 2), (2, 2, 1), (3, 1, 1).$$

Formula (9.21) gives

$$M_{3,5} = -5! \sum_{\substack{p_0+p_1+p_2=5 \\ p_j \geq 1}} \frac{2^{p_1+2p_2}}{p_0!p_1!p_2!} = -6720. \quad (20.12)$$

No recursion through lower moments is needed.

20.3 A Fourier sample

For $m = 4$, every even DFT frequency vanishes. At $k = 1$,

$$\widehat{\varepsilon}_4(1) = (2i)^4 e^{-15\pi i/16} \sin \frac{\pi}{2} \sin \frac{\pi}{4} \sin \frac{\pi}{8} \sin \frac{\pi}{16}. \quad (20.13)$$

The factorization exposes both the exact phase and the hierarchy of four dyadic scales, which direct summation of sixteen complex terms obscures.

20.4 Autocorrelation from the recursion

Starting with $\eta(0) = 1$, the fixed-point equation at one gives $\eta(1) = -1/3$. Then

$$\begin{aligned} \eta(2) &= \eta(1) = -\frac{1}{3}, \\ \eta(3) &= -\frac{\eta(1) + \eta(2)}{2} = \frac{1}{3}, \\ \eta(5) &= -\frac{\eta(2) + \eta(3)}{2} = 0. \end{aligned}$$

These are simultaneously combinatorial averages and Fourier coefficients of the singular continuous Thue–Morse measure.

20.5 A complete four-bit block

At $m = 4$,

$$\begin{aligned} P_4(z) &= 1 - z - z^2 + z^3 - z^4 + z^5 + z^6 - z^7 \\ &\quad - z^8 + z^9 + z^{10} - z^{11} + z^{12} - z^{13} - z^{14} + z^{15} \\ &= (1 - z)(1 - z^2)(1 - z^4)(1 - z^8). \end{aligned} \quad (20.14)$$

The factor $(1 - z)^4$ divides the product, so

$$\sum_{n < 16} \varepsilon_n n^r = 0 \quad (0 \leq r < 4), \quad \sum_{n < 16} \varepsilon_n n^4 = 4! 2^{\binom{4}{2}} = 1536, \quad (20.15)$$

the sign being positive because $(-1)^4 = 1$. The residue-class sums modulo three are, by [theorem 15.1](#) at level $m = 4$,

$$(A_{4,0}^{(3)}, A_{4,1}^{(3)}, A_{4,2}^{(3)}) = (6, -3, -3). \quad (20.16)$$

Their total is zero, as block balance requires, yet the distribution among residue classes is already strongly skewed toward the multiples of three.

20.6 A carry-corrected addition

Take $a = 13 = (1101)_2$ and $b = 7 = (0111)_2$. Then $\varepsilon_{13} = \varepsilon_7 = -1$, while $a + b = 20 = (10100)_2$ has $\varepsilon_{20} = +1$. Kummer’s formula [\(4.9\)](#) counts

$$\nu_2 \binom{20}{13} = \text{wt}_2(13) + \text{wt}_2(7) - \text{wt}_2(20) = 3 + 3 - 2 = 4 \quad (20.17)$$

carries, and the correction $(-1)^4 = 1$ in [\(4.10\)](#) restores multiplicativity: $\varepsilon_{20} = (-1)(-1)(+1)$. The example shows why a yes/no record of “was there a carry” cannot suffice: the sign needs the carry count with propagation, taken modulo two.

21 Concluding synthesis

The formula in Section 11.8 of the Fabius–Rvachev exposition is not an isolated trick. It lies at the intersection of four structures:

binary digits $\longleftrightarrow$ Frobenius/Pascal support $\longleftrightarrow$ Boolean-lattice characters $\longleftrightarrow$ dyadic finite differences
--

(21.1)

The signed Thue–Morse sequence is the common coefficient system of this translation dictionary.

The arithmetic formulas hide digit parity in factorial valuations, central binomial coefficients, Catalan numbers, and carry counts. The Pascal formulas recover it either globally, from the number of odd row entries, or sparsely, from the power-of-two columns. The modulo-three identity shows that the exact base-two logarithm is optional: the exponent parity can also be read from the cycle of powers of two in a finite residue field. Multinomial support turns the same mechanism into a whole $r^{\text{wt}_2(n)}$ family.

The product $P_m(z) = \prod_{j < m} (1 - z^{2^j})$ is the central unifying engine. At $z = 1$ it has an exact high-order zero and produces Prouhet cancellation. At $z = e^t$ it produces every finite moment. At roots of unity it produces the dyadic DFT. On the Boolean cube it is a pure Walsh character. After taking absolute squares it produces Riesz densities. After division by $(1 - z)^k$ it produces iterated-prefix grids and their terminal flatness. After the limit $m \rightarrow \infty$, it becomes a Mahler function, an Euler transform of the ruler sequence, and the Mellin kernel of an entire Dirichlet family.

The added Mellin theory sharpens the analytic picture: the infinite product is not merely small near one, but flat beyond every algebraic order, with a quadratic leading term in $\log(1/t)$. That single estimate explains why all potential Mellin poles disappear and why the shifted Thue–Morse Dirichlet series has zeros at every nonpositive integer. Its derivative at zero then recovers the logarithms of the gamma-like products generated by Thue–Morse exponents.

For the Fabius project, the most useful lesson is proof-interface diversity. No one formula dominates every task. Block multiplicativity is ideal for dyadic reindexing; the carry cocycle for ordinary sums; Pascal support for characteristic-two arguments; finite differences for cancellation; Bell polynomials for all moments; Walsh characters for finite harmonic analysis; and the Mellin product for analytic continuation. Formalizing several of these interfaces would not duplicate mathematics: it would let later Fabius proofs choose the representation adapted to the operation being performed.

A The first 64 terms

The rows display $\tau(n)$; replacing 0 by +1 and 1 by −1 gives ε_n .

Range	successive values							
0–7	0	1	1	0	1	0	0	1
8–15	1	0	0	1	0	1	1	0
16–23	1	0	0	1	0	1	1	0
24–31	0	1	1	0	1	0	0	1
32–39	1	0	0	1	0	1	1	0
40–47	0	1	1	0	1	0	0	1
48–55	0	1	1	0	1	0	0	1
56–63	1	0	0	1	0	1	1	0

Every row of length eight is a copy or complement according to the sign of its block index, as predicted by (2.6).

B A compact identity dictionary

For quick reference, the principal exact translations are

$$\begin{aligned}
\tau(n) &\longleftrightarrow \varepsilon_n = 1 - 2\tau(n), \\
\varepsilon_n &\longleftrightarrow (-1)^{\text{wt}_2(n)}, \\
\text{wt}_2(n) &\longleftrightarrow n - \nu_2(n!), \\
\text{wt}_2(n) &\longleftrightarrow \nu_2\binom{2n}{n}, \\
b_j(n) &\longleftrightarrow \binom{n}{2^j} \bmod 2, \\
2^{\text{wt}_2(n)} &\longleftrightarrow \#\{k : \binom{n}{k} \text{ odd}\}, \\
\varepsilon_n &\longleftrightarrow [z^n] \prod_{j \geq 0} (1 - z^{2^j}), \\
\sum_{n < 2^m} \varepsilon_n f(x + n) &\longleftrightarrow (-1)^m \Delta_1 \Delta_2 \cdots \Delta_{2^{m-1}} f(x),
\end{aligned}$$

$$P_m(e^{-2\pi i k/2^m}) \longleftrightarrow \widehat{e}_m(k),$$

$$E(e^{-t}) \longleftrightarrow \Gamma(s)D(s, a) \text{ under Mellin transformation.}$$

Every arrow denotes an exact identity or invertible conversion, not an asymptotic analogy.

C Draft-to-section concordance

The four source drafts overlap substantially. The following table records the main distinctive material retained from each, together with additions made in the unified treatment.

Source	Distinctive contributions represented in this atlas
Draft 1	Carry and XOR viewpoints; Boolean Möbius inversion and inverse Pascal parity matrix; ruler Euler transform; partition, Bell, and Hessenberg determinant formulae; finite-difference positivity and analytic transforms.
Draft 2	Sparse power-of-two Pascal columns; all-order dyadic moments; automaton/tensor/Walsh forms; exact DFT and Riesz products; master infinite products; base- q root-of-unity extension.
Draft 3	Multinomial support and exact r -nomial logarithms; sparse Prouhet theorem on arbitrary Pascal rows; autocorrelation recurrences; Artin–Schreier series over $\mathbb{F}_2$; Catalan and fractional-part formulae.
Draft 4	Logarithm-free modulo-three Pascal formula; sinc-product interval signs; evil/odious enumerators; integer algebraic lift; trigonometric product-to-sum identities; explicit iterated-prefix/Fabius bridge.
Unified additions	Dyadic weight-distribution chapter; centered-moment parity selection; all sparse Pascal-support moments; elementary boundary asymptotic for $E(e^{-t})$; entire shifted Dirichlet family, its nonpositive-integer zeros, and its derivative bridge to exponent products; sharp first moment for the general base- q character.
Parallel atlas 1	Self-contained natural-boundary proof; supertile arguments for uniform recurrence and the linear complexity bound $p(\ell) < 8\ell$; the literature panorama of arithmetic subsequences (primes, squares, cubes, level of distribution).
Parallel atlas 2	Gelfond’s sharp uniform exponent with the lower bound at cube roots of unity; the odd-modulus matrix recursion; the autocorrelation Mahler equation; subtraction/borrow cocycle; weighted digit enumerator; Hankel/Padé framing of irrationality measure.
Parallel atlas 3	Exact modulo-three rarefaction vectors with the two-scale constants; Newman positivity framing; complement/reversal closure of the factor language; critical exponent statement; Mahler value transcendence and $\mu = 2$ exposition; four-bit block and carry-corrected worked examples.
Second-pass additions	Even-modulus collapse of rarefied sums from exact prefix identities; the characteristic-zero/characteristic-two reading of the lacunary digit-sum series (6.20); Morse–Hedlund lower complexity bound paired with the elementary upper bound.

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